Group Theory with Applications in Solid State Physics

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Contents

1 Character Theory **2**
- 1.1 Linear algebra recap . . . 2
  - 1.1.1 Matrix representation of linear map . . . 2
  - 1.1.2 Direct sum . . . 3
  - 1.1.3 Inner product, adjoint, and unitarity . . . 3
- 1.2 Basic concepts . . . 5
  - 1.2.1 Basic concepts of groups . . . 5
  - 1.2.2 Group action . . . 6
  - 1.2.3 Basic concepts of representations . . . 7
  - 1.2.4 Complete reducibility of linear representations of finite groups . . . 7
  - 1.2.5 The canonical representation . . . 9
- 1.3 Character theory . . . 9
  - 1.3.1 The character of a representation . . . 9
  - 1.3.2 Schur's lemma and its basic applications . . . 10
  - 1.3.3 The space of class functions . . . 11
  - 1.3.4 Orthogonality relations for characters . . . 11
  - 1.3.5 Decomposition of the canonical representation . . . 14
  - 1.3.6 Example: the character table for D_4 . . . 14
- 1.4 An application of character theory: block diagonalization . . . 15
  - 1.4.1 Canonical decomposition of a representation . . . 15
  - 1.4.2 Block diagonalization of symmetric Hamiltonian . . . 16

2 Induced Representations **18**
- 2.1 The group algebra . . . 18
  - 2.1.1 The group algebra $\mathbb{C}[G]$ . . . 18
  - 2.1.2 Decomposition of $\mathbb{C}[G]$ . . . 19
  - 2.1.3 The center of $\mathbb{C}[G]$ . . . 20
  - 2.1.4 Basic properties of integers . . . 21
  - 2.1.5 Integrality properties of characters. Applications . . . 22
- 2.2 Induced representations . . . 23
  - 2.2.1 Tensor product of modules . . . 23
  - 2.2.2 Induced representations . . . 24
  - 2.2.3 The character of an induced representation; the reciprocity formula . . . 25
  - 2.2.4 Restriction to subgroups . . . 26
  - 2.2.5 Mackey's irreducibility criterion . . . 27
- 2.3 Examples of induced representations . . . 28
  - 2.3.1 Normal subgroups; applications to the degrees of the irreducible representations . . . 28
  - 2.3.2 Semidirect products by an abelian group . . . 30

3 Applications in solid state physics **32**
- 3.1 Translation group and the Brillouin zone . . . 32
- 3.2 Space group and irreducible Brillouin zone . . . 35
- 3.3 Magnetic groups and corepresentations . . . 41
- 3.4 The $\vec{k} \cdot \vec{p}$ expansion . . . 44

Chapter 1

Character Theory

1.1 Linear algebra recap

Unless stated otherwise, vector spaces are finite dimensional over $\mathbb{C}$ and the representation-theoretic results in Sections 1–7 concern finite groups. Some motivating examples involve infinite groups; Section 8 explicitly introduces infinite translation groups and Hilbert spaces. We assume the reader is familiar with matrix computations. In this section, we reintroduce linear algebra in an "intrinsic" way.

1.1.1 Matrix representation of linear map

Let V be an n dimensional vector space and $F : V \to V$ be a linear map over V. Given a basis $\{e_i\}_{i=1}^n$ of V, we can associate a matrix $M(F)$ to F in the following way. Since $\{e_i\}_{i=1}^n$ is a basis of V, $F(e_i)$ can be expressed as a linear combination of $\{e_i\}_{i=1}^n$:

$$F(e_i) = \sum_{j=1}^{n} a_{ji} e_j. \tag{1.1}$$

We defined the matrix $M(F)$ to be

$$M(F)_{ji} = a_{ji} \tag{1.2}$$

An important property of this association is the following:

Proposition 1.1.1. *Let $F : V \to V$ and $G : V \to V$ be two linear maps and $\{e_i\}_{i=1}^n$ be a basis of V, $M(F)$, $M(G)$ and $M(F \circ G)$ be the matrices of F, G and $F \circ G$ under basis $\{e_i\}_{i=1}^n$. Then we have*

$$M(F \circ G) = M(F)M(G), \tag{1.3}$$

where $F \circ G$ (the composition of F and G) on the left hand side is defined as the linear map $(F \circ G)(x) = F(G(x))$ and multiplication on the the right hand side is matrix multiplication.

Proof. Exercise. □

The following problem gives the relation between the matrix of F under different basis.

Exercise 1.1.1. *Let $\{f_i\}_{i=1}^n$ be another basis of V, $f_i = \sum_{j=1}^n T_{ji} e_j$. Let $M(F), N(F)$ be the matrices of F under basis $\{e_i\}_{i=1}^n$, $\{f_i\}_{i=1}^n$, show that*

$$N(F) = T^{-1} M(F) T \tag{1.4}$$

Definition 1.1.1. *Let M be an n by n matrix, the trace of M is $\operatorname{tr}(M) = \sum_{i=1}^n M_{ii}$.*

Exercise 1.1.2. *Let M, N be two n by n matrices, show that $\operatorname{tr}(MN) = \operatorname{tr}(NM)$.*

Exercise 1.1.3. *For the $N(F)$ and $M(F)$ defined above, show that $\operatorname{tr}(N(F)) = \operatorname{tr}(M(F))$.*

Definition 1.1.2. *Let F be a linear map on V, $M(F)$ be the matrix of F under basis $\{e_i\}_{i=1}^n$. The trace of F is defined as $\operatorname{tr} F \equiv \sum_{i=1}^n M(F)_{ii}$.*

The reader may find that we didn't specify which basis we should choose in Definition 1.1.2. However, by Exercise 1.1.3, $\operatorname{tr} F$ takes the same value under any basis, hence the definition of $\operatorname{tr} F$ has no ambiguity.

Exercise 1.1.4. *Let F, G be two linear maps on V, show that $\operatorname{tr}(FG) = \operatorname{tr}(GF)$.*

1.1.2 Direct sum

We begin by the direct sum of vector spaces.

Definition 1.1.3. *Let V_1 and V_2 be two vector spaces, we can define a vector space structure on the set $V_1 \times V_2$ by defining*

$$\begin{aligned}(v_1, v_2) + (w_1, w_2) &\equiv (v_1 + w_1, v_2 + w_2) \\ c(v_1, v_2) &\equiv (cv_1, cv_2)\end{aligned} \tag{1.5}$$

This vector space is called the direct sum of V_1 and V_2 and denoted as $V_1 \oplus V_2$.

Exercise 1.1.5. *Construct a isomorphism between $(V_1 \oplus V_2) \oplus V_3$ and $V_1 \oplus (V_2 \oplus V_3)$ that is independent of choice of basis. Because of the existence of this isomorphism, we can identify $(V_1 \oplus V_2) \oplus V_3$ and $V_1 \oplus (V_2 \oplus V_3)$ and simply write them as $V_1 \oplus V_2 \oplus V_3$.*

Exercise 1.1.6. *Let V be a vector space and V_1 and V_2 be two subspaces. Define*

$$\begin{aligned}\pi : V_1 \oplus V_2 &\to V \\ (v_1, v_2) &\mapsto v_1 + v_2\end{aligned} \tag{1.6}$$

1. Show that π is injective if and only if $V_1 \cap V_2 = 0$.
2. Show that π is surjective if and only if $\forall v \in V$, $\exists v_1 \in V_1, v_2 \in V_2$ such that $v = v_1 + v_2$.
If π is bijective, we call V a direct sum of V_1 and V_2 and write $V = V_1 \oplus V_2$.

The direct sum decomposition of V are in one to one correspondence with the projection maps on V.

Definition 1.1.4. *A linear map $p : V \to V$ is called a projection if $p^2 = p$.*

Proposition 1.1.2. *If p, q are projections, then*
1. The map

$$\begin{aligned}\pi : \ker(p) \oplus \mathrm{Im}(p) &\to V \\ (x, y) &\mapsto x + y\end{aligned} \tag{1.7}$$

is an isomorphism.
2. If $\ker(p) = \ker(q)$ and $\mathrm{Im}(p) = \mathrm{Im}(q)$, then $p = q$.
3. If $V = V_1 \oplus V_2$, where $V_1, V_2 \subset V$, then we can find a projection p such that $\ker(p) = V_1$ and $\mathrm{Im}(p) = V_2$.
1,2,3 together tell us that the direct sum decompositions of V are in one to one correspondence with projections on V.

Proof. 1. If $v \in \ker(p) \cap \mathrm{Im}(p)$, then $v = p(w)$ for some $w \in V$ and $0 = p(v) = p^2(w) = p(w) = v$. Hence π is injective. $\forall v \in V$, we have $v = v - p(v) + p(v)$. $v - p(v) \in \ker(p), p(v) \in \mathrm{Im}(p)$, hence π is surjective.
2. $\forall v \in V$, $v = v - p(v) + p(v)$. Since $v - p(v) \in \ker(p) = \ker(q)$, we have $q(v) = q(p(v))$. Since $p(v) \in \mathrm{Im}(p) = \mathrm{Im}(q)$, we have $p(v) = q(w)$ for some $w \in V$, hence $q(v) = q(p(v)) = q(q(w)) = q(w) = p(v)$ and $p = q$.
3. If $V = V_1 \oplus V_2$, then for any $v \in V$, we can write v uniquely as $v = v_1 + v_2$, where $v_1 \in V_1$ and $v_2 \in V_2$. Define p to be $p(v) = v_2$. Then p is a projection and $\ker(p) = V_1$, $\mathrm{Im}(p) = V_2$. □

Proposition 1.1.2 will be used in proving the complete reducibility theorem. We now define the direct sum of linear maps, which will be useful in the character theory.

Definition 1.1.5. *Let $F : V_1 \to V_1$ and $G : V_2 \to V_2$ be two linear maps, the direct sum of F and G is a linear map*

$$\begin{aligned}F \oplus G : V_1 \oplus V_2 &\to V_1 \oplus V_2 \\ (v_1, v_2) &\mapsto (Fv_1, Gv_2)\end{aligned} \tag{1.8}$$

Exercise 1.1.7. *Show that $\mathrm{tr}(F \oplus G) = \mathrm{tr}(F) + \mathrm{tr}(G)$ and $\det(F \oplus G) = \det(F)\det(G)$.*

1.1.3 Inner product, adjoint, and unitarity

Definition 1.1.6. *Let V be a vector space over $\mathbb{C}$, an inner product on V is a map $\langle , \rangle : V \times V \to \mathbb{C}$ such that*
1. $\langle v, v \rangle \geq 0$ and $\langle v, v \rangle = 0$ if and only if $v = 0$.
2. $\langle v, w \rangle = \langle w, v \rangle^*$
3. $\langle v, a_1 w_1 + a_2 w_2 \rangle = a_1 \langle v, w_1 \rangle + a_2 \langle v, w_2 \rangle$

Definition 1.1.7. *A linear map $\alpha : V \to \mathbb{C}$ is called a functional.*

Let α, β be two functionals on V, $c \in \mathbb{C}$ be a complex number. The addition of α and β, $\alpha + \beta$, is a map $(\alpha + \beta)(v) = \alpha(v) + \beta(v)$. The scalar product $c\alpha$ is a map $(c\alpha)(v) = c\alpha(v)$.

Proposition 1.1.3. *The set of all functionals on V together with the addition and scalar product defined above form a vector space $V^\vee$. If V is finite dimensional, then $V^\vee$ is of the same dimension of V.*

Proof. Exercise. □

The vector space $V^\vee$ is called the dual space of V. Inner product provides a bijective map between V and $V^\vee$.

Proposition 1.1.4. *Let V be a finite dimensional vector space equipped with inner product $\langle , \rangle$ and $V^\vee$ be its dual. For any $v \in V$, consider the map $\phi(v) : V \to \mathbb{C}$, $\phi(v)(w) = \langle v, w \rangle, \forall w \in V$. Then $\phi(v) \in V^\vee$ and the map $\phi : V \to V^\vee$ is bijective.*

Proof. Exercise. □

The bijection ϕ above is conjugate-linear: $\phi(cv) = c^*\phi(v)$. It is therefore not a complex-linear isomorphism between V and its linear dual; this distinction follows from our inner-product convention.

Proposition 1.1.5. *Let V be a vector space equipped with inner product $\langle , \rangle_V$, W be a vector space equipped with inner product $\langle , \rangle_W$, for any linear map $A : V \to W$, there is a unique linear map $A^\dagger : W \to V$, called the adjoint of A, such that $\langle w, Av \rangle_W = \langle A^\dagger w, v \rangle_V, \forall v \in V, w \in W$.*

Proof. Exercise. □

Proposition 1.1.6. *Let V be a vector space with inner product $\langle , \rangle$, $\{v_i\}_{i=1}^n$ be an orthonormal basis of V (i.e. $\langle v_i, v_j \rangle = \delta_{ij}$). Let $A : V \to V$ be a linear map on V, whose matrix under the basis $\{v_i\}_{i=1}^n$ is $M(A)$, show that the matrix of $A^\dagger : V \to V$ under the basis $\{v_i\}_{i=1}^n$ is the conjugate transpose of $M(A)$ (denoted as $M(A)^H$).*

Proof. Exercise. □

Definition 1.1.8. *Let $A : V \to V$ be a linear map, A is said to be self-adjoint if $A = A^\dagger$*

In general, the eigenvalues of an operator are complex. However, the eigenvalues of a self-adjoint operator are guaranteed to be real.

Proposition 1.1.7. *Let A be a self-adjoint operator, all its eigenvalues are real.*

Proof. Exercise. □

Eigenvectors of a self-adjoint operator with distinct eigenvalues are orthogonal, and in finite dimension a self-adjoint operator admits an orthonormal eigenbasis.

Proposition 1.1.8. *Let A be a self-adjoint operator on V, a_1, a_2 be two distinct eigenvalues of A and v_1, v_2 be the corresponding eigenvectors. Then $\langle v_1, v_2 \rangle = 0$. If V is n dimensional, then there exists orthonormal basis $\{v_1, ...v_n\}$ of V and $\{a_1, ...a_n\}$ (a_i not necessarily distinct) $Av_i = a_i v_i$.*

Proof. Exercise. □

Definition 1.1.9. *Let $\langle , \rangle$ be an inner product on V. A linear map U on V is said to be unitary if $\forall v, w \in V$, $\langle Uv, Uw \rangle = \langle v, w \rangle$.*

Proposition 1.1.9. *If U is a unitary operator, then $U^\dagger U = 1$. If $\{e_i\}_{i=1}^n$ is an orthonormal basis of V, and $M(U)$ is the matrix of U under $\{e_i\}_{i=1}^n$, then $M(U)^H M(U) = I_n$.*

Proof. Exercise. □

Definition 1.1.10. *An n by n matrix A is called a unitary matrix if $A^H A = AA^H = I_n$.*

Proposition 1.1.9 says unitary maps is represented by unitary matrices under orthonormal basis.

Proposition 1.1.10. *Let U be a unitary operator on V, there exists orthonormal basis $\{v_1, ...v_n\}$ of V and $\{a_1, ...a_n\}$ (a_i not necessarily distinct) with $|a_i| = 1$ and $Uv_i = a_i v_i$.*

Proof. Exercise. □

1.2 Basic concepts

1.2.1 Basic concepts of groups

Definition 1.2.1. *A set G together with a map (called multiplication)*

$$\begin{aligned} \pi : G \times G &\to G \\ (s,t) &\mapsto s \circ t \end{aligned} \tag{1.9}$$

is a group if
1. $\forall r,s,t \in G$ we have $r \circ (s \circ t) = (r \circ s) \circ t$.
2. $\exists e \in G$ such that $e \circ g = g \circ e = g, \forall g \in G$.
3. $\forall g \in G$, $\exists h \in G$ such that $g \circ h = h \circ g = e$, h is usually denoted as g^{-1}.

We usually omit $\circ$ and write $s \circ t$ as st. The definition above is abstract. In real applications, as the following examples show, groups always come from the set of operations that keep a certain object unchanged.

Example 1.2.1. *Let $\mathbb{R}^2$ be the Euclidean plane, $d : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ be the distance function on $\mathbb{R}^2$, the set of distance preserving maps (a map $F : \mathbb{R}^2 \to \mathbb{R}^2$ is called distance preserving if $\forall x, y \in \mathbb{R}^2, d(F(x), F(y)) = d(x,y)$) forms a group, whose multiplication is the composition of maps. This group, denoted as $E(2)$, is called the Euclidean group. This group plays an important role in solid state physics. The elements of $E(2)$ are translations, rotations, reflections, and glide reflections, with the identity included as a zero translation or zero-angle rotation.*

Example 1.2.2. *Let $A = \{A_1, A_2, A_3, A_4\} \subset \mathbb{R}^2$ be the vertices of a square centered at the origin, with sides parallel to the coordinate axes. Let D_4 be the set of distance preserving maps on $\mathbb{R}^2$ that keeps the set A invariant (i.e. $F(A) = A$, where $F(A) \equiv \{F(x), x \in A\}$). Then D_4 forms a group under composition of maps.*

Example 1.2.3. *Let $\mathbb{Z}^2$ be the square lattice on $\mathbb{R}^2$, the set of distance preserving maps on $\mathbb{R}^2$ that keeps the $\mathbb{Z}^2$ invariant forms a group under composition of maps. This group is called the space group of the square lattice.*

Definition 1.2.2. *A subset $H \subset G$ is called a subgroup of G if H itself forms a group under the group multiplication of G.*

Proposition 1.2.1. *A nonempty subset H is a subgroup of G if and only if $\forall s, t \in H$, $s \circ t \in H$ and $s^{-1} \in H$.*

Proof. Exercise. □

Example 1.2.4. *The space group of the square lattice is a subgroup of $E(2)$, D_4 is also a subgroup of $E(2)$.*

Definition 1.2.3. *Let H be a subgroup of G. A subset T of G is said to be a coset of H if $\exists g \in G$ such that $T = gH \equiv \{gh, h \in H\}$.*

Proposition 1.2.2. *Let R and T be two cosets of H, then either $R = T$ or $R \cap T = \emptyset$.*

Proof. Exercise. □

Proposition 1.2.3. *Let R and T be two cosets of H, there is a bijective map between R and T. (Roughly speaking, this tells us that R and T have the same number of elements.)*

Proof. Exercise. □

Proposition 1.2.4. *Cosets of H gives a partition of G, i.e., let Q be the collection of cosets of H, then the union $\cup_{T \in Q} T$ is disjoint and $\cup_{T \in Q} T = G$.*

Proof. Use Proposition 1.2.2. □

Definition 1.2.4. *Two elements $s, t \in G$ are said to be conjugate if there exists $g \in G$ such that $s = g^{-1}tg$.*

The following examples tell that you should think of conjugate elements as elements of the same type.

Example 1.2.5. *Let $GL(n, \mathbb{C})$ be the set of $n \times n$ invertible matrix. Then $GL(n, \mathbb{C})$ forms a group under matrix multiplication. If two matrices M, N are conjugate, then by definition, we can find another matrix $T \in GL(n, \mathbb{C})$ such that $N = T^{-1}MT$. From Eq. (1.4), we know that M and N can be viewed as matrix representation of the same linear map under different basis, hence we should really think M and N are essentially the same.*

Example 1.2.6. *Conjugation in $E(2)$ changes the position and orientation of a geometric operation while preserving its metric data. All reflections are conjugate. Two nonzero translations are conjugate precisely when their translation vectors have the same length, and two nonidentity rotations are conjugate precisely when their angles agree up to sign modulo 2π. Thus operations of the same broad type need not be conjugate; translations of different lengths provide a simple counterexample. Glide reflections must also be included among Euclidean isometries.*

The following proposition tells the reason why conjugate elements can be viewed as the same.

Proposition 1.2.5. *Conjugacy is an equivalence relation, i.e.*
1. g is conjugate to g.
2. If g is conjugate to h then h is conjugate to g.
3. If g is conjugate to h, h is conjugate to k, then g is conjugate to k.

Proof. Exercise. □

Definition 1.2.5. *Let G and H be two groups. A map $\varphi : G \to H$ is called a homomorphism if $\varphi(st) = \varphi(s)\varphi(t), \forall s, t \in G$.*

Example 1.2.7. *The determinant map $\det : GL(n, \mathbb{C}) \to \mathbb{C}^\times$ is a homomorphism, where $\mathbb{C}^\times$ is the multiplicative group of nonzero complex numbers.*

Definition 1.2.6. *Let $\varphi : G \to H$ be a homomorphism, the kernel of φ is defined as $\ker(\varphi) \equiv \{g \in G, \varphi(g) = e\}$.*

Definition 1.2.7. *Let N be a subgroup of G, N is called a normal subgroup of G if $\forall g \in G, n \in N$, we have $g^{-1}ng \in N$.*

Proposition 1.2.6. *Let $\varphi : G \to H$ be a homomorphism, $\ker(\varphi)$ is a normal subgroup of G.*

Proof. Exercise. □

Proposition 1.2.7. *If N is a normal subgroup of G, let G/N be the collection of cosets of N. Let $\pi : G \to G/N$ be the map that sends $g \in G$ to the coset of N that contains g. There exists a map*

$$\begin{aligned} \varphi : G/N \times G/N &\to G/N \\ (S, T) &\mapsto ST \end{aligned} \tag{1.10}$$

such that G/N forms a group under the map φ and π gives a homomorphism from G to G/N. The group G/N is called the quotient group of G by N.

Proof. Exercise. □

Definition 1.2.8. *An isomorphism is a bijective homomorphism whose inverse is also a homomorphism.*

Proposition 1.2.8. *Let $\varphi : G \to H$ be a homomorphism. Let $\mathrm{Im}(\varphi) = \{\varphi(g), g \in G\}$, then there exists an isomorphism $\eta : G/\ker(\varphi) \to \mathrm{Im}(\varphi)$ such that $\varphi = \eta \circ \pi$, where π is the homomorphism defined in Proposition 1.2.7.*

Proof. Exercise. □

1.2.2 Group action

Here we introduce the most important concept in group theory, the group action.

Definition 1.2.9. *Let S be a set, a left action of a group G on S is a map*

$$\begin{aligned} \pi : G \times S &\to S \\ (g, s) &\mapsto gs \end{aligned} \tag{1.11}$$

with the property
1. $es = s, \forall s \in S$
2. $g(hs) = (gh)s, \forall g, h \in G, s \in S$.

Example 1.2.8. *Let $E(2)$ be the Euclidean group, the map defined by*

$$\begin{aligned} \pi : E(2) \times \mathbb{R}^2 &\to \mathbb{R}^2 \\ (F, x) &\mapsto F(x) \end{aligned} \tag{1.12}$$

is an action of $E(2)$ on $\mathbb{R}^2$.

Example 1.2.9. *Let $SO(3)$ be the set of real orthogonal matrices ($g^T g = I_3$) with determinant 1. Let V be the set consisting of all functions $f : \mathbb{R}^3 \to \mathbb{C}$. $\forall g \in SO(3), f \in V$, define the function gf to be*

$$(gf)(x) = f(g^{-1}x), \forall x \in \mathbb{R}^3. \tag{1.13}$$

We have

$$(g(hf))(x) = (hf)(g^{-1}x) = f(h^{-1}g^{-1}x) = f((gh)^{-1}x) = ((gh)f)(x). \tag{1.14}$$

Hence, $g(hf) = (gh)f$. $(gf)(x) = f(g^{-1}x)$ is a action of $SO(3)$ on V.

Definition 1.2.10. *Let V be a vector space, $GL(V)$ be the set of invertible linear maps on V, G be a group. A linear representation of G is a homomorphism $\rho : G \to GL(V)$.*

Example 1.2.10. *Once we have a representation ρ of G, we can define the action of G on V by*

$$\begin{aligned} \pi : G \times V &\to V \\ (g, v) &\mapsto \rho(g)v \end{aligned} \tag{1.15}$$

Definition 1.2.11. *A vector space V together with an action of G on V is called a G-module if for any $g \in G$, the map*

$$\begin{aligned} \pi_g : V &\to V \\ v &\mapsto gv \end{aligned} \tag{1.16}$$

is a linear map.

G-modules and representations are in one to one correspondence.

Exercise 1.2.1. *Let $\mathcal{M}$ be the set of G-modules and $\mathcal{R}$ be the set of linear representations of G. Construct a bijective map between $\mathcal{M}$ and $\mathcal{R}$.*

Because of Exercise 1.2.1, we will identify a G-module with a representation of G and simply say V is a representation of G.

1.2.3 Basic concepts of representations

We begin by defining isomorphic representations.

Definition 1.2.12. *Let $\rho_1 : G \to GL(V)$ and $\rho_2 : G \to GL(W)$ be two representations of G, we say ρ_1 and ρ_2 are isomorphic if there is an isomorphism $F : V \to W$ so that $\forall g \in G, v \in V$, we have $F(\rho_1(g)v) = \rho_2(g)(Fv)$.*

The following exercise tells us why we should really think of ρ_1 and ρ_2 as "the same" representations.

Exercise 1.2.2. *If ρ_1 and ρ_2 are isomorphic, and V and W are n dimensional, we can find a base $\{v_i\}_{i=1}^n$ of V and a base $\{w_i\}_{i=1}^n$ of W so that $\forall g \in G$, the matrix of $\rho_1(g)$ under the base $\{v_i\}_{i=1}^n$ and the matrix of $\rho_2(g)$ under the base $\{w_i\}_{i=1}^n$ are the same.*

Definition 1.2.13. *Let V be a G-module, $W \subset V$ be a subspace, W is said to be a submodule (subrepresentation) if $\forall g \in G, w \in W$, $gw \in W$.*

Exercise 1.2.3. *If W is a submodule of V, then we can define a map*

$$\begin{aligned} G \times W &\to W \\ (g, w) &\mapsto gw \end{aligned} \tag{1.17}$$

This map makes W a G-module. The corresponding representation is denoted as $\rho|_W$.

1.2.4 Complete reducibility of linear representations of finite groups

The goal of linear representation theory is to classify all possible vector spaces that "has the symmetry" of G, i.e., classify all G-modules. For general groups, this is almost impossible. Here we begin to introduce the methods classifying the modules of finite groups. We begin by introducing the most important concept: the irreducible representation.

Definition 1.2.14. *A G-module V is said to be irreducible if $V \neq 0$ and V has no submodule except 0 and itself. Such module is also called an irreducible representation of G.*

We now show that every finite dimensional G-module V can be decomposed into direct sum of irreducible representations of G. First of all, we define the direct sum of representations.

Definition 1.2.15. *Let (ρ_1, W_1),..., (ρ_k, W_k) be representations of G. We define the direct sum representation of W_1,...,W_k by*

$$\begin{aligned} \rho_1 \oplus ... \oplus \rho_k : G \times (W_1 \oplus ... \oplus W_k) &\to W_1 \oplus ... \oplus W_k \\ (g, (w_1, ..., w_k)) &\mapsto (\rho_1(g)w_1, ..., \rho_k(g)w_k) \end{aligned} \tag{1.18}$$

Proposition 1.2.9. *Let (ρ, V) be a representation of G, $(\rho|_W, W)$ be a subrepresentation of G, then there exists a subrepresentation $(\rho|_{W'}, W')$ of V such that the representation $\rho|_W \oplus \rho|_{W'}$ is isomorphic to ρ.*

Proof. Let $\{v_1, ..., v_m\}$ be a base of W, we can extend it into a base $\{v_1, ..., v_m, ..., v_n\}$ of V. Let $U = \mathrm{Span}\{v_{m+1}, ..., v_n\}$. Then it is easy to check that $V = W \oplus U$. By Proposition 1.1.2, we know there exists a projection P whose kernel is U, and whose image is W. Now we introduce the most important trick: **average over group**. Define

$$Q = \frac{1}{|G|} \sum_{g \in G} \rho(g^{-1}) P \rho(g). \tag{1.19}$$

We claim that Q is a projection. For every $v \in V$ and $g \in G$, the vector $P\rho(g)v$ lies in W, and W is invariant under $\rho(g^{-1})$. Thus every term of the average defining Qv lies in W, so $\mathrm{Im}\, Q \subset W$. If $w \in W$, then $P\rho(g)w = \rho(g)w$ for every g, and consequently $Qw = w$. This proves $\mathrm{Im}\, Q = W$. Applying this identity to $Qv \in W$ gives $Q^2 v = Qv$, so $Q^2 = Q$. Let $W' = \ker Q$. We claim that W' is a subrepresentation of V. $\forall w' \in W', h \in G$, we want to show $\rho(h)w' \in W'$, i.e., $Q\rho(h)w' = 0$. Indeed,

$$\begin{aligned} Q\rho(h)w' &= \frac{1}{|G|} \sum_{g \in G} \rho(g^{-1}) P \rho(g) \rho(h) w' \\ &= \frac{1}{|G|} \sum_{g \in G} \rho(h) \rho(h^{-1}) \rho(g^{-1}) P \rho(gh) w' \\ &= \frac{1}{|G|} \sum_{g \in G} \rho(h) \rho(h^{-1} g^{-1}) P \rho(gh) w' \\ &= \frac{1}{|G|} \sum_{g \in G} \rho(h) \rho((gh)^{-1}) P \rho(gh) w', \end{aligned} \tag{1.20}$$

where we used $h^{-1}g^{-1} = (gh)^{-1}$ in the last step. Now, the **key observation** is that if g runs over G, gh also runs over G. We actually have

$$\begin{aligned} Q\rho(h)w' &= \frac{1}{|G|} \sum_{g \in G} \rho(h) \rho((gh)^{-1}) P \rho(gh) w' \\ &= \rho(h) \frac{1}{|G|} \sum_{g \in G} \rho((gh)^{-1}) P \rho(gh) w' \\ &= \rho(h) \frac{1}{|G|} \sum_{t \in G} \rho(t^{-1}) P \rho(t) w' \\ &= \rho(h) Q w' \\ &= 0. \end{aligned} \tag{1.21}$$

The last step is because $w' \in W' = \ker Q$. Hence W' is also a subrepresentation of G. Now consider the map

$$\begin{aligned} \pi : W \oplus W' &\to V \\ (w, w') &\mapsto w + w' \end{aligned} \tag{1.22}$$

From Proposition 1.1.2 we know π is an isomorphism between vector spaces. $\forall w \in W, w' \in W', g \in G$, we have

$$\begin{aligned} \pi((\rho|_W \oplus \rho|_{W'})(g)((w, w'))) &= \pi(\rho(g)w, \rho(g)w') \\ &= \rho(g)w + \rho(g)w' \\ &= \rho(g)(w + w') \\ &= \rho(g)\pi((w, w')). \end{aligned} \tag{1.23}$$

Hence, $\pi(\rho|_W \oplus \rho|_{W'})(g) = \rho(g)\pi$, and $\rho|_W \oplus \rho|_{W'}$ is isomorphic to ρ. □

The following exercise is an application of the trick of average over group.

Exercise 1.2.4. *Let (ρ, V) be a representation of G, the goal of this exercise is to show that we can always find a base $\{v_i\}_{i=1}^n$ of V such that $\forall g \in G$, the matrix of $\rho(g)$ under $\{v_i\}_{i=1}^n$ is unitary.*
Let $\{w_i\}_{i=1}^n$ be an arbitrary base of V, define a inner product $\langle , \rangle$ on V by

$$\langle \sum_{i=1}^n a_i w_i, \sum_{i=1}^n b_i w_i \rangle \equiv \sum_{i=1}^n a_i^* b_i. \tag{1.24}$$

Let $|G|$ be the number of elements in G and define

$$\langle u, v \rangle_1 \equiv \frac{1}{|G|} \sum_{s \in G} \langle \rho(s)u, \rho(s)v \rangle. \tag{1.25}$$

1. Show that $\langle , \rangle_1$ is also an inner product.
2. Show that $\langle \rho(h)u, \rho(h)v \rangle_1 = \langle u, v \rangle_1, \forall h \in G, u, v \in V$.
3. Let $\{v_i\}_{i=1}^n$ be an orthonormal base of V under the inner product $\langle , \rangle_1$, show that $\forall g \in G$, the matrix of $\rho(g)$ under $\{v_i\}_{i=1}^n$ is unitary.

Theorem 1.2.1. *Let V be a finite dimensional representation of a finite group G. Then V is isomorphic to a direct sum irreducible representations of G.*

Proof. We prove by doing induction on the dimension of V. $\dim V = 0$ is obvious as it is the direct sum of empty family of irreducible representations. Suppose the theorem is true for $\dim V \leq n$. Then for $\dim V = n + 1$, if V is already irreducible, we are done. If V is not irreducible, by definition, we can always find a nonzero proper subrepresentation $W \subset V$. Then use Proposition 1.2.9, we can find a subrepresentation $W' \subset V$ so that the representation $W \oplus W'$ is isomorphic to V (we will simply write it as $V = W \oplus W'$). Since $\dim W \leq n, \dim W' \leq n$, we can apply the induction hypothesis to W and W' and prove that V is isomorphic to a direct sum of irreducible representations. □

Because of this theorem, we only need to focus on the irreducible representations, and we say linear representation of finite groups is completely reducible.

1.2.5 The canonical representation

In this section we introduce the most important example in the theory of linear representation of finite group: the canonical representation.

Example 1.2.11. *Let V be the set consisting of all functions $f : G \to \mathbb{C}$. $\forall f, h \in V$, define $(f + h)(g) = f(g) + h(g), \forall g \in G$ and $(cf)(g) = cf(g), \forall g \in G, c \in \mathbb{C}$. These operations make V a vector space. Now, $\forall g \in G, f \in V$, define gf to be the function*

$$(gf)(h) = f(g^{-1}h), \forall h \in G. \tag{1.26}$$

Using the same argument in Example 1.2.9, we know that this makes V a G-module. This representation is called the canonical representation of G.

Exercise 1.2.5. *Show that V is $|G|$ dimensional, where $|G|$ is the number of elements in G by showing that the functions $\{\delta_s\}_{s \in G}$, where $\delta_s(g) = 1$ if $g = s$ and $\delta_s(g) = 0$ if $g \neq s$ is a base of V.*

Exercise 1.2.6. *Show that $g\delta_s = \delta_{gs}$.*

Example 1.2.11 is of fundamental importance in the representation theory a finite groups. As will be shown in the next chapter, the canonical representation contains all irreducible representations of a finite group G.

1.3 Character theory

1.3.1 The character of a representation

Definition 1.3.1. *Let $\rho : G \to GL(V)$ be a representation, the character of ρ is a map*

$$\begin{aligned} \chi_\rho : & G \to \mathbb{C} \\ & g \mapsto \mathrm{tr}(\rho(g)) \end{aligned} \tag{1.27}$$

Proposition 1.3.1. *Let $\rho : G \to GL(V)$ be an n dimensional representation ($\dim V = n$). Then we have,*
1. $\chi_\rho(e) = n$, where e is the identity element in G.
2. $\chi_\rho(s^{-1}) = \chi_\rho(s)^*$.
3. $\chi_\rho(t^{-1}st) = \chi_\rho(s)$.

Proof. 1. $\rho(e)\rho(e) = \rho(ee) = \rho(e)$, since $\rho(e) \in GL(V)$, we can multiply $\rho(e)^{-1}$ on both side and obtain $\rho(e) = \mathrm{Id}$, where Id is the identity map on V. The matrix of Id under any base is the unit n by n matrix I_n. Hence $\chi_\rho(e) = \mathrm{tr}(\mathrm{Id}) = \sum_{i=1}^n (I_n)_{ii} = \sum_{i=1}^n 1 = n$.
2. From Exercise 1.2.4, we know that we can find a base $\{v_i\}_{i=1}^n$ of V such that the matrix of $\rho(s)$, denoted by $M(\rho(s))$ is a unitary matrix. Using Proposition 1.1.1, we know that

$$M(\rho(s^{-1}))M(\rho(s)) = M(\rho(s^{-1})\rho(s)) = M(\rho(s^{-1}s)) = M(\rho(e)) = I_n, \tag{1.28}$$

Hence, $M(\rho(s^{-1})) = M(\rho(s))^{-1} = M(\rho(s))^H$, where the last step is from Proposition 1.1.9. Now we can compute

$$\chi_\rho(s^{-1}) = \mathrm{tr}(\rho(s^{-1})) = \sum_{i=1}^n M(\rho(s^{-1}))_{ii} = \sum_{i=1}^n (M(\rho(s))^H)_{ii} = \sum_{i=1}^n (M(\rho(s))_{ii})^* = \chi_\rho(s)^*. \tag{1.29}$$

3. Using $\mathrm{tr}(FG) = \mathrm{tr}(GF)$, we see that

$$\chi_\rho(t^{-1}st) = \mathrm{tr}(\rho(t^{-1})\rho(s)\rho(t)) = \mathrm{tr}(\rho(s)\rho(t)\rho(t^{-1})) = \mathrm{tr}(\rho(s)) = \chi_\rho(s). \tag{1.30}$$

□

Proposition 1.3.2. *Let (ρ, V) be a representation of G, and (ρ_1, V_1) and (ρ_2, V_2) be two subrepresentations of V and $V = V_1 \oplus V_2$. The characters of ρ, ρ_1, ρ_2 are χ, χ_1, χ_2. We have $\chi = \chi_1 + \chi_2$*

Proof. Let $\{e_i\}_{i=1}^m$ be a base of V_1 and $\{f_i\}_{i=1}^n$ be a base of V_2. Then $\{e_1, ..., e_m, f_1, ..., f_n\}$ form a base of V. $\forall s \in G$, write down the matrix of $\rho(s)$ with respect to $\{e_1, ..., e_m, f_1, ..., f_n\}$ and the result follows. □

1.3.2 Schur's lemma and its basic applications

In this section, we first introduce the Schur's lemma, which is of **fundamental importance** in group representation theory. We then talk about its basic applications.

Proposition 1.3.3. *Let $\rho_1 : G \to GL(V_1)$, $\rho_2 : G \to GL(V_2)$ be two irreducible representations. Let $F : V_1 \to V_2$ be a linear map such that $\forall s \in G$, $F\rho_1(s) = \rho_2(s)F$.*
1. If ρ_1 and ρ_2 are not isomorphic, then $F = 0$.
2. If $V_1 = V_2 = V$, $\rho_1 = \rho_2 = \rho$, then $F = \lambda\,\mathrm{Id}$.

Proof. 1. We begin by showing that $\ker(F)$ is a subrepresentation of V_1 and $\mathrm{Im}(F)$ is a subrepresentation of V_2. $\forall v_1 \in \ker(F), s \in G$, $F\rho_1(s)v_1 = \rho_2(s)Fv_1 = 0$, hence $\rho_1(s)v_1 \in \ker(F)$ and $\ker(F)$ is a subrepresentation of V_1. Since V_1 is irreducible, $\ker(F) = 0$ or $\ker(F) = V_1$. $\forall Fv_1 \in \mathrm{Im}(F), s \in G$, $\rho_2(s)Fv_1 = F\rho_1(s)v_1 \in \mathrm{Im}(F)$, hence $\mathrm{Im}(F)$ is a subrepresentation of V_2. Since V_2 is irreducible, $\mathrm{Im}(F) = 0$ or $\mathrm{Im}(F) = V_2$. If $\ker(F) = 0$, then we must have $\mathrm{Im}(F) = V_2$, this implies F is an isomorphism between ρ_1 and ρ_2, contradicts with the assumption that ρ_1 and ρ_2 are not isomorphic. So we must have $\ker(F) = V_1$ and hence $F = 0$.
2. Let λ be an eigenvalue of F (we know it must exist as we are working on vector spaces over $\mathbb{C}$), consider $G = F - \lambda\,\mathrm{Id}$. Then $G\rho(s) = \rho(s)G$. Repeating the argument used in the first part we know either $\ker(G) = 0$ or $\ker(G) = V$. However, G by construction has zero eigenvalue and $\ker(G) \neq 0$. Therefore, $\ker(G) = V$, $G = 0$ and $F = \lambda\,\mathrm{Id}$. □

Corollary 1.3.1. *Let $\rho_1 : G \to GL(V_1)$, $\rho_2 : G \to GL(V_2)$ be two irreducible representations, H be a linear map from V_1 to V_2, and $|G|$ be the number of elements in G. Let*

$$J = \frac{1}{|G|}\sum_{s\in G} \rho_2(s^{-1})H\rho_1(s). \tag{1.31}$$

1. If ρ_1 and ρ_2 are not isomorphic, $J = 0$.
2. If $V_1 = V_2 = V$, $\dim V = n$, and $\rho_1 = \rho_2 = \rho$, then

$$J = \frac{1}{n}\,\mathrm{tr}(H)\,\mathrm{Id}\,. \tag{1.32}$$

Proof. 1. Using the same trick as that in Proposition 1.2.9, we can show that $\rho_2(s)J = J\rho_1(s), \forall s \in G$. Then use Proposition 1.3.3 we know that $J = 0$.
2. Use Proposition 1.3.3 we know that $J = \lambda\,\mathrm{Id}$, take trace on both side we obtain $\mathrm{tr}(J) = \lambda n$. From the definition of J we know $\mathrm{tr}(J) = \mathrm{tr}(H)$ and the result follows. □

Corollary 1.3.2. *Under the same set up of Corollary 1.3.1,*
1. If ρ_1 and ρ_2 are not isomorphic, pick arbitrary basis $\{u_i\}_{i=1}^m$ and $\{v_i\}_{i=1}^n$ of V_1 and V_2, we can write down the matrix of $\rho_1(s)$ as $M_1(s)$ and the matrix of $\rho_2(s)$ as $M_2(s)$, we have

$$\frac{1}{|G|}\sum_{s\in G} M_2(s^{-1})_{i_2j_2}M_1(s)_{j_1i_1} = 0, \forall i_1, j_1, i_2, j_2. \tag{1.33}$$

2. If $V_1 = V_2 = V$, $\dim V = n$, and $\rho_1 = \rho_2 = \rho$, pick arbitrary basis $\{v_i\}_{i=1}^n$ of V, we can write down the matrix of $\rho(s)$ as $M(s)$. We have

$$\frac{1}{|G|}\sum_{s\in G} M(s^{-1})_{i_2j_2}M(s)_{j_1i_1} = \frac{1}{n}\delta_{i_2i_1}\delta_{j_2j_1}, \forall i_1, j_1, i_2, j_2. \tag{1.34}$$

Proof. 1. Let the matrix of H, J under basis $\{u_i\}_{i=1}^m$ and $\{v_i\}_{i=1}^n$ be $M(H)$ and $M(J)$. Then we have

$$0 = M(J) = \frac{1}{|G|}\sum_{g\in G} M_2(g^{-1})M(H)M_1(g). \tag{1.35}$$

Take the i_2i_1 component of $M(J)$ we get

$$0 = \frac{1}{|G|}\sum_{j_2=1}^{n}\sum_{j_1=1}^{m}\sum_{g\in G} M_2(g^{-1})_{i_2j_2}M(H)_{j_2j_1}M_1(g)_{j_1i_1} \tag{1.36}$$

Since $M(H)_{j_2j_1}$ is arbitrary, the coefficient of $M(H)_{j_2j_1}$ must be zero and we obtain the desired result.
2. Let the matrix of H, J under base $\{v_i\}_{i=1}^n$ be $M(H)$ and $M(J)$. Take the i_2i_1 component of $M(J)$, we have

$$\frac{1}{n}\sum_{k=1}^{n} M(H)_{kk}\delta_{i_2i_1} = \frac{1}{|G|}\sum_{j_2=1}^{n}\sum_{j_1=1}^{n}\sum_{g\in G} M(g^{-1})_{i_2j_2}M(H)_{j_2j_1}M(g)_{j_1i_1} \tag{1.37}$$

The left hand side can be written as

$$\frac{1}{n}\sum_{k=1}^{n} M(H)_{kk}\delta_{i_2i_1} = \frac{1}{n}\sum_{j_2=1}^{n}\sum_{j_1=1}^{n} M(H)_{j_2j_1}\delta_{j_1j_2}\delta_{i_2i_1} \tag{1.38}$$

Since $M(H)_{j_2j_1}$ is arbitrary, the coefficient of $M(H)_{j_2j_1}$ on the left hand side and right hand side must be the same, and we obtain the desired formula. □

1.3.3 The space of class functions

Definition 1.3.2. *A function $f : G \to \mathbb{C}$ is a class function if $f(t^{-1}gt) = f(g), \forall t, g \in G$, i.e., f take the same value on elements in the same conjugate class.*

Characters are class functions. Let H be the set of class functions. H forms a vector space under the usual addition of functions and scalar multiplication. Let f_s be the function that equals 1 on conjugate class s and equals 0 on other conjugate classes. $\{f_s\}$ forms a base of H and the dimension of H is the number of conjugate classes in G. We can define an inner product in the following way. $\forall \psi, \chi \in H$ define their inner product by

$$\langle \psi, \chi \rangle \equiv \frac{1}{|G|}\sum_{g\in G} \psi(g)^*\chi(g). \tag{1.39}$$

In the following section, we will show that the set of characters of irreducible representations form an orthonormal base of H.

1.3.4 Orthogonality relations for characters

We first show that characters of irreducible representations are orthogonal to each other under the inner product defined in Eq. (1.39).

Proposition 1.3.4. *If χ_1, χ_2 are the characters of two non-isomorphic irreducible representation (of degree n and m respectively) of G, then*

$$\begin{aligned} \langle \chi_1, \chi_1 \rangle &= 1 \\ \langle \chi_1, \chi_2 \rangle &= 0 \end{aligned} \tag{1.40}$$

Proof. Let $j_1 = i_1$, $j_2 = i_2$ in Eq. (1.34) and sum over i_1 and i_2 we have

$$\sum_{i_1=1}^{n}\sum_{i_2=1}^{n}\frac{1}{|G|}\sum_{s\in G}M(s^{-1})_{i_2i_2}M(s)_{i_1i_1} = \sum_{i_1=1}^{n}\sum_{i_2=1}^{n}\frac{1}{n}\delta_{i_2i_1}\delta_{i_2i_1} = 1 \tag{1.41}$$

The left hand side is

$$\begin{aligned}\sum_{i_1=1}^{n}\sum_{i_2=1}^{n}\frac{1}{|G|}\sum_{s\in G}M(s^{-1})_{i_2i_2}M(s)_{i_1i_1} &= \frac{1}{|G|}\sum_{s\in G}\sum_{i_2=1}^{n}M(s^{-1})_{i_2i_2}\sum_{i_1=1}^{n}M(s)_{i_1i_1}\\ &= \frac{1}{|G|}\sum_{g\in G}\chi_1(g^{-1})\chi_1(g)\\ &= \frac{1}{|G|}\sum_{g\in G}\chi_1(g)^*\chi_1(g)\\ &= \langle\chi_1,\chi_1\rangle.\end{aligned} \tag{1.42}$$

The right hand side is 1. Hence $\langle\chi_1,\chi_1\rangle = 1$. Let $j_1 = i_1$, $j_2 = i_2$ in Eq. (1.33) and sum over i_1 and i_2 we have

$$\sum_{i_1=1}^{n}\sum_{i_2=1}^{m}\frac{1}{|G|}\sum_{s\in G}M_2(s^{-1})_{i_2i_2}M_1(s)_{i_1i_1} = 0. \tag{1.43}$$

Using the same argument, the left hand side is $\langle\chi_2,\chi_1\rangle$, so this inner product vanishes. Taking its complex conjugate also gives $\langle\chi_1,\chi_2\rangle = 0$. □

An immediate consequence of Proposition 1.3.4 is that different irreducible representations must have different characters. Hence a finite group G can only have finite number of irreducible representations (as the characters of them form an orthogonal system of a finite dimensional vector space H). Now we give some applications of Proposition 1.3.4. First, we study the irreducible decomposition. Theorem 1.2.1 shows that any representation can be decomposed into direct sum of irreducible representations. This decomposition, however, is in general not unique. Here we show that the number of appearence of an irreducible representation in an irreducible decomposition is independent of the irreducible decomposition you choose. Let V be a representation of G and χ be its character. Let the irreducible representations of G be $(\rho_1, V_1), ..., (\rho_h, V_h)$ and $\chi_1, ..., \chi_h$ be the corresponding characters. Let $V = W_1 \oplus ... \oplus W_k$ be an arbitrary irreducible decomposition of V (note this decomposition is in general not unique).

Theorem 1.3.1. *Let m_k be the number of irreducible representation isomorphic to V_k in the irreducible decomposition above, we have $m_k = \langle\chi_k, \chi\rangle$, and m_k is independent of the decomposition.*

Proof. This is an immediate consequence of Proposition 1.3.4 and Proposition 1.3.2. □

Because of Theorem 1.3.1, we usually write the irreducible decomposition as $V = m_1V_1 \oplus ... \oplus m_hV_h$. Now we give a criterion for determining whether two representations are isomorphic.

Theorem 1.3.2. *Two representations are isomorphic if and only if they have the same characters.*

Proof. The only if part is easy. For the if part, note that if they have the same characters, by Theorem 1.3.1, they have the same number of copies of irreducible representation V_k for each $1 \le k \le h$ and hence they are isomorphic. □

Finally, we give a criterion for irreducibility.

Theorem 1.3.3. *Let (ρ, V) be a representation of G and χ be its character, then ρ is irreducible if and only if $\langle\chi,\chi\rangle = 1$.*

Proof. The only if part if easy. For the if part, let $V = m_1V_1 \oplus ... \oplus m_hV_h$, then $1 = \langle\chi,\chi\rangle = \sum_{i=1}^{h} m_i^2$. Hence only one m_i is 1 and all other m_j are zero, and the representation is irreducible. □

Proposition 1.3.4 says the irreducible characters form an orthogonal system of H. Now we further show that they form an orthonormal basis of H. We begin by the trick of averaging over group.

Proposition 1.3.5. *Let $f \in H$ be a class function, $\rho : G \to GL(V)$ be an irreducible representation of degree n (meaning $\dim V = n$) with character χ. Let ρ_f be the linear map on V defined by*

$$\rho_f = \frac{1}{|G|}\sum_{g\in G}f(g)\rho(g). \tag{1.44}$$

Then $\rho_f = \lambda\,\mathrm{Id}$, with

$$\lambda = \frac{1}{n}\langle\chi^*, f\rangle. \tag{1.45}$$

Proof. $\forall s \in G$, we have

$$\begin{aligned}\rho(s^{-1})\rho_f\rho(s) &= \frac{1}{|G|}\sum_{g\in G} f(g)\rho(s^{-1})\rho(g)\rho(s)\\ &= \frac{1}{|G|}\sum_{g\in G} f(g)\rho(s^{-1}gs)\\ &= \frac{1}{|G|}\sum_{g\in G} f(s^{-1}gs)\rho(s^{-1}gs)\\ &= \frac{1}{|G|}\sum_{t\in G} f(t)\rho(t)\\ &= \rho_f.\end{aligned} \tag{1.46}$$

Third equation uses $f(s^{-1}gs) = f(g)$. The fourth equation uses the fact that for any fixed $s \in G$, $s^{-1}gs$ runs over G when g runs over G. Hence we have $\rho_f\rho(s) = \rho(s)\rho_f$. We may then use Proposition 1.3.3 to conclude that $\rho_f = \lambda\,\mathrm{Id}$. Take trace on both side we have

$$n\lambda = \frac{1}{|G|}\sum_{g\in G} f(g)\chi(g) = \langle\chi^*, f\rangle, \tag{1.47}$$

and we have $\lambda = \langle\chi^*, f\rangle/n$. □

With Proposition 1.3.5, we are ready to prove that irreducible characters forms orthonormal base of H.

Theorem 1.3.4. *Irreducible characters of G form an orthonormal base of H.*

Proof. Let $\chi_1, ..., \chi_h$ be all irreducible characters of G. First we prove that if a class function f is orthogonal to all χ_k^* (where $\chi_k^*(s) \equiv \chi_k(s)^*$), then $f = 0$. For this, let ρ be the canonical representation of G (Example 1.2.11) and χ be its character. By Theorem 1.2.1, χ is a linear combination of $\chi_1, ..., \chi_h$. If f is orthogonal to every χ_k^*, Proposition 1.3.5 says that the averaged operator vanishes on each irreducible summand of the canonical representation. Therefore $\rho_f = 0$ on their direct sum. The vanishing of the total trace alone would not suffice for this conclusion. Using Exercise 1.2.6, acting ρ_f on δ_e, where e is the unit element of G, we have

$$\rho_f\delta_e = \frac{1}{|G|}\sum_{g\in G} f(g)\rho(g)\delta_e = \frac{1}{|G|}\sum_{g\in G} f(g)\delta_g = 0. \tag{1.48}$$

Exercise 1.2.5 tells us that δ_g are linearly independent. Hence $f(g) = 0, \forall g \in G$ and $f = 0$.
Now we prove $\chi_1, ..., \chi_h$ form a orthonormal base of H. It's enough to prove that $\forall f \in H$, f can be written as a linear combination of $\chi_1, ..., \chi_h$. Let $q = f - \sum_{k=1}^h \langle\chi_k, f\rangle\chi_k$. Then by the definition of $\langle\chi_k, f\rangle$, we have

$$\langle\chi_k, f\rangle^* = \langle\chi_k^*, f^*\rangle. \tag{1.49}$$

Hence $q^* = f^* - \sum_{k=1}^h \langle\chi_k^*, f^*\rangle\chi_k^*$. Using $\langle\chi_k^*, \chi_j^*\rangle = \langle\chi_k, \chi_j\rangle^* = \delta_{kj}$, we see $\langle\chi_r^*, q^*\rangle = 0, \forall 1 \le r \le h$, and $q^* = 0$. Hence $q = 0$ and

$$f = \sum_{k=1}^h \langle\chi_k, f\rangle\chi_k, \tag{1.50}$$

and $\chi_1, ..., \chi_h$ form a orthonormal base of H. □

Theorem 1.3.5. *The number of irreducible representations of G equals the number of conjugate classes of G.*

Proof. The number of irreducible representations of G equals the dimension of H, which equal the number of conjugate classes of G. □

We now provide two more relations between the irreducible characters.

Proposition 1.3.6. *Let $s, t \in G$ be in different conjugate class, and $c(s)$ be the number of elements in the conjugate class of s, we have*

$$\begin{aligned}\sum_{k=1}^h \chi_k(s)^*\chi_k(s) &= \frac{|G|}{c(s)}\\ \sum_{k=1}^h \chi_k(s)^*\chi_k(t) &= 0\end{aligned} \tag{1.51}$$

Proof. Let f_s be the class function that equals 1 on the conjugate class of s and equals 0 on the other elements. Using Eq. (1.50), we have

$$f_s = \sum_{k=1}^{h} \langle \chi_k, f_s \rangle \chi_k. \tag{1.52}$$

By definition,

$$\langle \chi_k, f_s \rangle = \frac{1}{|G|} \sum_{g \in G} \chi_k(g)^* f_s(g) = \frac{c(s)}{|G|} \chi_k^*(s). \tag{1.53}$$

Hence we have

$$f_s = \frac{c(s)}{|G|} \sum_{k=1}^{h} \chi_k(s)^* \chi_k. \tag{1.54}$$

Using $f_s(s) = 1, f_s(t) = 0$ we obtain the desired results. □

1.3.5 Decomposition of the canonical representation

Proposition 1.3.7. *Let (ρ, V) be the canonical representation of G (Example 1.2.11), $(\rho_1, V_1), ..., (\rho_h, V_h)$ be the irreducible representations of G of degree $n_1, ..., n_h$, and $\chi_1, ..., \chi_h$ be the corresponding characters. Then we have*

$$V = n_1 V_1 \oplus ... \oplus n_h V_h \tag{1.55}$$

Proof. Let χ be the character of ρ. It's easy to compute

$$\begin{aligned} \chi(e) &= |G| \\ \chi(s) &= 0, \forall s \neq e \end{aligned} \tag{1.56}$$

Let $V = m_1 V_1 \oplus ... \oplus m_h V_h$, then from Theorem 1.3.1, we know

$$m_k = \langle \chi_k, \chi \rangle = \frac{1}{|G|} \sum_{g \in G} \chi_k(g)^* \chi(g) = \frac{1}{|G|} \chi_k(e)^* \chi(e) = \chi_k(e)^* = n_k. \tag{1.57}$$

□

Proposition 1.3.7 tells us that the canonical representation contains all the irreducible representations, it also provides us with two more identities for the irreducible characters.

Corollary 1.3.3. *We have the following identities*

$$\begin{aligned} \sum_{k=1}^{h} n_k^2 &= |G| \\ \sum_{k=1}^{h} n_k \chi_k(s) &= 0, \forall s \neq e. \end{aligned} \tag{1.58}$$

Proof. By Eq. (1.50), we have $\chi = \sum_{k=1}^{h} n_k \chi_k$. $\chi(e) = |G|$ gives us the first identity and $\chi(s) = 0$ gives us the second identity. □

Note that Corollary 1.3.3 can also be obtained from Eq. (1.51).

1.3.6 Example: the character table for D_4

In this section we work the character table of D_4 (Example 1.2.2). D_4 contains 8 elements: the identity element Id, rotation by $\frac{\pi}{2}, \pi, \frac{3\pi}{2}$ and four reflections. Let $Ro(\theta)$ be the rotation (counterclockwise) by θ and $Re(\theta)$ be the reflection along the line whose angle (counterclockwise) between the x-axis is θ. We have

$$\begin{aligned} Re(\theta)^{-1} Ro(\alpha) Re(\theta) &= Ro(-\alpha) \\ Ro(\theta)^{-1} Re(\alpha) Ro(\theta) &= Re(\alpha - \theta) \end{aligned} \tag{1.59}$$

Hence D_4 is divided into 5 conjugate classes

$$D_4 = \{\mathrm{Id}\} \sqcup \{Ro(\pi)\} \sqcup \{Ro(\frac{\pi}{2}), Ro(\frac{3\pi}{2})\} \sqcup \{Re(0), Re(\frac{\pi}{2})\} \sqcup \{Re(\frac{\pi}{4}), Re(\frac{3\pi}{4})\}. \tag{1.60}$$

Let the classes be c_1, c_2, c_3, c_4, c_5 respectively. From Theorem 1.3.5, D_4 has 5 irreducible representations $\chi_1, ..., \chi_5$. Using Corollary 1.3.3, we know one of the irreducible representation has degree 2 and all the other 4 has degree 1. Suppose χ_5 has degree 2. First of all, we have two degree 1 representations: ρ_1 that maps every group element to 1 and ρ_2 that maps $g \in D_4$ to $\det(g)$. Since by definition, representations of degree 1 are irreducible, these two irreducible representations gives us irreducible characters χ_1 and χ_2. Second, notice that D_4 is already a group consisting of linear maps on $\mathbb{R}^2$, i.e., $D_4 \subset GL(\mathbb{R}^2)$, we have a tautological representation $\rho_5 : D_4 \to GL(\mathbb{C}^2)$, where ρ_5 acts by the same real matrices on the complexified vector space. Using Theorem 1.3.3, we see this is an irreducible representation and we have the last row in Table 1.1. Third, write $r = Ro(\pi/2)$ and $s = Re(0)$, with $r^4 = s^2 = e$ and $srs = r^{-1}$. In a one-dimensional representation these relations force $\rho(r) = \pm 1$ and $\rho(s) = \pm 1$. The two choices with $\rho(r) = 1$ give χ_1, χ_2, and the choices $\rho(r) = -1$, $\rho(s) = 1$ or -1 give χ_3, χ_4. Computing their values on the five classes gives Table 1.1; character orthogonality checks the result.

	c_1	c_2	c_3	c_4	c_5
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	1	-1	1	-1
χ_4	1	1	-1	-1	1
χ_5	2	-2	0	0	0

Table 1.1: The character table of D_4

1.4 An application of character theory: block diagonalization

1.4.1 Canonical decomposition of a representation

Theorem 1.3.1 tells us that the multiplicity of an irreducible representation is independent of the chosen irreducible decomposition. The individual irreducible subspaces, however, need not be unique. In this section, we show that the sum of all subrepresentations of a fixed irreducible type is unique, and construct the projection onto this sum directly from the character table. This is the decomposition that a symmetry group determines before we know the Hamiltonian.

Proposition 1.4.1. *Let (ρ, V) be a representation of G, and let $\{(\rho_j, V_j)\}_{j=1}^h$ be representatives of the isomorphism classes of irreducible representations, with degrees n_j and characters χ_j. In any irreducible decomposition of V, let U_j be the sum of the m_j summands isomorphic to V_j. Then*

$$p_j \equiv \frac{n_j}{|G|} \sum_{g \in G} \chi_j(g)^* \rho(g) \tag{1.61}$$

is the projection onto U_j along the sum of the other U_i. In particular,

$$p_i p_j = \delta_{ij} p_j, \qquad \sum_{j=1}^{h} p_j = \mathrm{Id}_V, \qquad V = \bigoplus_{j=1}^{h} U_j, \qquad U_j = p_j(V). \tag{1.62}$$

Each U_j is independent of the chosen irreducible decomposition. We call U_j the V_j-isotypic component and the resulting direct sum the canonical decomposition of V.

Proof. The idea is to examine p_j on one irreducible summand at a time. Since a character is a class function, p_j commutes with the representation; Schur's lemma therefore makes its restriction a scalar. Character orthogonality determines that scalar.

Let $W \subset V$ be an irreducible subrepresentation, with degree n and character χ, and write $\rho_W = \rho|_W$ and $p_j^W = p_j|_W$. This restriction is well defined because every $\rho(g)$ preserves W. For $s \in G$, change variables from g to $t = s^{-1}gs$ to obtain

$$\rho_W(s)^{-1} p_j^W \rho_W(s) = \frac{n_j}{|G|} \sum_{g \in G} \chi_j(g)^* \rho_W(s^{-1}gs) = \frac{n_j}{|G|} \sum_{t \in G} \chi_j(t)^* \rho_W(t) = p_j^W. \tag{1.63}$$

By Proposition 1.3.3, $p_j^W = \lambda \, \mathrm{Id}_W$. Taking the trace gives

$$\lambda n = \operatorname{tr} p_j^W = \frac{n_j}{|G|} \sum_{g \in G} \chi_j(g)^* \chi(g) = n_j \langle \chi_j, \chi \rangle. \tag{1.64}$$

If $W \cong V_j$, then $n = n_j$ and the inner product is 1, so $\lambda = 1$. Otherwise the inner product is 0, so $\lambda = 0$. Thus p_j is the identity on every summand of type V_j and zero on every other summand. This proves all the identities in Eq. (1.62). Finally, Eq. (1.61) only involves the original representation and the irreducible character χ_j, so its image cannot depend on how we decomposed V. □

The distinction between an isotypic component and an irreducible subspace is important. If $m_j > 1$, the projector p_j selects a space of dimension $m_j n_j$, not one particular copy of V_j. Characters determine the type of a state, but do not distinguish equivalent copies of that type. Also, the word "canonical" here refers to independence from the choice of irreducible summands; it does not refer to the canonical representation on functions on G introduced in Example 1.2.11.

Exercise 1.4.1. *Equip V with a G-invariant inner product, so that every $\rho(g)$ is unitary. Show directly from Eq. (1.61), using $\chi_j(g^{-1}) = \chi_j(g)^*$, that $p_j^\dagger = p_j$. Conclude that the canonical decomposition is orthogonal. Show also that $\operatorname{tr} p_j = m_j n_j$, and explain how applying p_j to a spanning set produces a basis for U_j after zero vectors and linear dependencies are removed.*

1.4.2 Block diagonalization of symmetric Hamiltonian

Let $\mathbb{H}$ be a finite dimensional Hilbert space and H its Hamiltonian. A finite group of unitary symmetries gives a representation $\rho : G \to GL(\mathbb{H})$ such that $H\rho(g) = \rho(g)H$ for every $g \in G$. This setting includes a finite tight-binding model and a finite orbital space at a momentum fixed by the relevant symmetries. The infinite translation group and antiunitary symmetries require the additional constructions in Section 8. We first explain what follows from ordinary finite group representations alone.

Theorem 1.4.1. *Let (ρ, V) be a representation of G, let $V = \bigoplus_{j=1}^h U_j$ be its canonical decomposition, and let $H : V \to V$ satisfy $[H, \rho(g)] \equiv H\rho(g) - \rho(g)H = 0$ for every $g \in G$. Then each U_j is invariant under H.*

Proof. Since p_j is a linear combination of the operators $\rho(g)$, we have $Hp_j = p_j H$. If $w \in U_j$, then $p_j w = w$, and hence

$$p_j H w = H p_j w = H w. \tag{1.65}$$

Thus Hw belongs to the image of p_j, which is U_j. □

Theorem 1.4.1 allows us to solve the eigenvalue problem separately in different symmetry sectors. It does not say that every irreducible summand in an arbitrarily chosen decomposition is invariant under H: equivalent copies may mix. Schur's lemma gives a more precise description of this remaining freedom.

Proposition 1.4.2. *Choose an irreducible decomposition $U_j = W_{j1} \oplus \cdots \oplus W_{jm_j}$ and isomorphisms $T_{ja} : V_j \to W_{ja}$. Transport a fixed basis of V_j to each copy by these isomorphisms. In the resulting basis of U_j, every operator commuting with G has the form*

$$H|_{U_j} = \begin{pmatrix} a_{11}^{(j)} I_{n_j} & \cdots & a_{1m_j}^{(j)} I_{n_j} \\ \vdots & \ddots & \vdots \\ a_{m_j 1}^{(j)} I_{n_j} & \cdots & a_{m_j m_j}^{(j)} I_{n_j} \end{pmatrix}, \qquad A_j = (a_{ab}^{(j)})_{a,b=1}^{m_j}. \tag{1.66}$$

If ρ is unitary and H is Hermitian, the choices can be made orthonormally and A_j is Hermitian. Each eigenvalue of A_j with multiplicity r then contributes an eigenspace of dimension rn_j in U_j.

Proof. We reduce the statement to Schur's lemma by examining the map from one irreducible copy to another. Let $q_{ja} : U_j \to W_{ja}$ be the projection along the other summands. Since each summand is invariant under G, q_{ja} commutes with the group action. Consequently, the map $T_{ja}^{-1} q_{ja} H T_{jb} : V_j \to V_j$ commutes with $\rho_j(g)$ for every $g \in G$. Proposition 1.3.3 gives $T_{ja}^{-1} q_{ja} H T_{jb} = a_{ab}^{(j)} \operatorname{Id}_{V_j}$, which is exactly Eq. (1.66).

For a unitary representation, we may choose the irreducible summands orthogonally. An intertwining isomorphism T between two unitary irreducible representations satisfies $T^\dagger T = c \operatorname{Id}$ by Schur's lemma, with $c > 0$; rescaling T therefore makes it unitary. Hence the transported bases can be chosen orthonormally. Hermiticity of H now implies $a_{ab}^{(j)} = (a_{ba}^{(j)})^*$, so A_j is Hermitian. If $x = (x_1, \ldots, x_{m_j})$ is an eigenvector of A_j with eigenvalue E, then for every $v \in V_j$,

$$H \left(\sum_{b=1}^{m_j} x_b T_{jb} v \right) = \sum_{a=1}^{m_j} (A_j x)_a T_{ja} v = E \sum_{a=1}^{m_j} x_a T_{ja} v. \tag{1.67}$$

Choose a basis of the r dimensional E-eigenspace of A_j and a basis of V_j. Their images in Eq. (1.67) give rn_j independent eigenvectors. Since an orthonormal eigenbasis of A_j accounts for all $m_j n_j$ dimensions of U_j, these span the whole E-eigenspace in this sector. □

This proposition explains both the use and the limitation of symmetry in a band calculation. A degree n_j irreducible representation forces an n_j-fold multiplet, while the energies and the mixing of repeated copies are determined by the smaller matrix A_j. If $m_j = 1$, H is a scalar on U_j. If $m_j > 1$, symmetry generally leaves an m_j by m_j eigenvalue problem. Equal energies in inequivalent sectors can occur accidentally; symmetry alone does not require them. The matrix A_j depends on a choice of equivalent copies, whereas the sector U_j does not.

Example 1.4.1. *Consider four identical orbitals at $(a, 0)$, $(0, a)$, $(-a, 0)$, and $(0, -a)$, in this order. The square group D_4 acts by permuting these orbitals. Its character is the number of orbitals fixed by a group element, so on the five classes in Table 1.1 it equals $(4, 0, 0, 2, 0)$. Character orthogonality gives $V \cong V_1 \oplus V_3 \oplus V_5$. Thus a Hermitian D_4-symmetric Hamiltonian has two singlet sectors and one doublet sector. A concrete example, with real parameters ϵ, t, u, is*

$$H = \begin{pmatrix} \epsilon & t & u & t \\ t & \epsilon & t & u \\ u & t & \epsilon & t \\ t & u & t & \epsilon \end{pmatrix}. \tag{1.68}$$

The singlet vectors are $(1, 1, 1, 1)/2$ and $(1, -1, 1, -1)/2$, with energies $\epsilon + 2t + u$ and $\epsilon - 2t + u$. The doublet is spanned by $(1, 0, -1, 0)/\sqrt{2}$ and $(0, 1, 0, -1)/\sqrt{2}$, both with energy $\epsilon - u$. The equality of the last two energies follows from the degree 2 irreducible representation, independently of the values of the three parameters.

Exercise 1.4.2. *Use Eq. (1.61) and Table 1.1 to construct the three nonzero projectors for the four-orbital example and recover its symmetry-adapted vectors. Add the perturbation $\delta\,\mathrm{diag}(1, -1, 1, -1)$ to Eq. (1.68). Which rotations and reflections remain symmetries? Show that the doublet splits, and explain why this does not contradict Proposition 1.4.2.*

Exercise 1.4.3. *Suppose V contains two copies of the degree 2 representation V_5 of D_4. Write down the most general Hermitian operator commuting with D_4 on this four dimensional isotypic component. Show explicitly that a nonzero off-diagonal entry of A_5 mixes the two chosen copies while each energy remains at least doubly degenerate.*

Chapter 2

Induced Representations

2.1 The group algebra

2.1.1 The group algebra $\mathbb{C}[G]$

The projector in Eq. (1.61) is a linear combination of symmetry operators. We now introduce an algebra in which we can form such combinations before choosing any particular representation. This gives a common language for projectors, class sums, and induced representations. Throughout this section, G is finite and all representations are over $\mathbb{C}$.

Definition 2.1.1. *An algebra over $\mathbb{C}$ is a complex vector space A with a bilinear multiplication $(a,b)\mapsto ab$ that is associative, $(ab)c=a(bc)$, and has a unit 1_A satisfying $1_Aa=a1_A=a$ for every $a\in A$. We use the word algebra to mean an associative algebra with a unit throughout this note.*

Definition 2.1.2. *The group algebra $\mathbb{C}[G]$ is the complex vector space with basis indexed by G. Its elements are formal sums $\sum_{g\in G}c_g g$, with $c_g\in\mathbb{C}$; addition and scalar multiplication are defined coefficient-wise. Multiplication extends the group multiplication bilinearly:*

$$\left(\sum_{g\in G}c_g g\right)\left(\sum_{s\in G}d_s s\right)=\sum_{g,s\in G}c_g d_s (gs). \tag{2.1}$$

The unit is the basis element e, where e is the identity of G.

Proposition 2.1.1. *The multiplication in Eq. (2.1) makes $\mathbb{C}[G]$ an algebra over $\mathbb{C}$ of dimension $|G|$.*

Proof. Bilinearity holds by construction. Associativity holds on basis elements because $(g_1g_2)g_3=g_1(g_2g_3)$ in G, and extends to all elements by linearity. Similarly, $eg=ge=g$ on the basis proves that e is the unit. The dimension is the number of basis elements. □

Definition 2.1.3. *A left module over an algebra A is a complex vector space V with a bilinear action $(a,v)\mapsto av$ such that $(ab)v=a(bv)$ and $1_Av=v$ for all $a,b\in A$ and $v\in V$. A map between left A-modules is a module homomorphism if it is complex linear and satisfies $F(av)=aF(v)$.*

Proposition 2.1.2. *Giving a representation of G on V is equivalent to giving a left $\mathbb{C}[G]$-module structure on V. Under this correspondence, invariant subspaces are submodules and intertwining maps are module homomorphisms.*

Proof. Let $\rho: G\to GL(V)$ be a representation. Extend it linearly by defining

$$\rho\left(\sum_{g\in G}c_g g\right)=\sum_{g\in G}c_g\rho(g),\qquad av\equiv\rho(a)v. \tag{2.2}$$

For $a=\sum_g c_g g$ and $b=\sum_s d_s s$, the representation identity gives

$$\rho(a)\rho(b)=\sum_{g,s\in G}c_g d_s\rho(g)\rho(s)=\sum_{g,s\in G}c_g d_s\rho(gs)=\rho(ab). \tag{2.3}$$

Together with $\rho(e)=\mathrm{Id}_V$, this proves the module axioms. Conversely, restrict a module action to the basis elements $g\in G$. The module axioms give $\rho(gs)=\rho(g)\rho(s)$ and $\rho(e)=\mathrm{Id}_V$; in particular, $\rho(g^{-1})$ is the inverse of $\rho(g)$. Thus this restriction is a representation, and bilinearity shows that its linear extension is the original module action. Finally, a subspace is stable under all group elements if and only if it is stable under all their linear combinations, and the same linearity argument proves the assertion about homomorphisms. □

We will therefore freely pass between representations and modules. This change of language introduces no new kind of representation: it permits us to manipulate sums and products of symmetries within one algebra. In particular, an element $a \in \mathbb{C}[G]$ acts on every representation, but its action can be different on different representations.

Example 2.1.1. *The group algebra is a left module over itself by left multiplication. For a basis element $g \in G$, this action is*

$$g\left(\sum_{s\in G} c_s s\right) = \sum_{s\in G} c_s(gs). \tag{2.4}$$

The map sending the basis element s to the function δ_s identifies Eq. (2.4) with the canonical representation in Example 1.2.11. We also call it the left regular representation.

Exercise 2.1.1. *For $a = \sum_g c_g g$, define $a^* = \sum_g c_g^* g^{-1}$. Show that $(ab)^* = b^*a^*$ and that $\rho(a^*) = \rho(a)^\dagger$ whenever ρ is unitary. Conclude that $a = a^*$ produces a Hermitian operator in every unitary representation. Is $\rho(a)$ necessarily a symmetry-invariant Hamiltonian for an arbitrary such a? Identify a sufficient additional condition on a.*

2.1.2 Decomposition of $\mathbb{C}[G]$

Let $(\rho_1, V_1), \ldots, (\rho_h, V_h)$ represent all irreducible representations of G, with characters χ_i and degrees $n_i = \dim V_i$. Proposition 1.3.7 gives the decomposition of the left regular representation,

$$\mathbb{C}[G] \cong n_1 V_1 \oplus \cdots \oplus n_h V_h. \tag{2.5}$$

Here the isomorphism respects the left G-action, and $n_i V_i$ means a sum of n_i copies. The following theorem is stronger: it also describes multiplication in the group algebra.

Theorem 2.1.1. *Extend each ρ_i to $\mathbb{C}[G]$ as in Eq. (2.2). The map*

$$\Phi : \mathbb{C}[G] \longrightarrow \bigoplus_{i=1}^{h} \mathrm{End}(V_i), \qquad a \longmapsto \big(\rho_1(a), \ldots, \rho_h(a)\big) \tag{2.6}$$

is an isomorphism of algebras, where multiplication on the right is component-wise composition.

Proof. The proof has two steps. First, the regular representation detects whether an element of $\mathbb{C}[G]$ is zero, so an element acting as zero on every irreducible representation must itself vanish. Second, the sum-of-squares formula shows that an injective map already has the correct dimension to be surjective.

Eq. (2.3) shows that Φ preserves multiplication, and it plainly preserves addition, scalar multiplication, and the unit. Suppose $\Phi(a) = 0$. Then a acts as zero on every V_i, hence also on every copy of V_i in Eq. (2.5). Therefore left multiplication by a on $\mathbb{C}[G]$ is the zero map. Applying it to e gives $a = ae = 0$, proving injectivity. On the other hand, Corollary 1.3.3 gives

$$\dim \mathbb{C}[G] = |G| = \sum_{i=1}^{h} n_i^2 = \dim\left(\bigoplus_{i=1}^{h} \mathrm{End}(V_i)\right). \tag{2.7}$$

An injective linear map between finite dimensional spaces of equal dimension is surjective, so Φ is an algebra isomorphism. □

Eq. (2.6) is the finite-group version of a Fourier transform: it replaces coefficients indexed by group elements with matrix coefficients indexed by irreducible representations. For an abelian group every irreducible representation has degree 1, and the matrices reduce to scalars. For a nonabelian group, the matrix structure records the internal dimension of an irreducible multiplet. The two decompositions should not be confused: $\mathrm{End}(V_i)$ has dimension n_i^2 and, under left multiplication by $\rho_i(g)$, consists of n_i copies of V_i, one for each matrix column.

Definition 2.1.4. *For each irreducible representation V_i, define the element*

$$p_i = \frac{n_i}{|G|} \sum_{g\in G} \chi_i(g)^* g \in \mathbb{C}[G]. \tag{2.8}$$

We call these elements the primitive central idempotents of $\mathbb{C}[G]$. Here an idempotent satisfies $p_i^2 = p_i$, and "primitive central" means that it cannot be expressed as a sum of two nonzero central idempotents whose product is zero.

Proposition 2.1.3. *The elements in Eq.* (2.8) *are central and satisfy*

$$p_i p_j = \delta_{ij} p_i, \qquad \sum_{i=1}^{h} p_i = e, \qquad \rho_k(p_i) = \delta_{ki} \operatorname{Id}_{V_k} . \tag{2.9}$$

For any representation (ρ, V), $\rho(p_i)$ is its canonical projector onto the V_i-isotypic component.

Proof. The formula for $\rho(p_i)$ is Eq. (1.61), so Proposition 1.4.1 gives both the last assertion and $\rho_k(p_i) = \delta_{ki} \operatorname{Id}_{V_k}$. Under Φ, p_i is therefore the tuple with Id_{V_i} in position i and zero in all other positions. Such tuples are central, multiply as in Eq. (2.9), and sum to the unit. Since Φ is an algebra isomorphism, the same statements hold in $\mathbb{C}[G]$. Finally, a matrix commuting with every endomorphism is a scalar matrix; a scalar idempotent is either 0 or Id. Thus the tuple associated with p_i cannot split into two nonzero orthogonal central idempotents, proving the asserted primitivity. □

The same symbol p_i now denotes an element of the group algebra; its action on a particular representation is $\rho(p_i)$. This separates the universal character formula from the projector matrix in a chosen physical Hilbert space. It also yields the algebra decomposition $\mathbb{C}[G] = \bigoplus_i p_i \mathbb{C}[G]$, with $p_i \mathbb{C}[G] \cong \operatorname{End}(V_i)$.

Exercise 2.1.2. *Show directly, without using Φ, that $g p_i g^{-1} = p_i$ for every $g \in G$. Then prove that $p_i^* = p_i$ for the involution in the preceding exercise. Explain how these two identities become commutation with the symmetry operators and orthogonality of the canonical projectors in a unitary representation.*

2.1.3 The center of $\mathbb{C}[G]$

Definition 2.1.5. *The center of an algebra A is the subalgebra $Z(A) = \{a \in A : ab = ba \text{ for every } b \in A\}$. For a conjugacy class $C \subset G$, its class sum is*

$$\Omega_C = \sum_{g \in C} g. \tag{2.10}$$

Proposition 2.1.4. *The class sums Ω_C, with C running over the conjugacy classes of G, form a basis of $Z(\mathbb{C}[G])$.*

Proof. Write $a = \sum_g a_g g$. It suffices to test whether a commutes with each basis element $s \in G$. This is equivalent to $sas^{-1} = a$. Since conjugation permutes the basis elements, comparing coefficients shows that this identity holds for every s precisely when $a_{sgs^{-1}} = a_g$ for every $s, g \in G$. Thus a is central exactly when its coefficients are constant on each conjugacy class, or equivalently when it is a linear combination of the class sums. Distinct class sums have disjoint supports in the basis G, so they are linearly independent. □

Corollary 2.1.1. *The dimension of $Z(\mathbb{C}[G])$ equals the number of conjugacy classes of G.*

Class sums make central elements concrete: we can form them from the group multiplication table without knowing any representations. The irreducible representations provide a second description, in which a central element is specified by one scalar for each irreducible type.

Proposition 2.1.5. *For each irreducible representation (ρ_i, V_i) and each $z \in Z(\mathbb{C}[G])$, there is a scalar $\omega_i(z)$ such that*

$$\rho_i(z) = \omega_i(z) \operatorname{Id}_{V_i} . \tag{2.11}$$

The map $\omega_i : Z(\mathbb{C}[G]) \to \mathbb{C}$ is an algebra homomorphism. If C is a conjugacy class and $c \in C$, then

$$\omega_i(\Omega_C) = \frac{|C| \chi_i(c)}{n_i}. \tag{2.12}$$

Proof. Since z is central, $\rho_i(z)$ commutes with every $\rho_i(g)$. Schur's lemma gives Eq. (2.11). Linearity, multiplicativity, and preservation of the unit follow from the corresponding properties of ρ_i. For $z = \Omega_C$, taking the trace gives

$$n_i \omega_i(\Omega_C) = \operatorname{tr}\left(\sum_{g \in C} \rho_i(g) \right) = \sum_{g \in C} \chi_i(g) = |C| \chi_i(c), \tag{2.13}$$

where the last equality uses that characters are constant on conjugacy classes. Division by n_i proves Eq. (2.12). □

Definition 2.1.6. *The homomorphism ω_i is called the central character associated with V_i; $\omega_i(z)$ is its value on the central element z.*

Proposition 2.1.6. *The elements $p_1, \dots, p_h$ form another basis of the center. More precisely, every $z \in Z(\mathbb{C}[G])$ has the unique expansion*

$$Z(\mathbb{C}[G]) = \bigoplus_{i=1}^{h} \mathbb{C}p_i, \qquad z = \sum_{i=1}^{h} \omega_i(z) p_i. \tag{2.14}$$

Proof. Under Φ, a central element z becomes the tuple $(\omega_1(z)\,\mathrm{Id}_{V_1}, \dots, \omega_h(z)\,\mathrm{Id}_{V_h})$. Proposition 2.1.3 identifies this tuple with the image of $\sum_i \omega_i(z) p_i$. Injectivity of Φ proves the expansion and its uniqueness. □

Comparing the two bases recovers Theorem 1.3.5. It also clarifies a limitation of class sums in physics: a central element acts as a scalar on a whole isotypic component, including all repeated copies. A general symmetry-invariant Hamiltonian can contain the nontrivial multiplicity matrices in Eq. (1.66), so it need not be the action of a central group-algebra element.

Exercise 2.1.3. *For conjugacy classes C, D, write*

$$\Omega_C \Omega_D = \sum_E N_{CD}^E \Omega_E, \tag{2.15}$$

where E runs over all conjugacy classes. Show that N_{CD}^E is the number of pairs $(c, d) \in C \times D$ with $cd = x$, for any fixed $x \in E$. Prove that this count is independent of the choice of $x \in E$ by conjugating both factors. Thus all N_{CD}^E are nonnegative integers. These integer multiplication coefficients will explain the arithmetic restriction on irreducible degrees below.

2.1.4 Basic properties of integers

We next introduce algebraic integers. They enter character theory because the eigenvalues of a finite-order symmetry are roots of unity, and character values are sums of these eigenvalues. We will keep the distinction between an algebraic integer and an ordinary integer in $\mathbb{Z}$ explicit. The goal is to show that $|G|/n_i$ is both rational and an algebraic integer, which forces it to be an ordinary integer.

Definition 2.1.7. *A complex number α is an algebraic integer if it satisfies a monic polynomial with coefficients in $\mathbb{Z}$, that is,*

$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 = 0, \qquad a_0, \dots, a_{n-1} \in \mathbb{Z}, \qquad n \geq 1. \tag{2.16}$$

Every $m \in \mathbb{Z}$ is an algebraic integer because it satisfies $x - m = 0$. The requirement that the polynomial be monic is essential: for example, $1/2$ satisfies $2x - 1 = 0$, but is not an algebraic integer, as we will prove below.

Proposition 2.1.7. *A complex number α is an algebraic integer if and only if there is a nonzero finitely generated additive subgroup $M \subset \mathbb{C}$ with $\alpha M \subset M$.*

Proof. The idea is to translate a polynomial equation into a finite set of additive generators, and then reverse the argument using an integer matrix. If α satisfies Eq. (2.16), take $M = \mathbb{Z} + \mathbb{Z}\alpha + \cdots + \mathbb{Z}\alpha^{n-1}$. It is nonzero because it contains 1, and is finitely generated by construction. Multiplication by α sends every displayed generator except the last to the next one; it sends the last to α^n, which Eq. (2.16) expresses as an integer linear combination of the generators. Therefore $\alpha M \subset M$.

Conversely, let $m_1, \dots, m_r$ generate such a nonzero subgroup. Stability gives integers a_{ij} with $\alpha m_j = \sum_{i=1}^{r} a_{ij} m_i$. Put $A = (a_{ij})$ and regard $m = (m_1, \dots, m_r)$ as a row vector. At least one generator is nonzero, so

$$m(\alpha I_r - A) = 0, \qquad m \neq 0, \qquad \det(\alpha I_r - A) = 0. \tag{2.17}$$

Thus α is a root of the monic integer polynomial $\det(xI_r - A)$. Notice that the additive generators need not be linearly independent over $\mathbb{C}$; the existence of the nonzero left null vector in Eq. (2.17) is sufficient. □

Proposition 2.1.8. *The algebraic integers form a subring of $\mathbb{C}$: they contain 0 and 1, and are closed under sums, products, and additive inverses.*

Proof. The assertion about 0 and 1 follows from their being ordinary integers. Let α, β be algebraic integers. By Proposition 2.1.7, choose nonzero finitely generated additive subgroups $M, N \subset \mathbb{C}$ stable under multiplication by α, β, respectively. If $m_1, \dots, m_r$ and $n_1, \dots, n_s$ are generators, let L be generated by the finitely many products $m_i n_j$. It is nonzero because a product of nonzero elements of M and N is nonzero. Multiplication by α sends $m_i n_j$ to an integer linear combination of the $m_k n_j$, and multiplication by β sends it to an integer linear combination of the $m_i n_\ell$. Hence $\alpha L \subset L$ and $\beta L \subset L$, giving $(\alpha + \beta)L \subset L$ and $\alpha\beta L \subset L$. The criterion proves that $\alpha + \beta$ and $\alpha\beta$ are algebraic integers. Finally, $(-\alpha)M \subset M$ proves the same for $-\alpha$. □

Proposition 2.1.9. *If a rational number is an algebraic integer, then it belongs to* $\mathbb{Z}$.

Proof. Write $\alpha = p/q$ with $p, q \in \mathbb{Z}$, $q > 0$, and $\gcd(p, q) = 1$. Substitute into Eq. (2.16) and multiply by q^n to obtain

$$p^n + a_{n-1}p^{n-1}q + \cdots + a_1 pq^{n-1} + a_0 q^n = 0. \tag{2.18}$$

Every term except the first is divisible by q, so q divides p^n. Coprimality of p and q forces $q = 1$, and hence $\alpha \in \mathbb{Z}$. □

Proposition 2.1.10. *Every eigenvalue of a square matrix with integer entries is an algebraic integer.*

Proof. Its characteristic polynomial is monic with integer coefficients, and every eigenvalue is a root of that polynomial. □

Exercise 2.1.4. *Show that every root of unity is an algebraic integer. If $g \in G$ has order r, use $\rho(g)^r = \mathrm{Id}$ to show that every eigenvalue of $\rho(g)$ is an rth root of unity. Deduce from Proposition 2.1.8 that every character value $\chi_\rho(g)$ is an algebraic integer. Finally, show that a rational character value must be an ordinary integer. Does this imply that every character value is rational? Test the degree* 1 *representations of the cyclic group of order* 3.

2.1.5 Integrality properties of characters. Applications

Exercise 2.1.4 proves integrality of character values, but this does not immediately prove that the quotient $|C|\chi_i(c)/n_i$ is integral: division need not preserve algebraic integers. The class sums supply the missing argument. Their multiplication has integer coefficients, and their eigenvalues on irreducible representations are exactly these quotients.

Proposition 2.1.11. *Let C be a conjugacy class, let $c \in C$, and let (ρ_i, V_i) be an irreducible representation with degree n_i and character χ_i. Then $|C|\chi_i(c)/n_i$ is an algebraic integer.*

Proof. Consider left multiplication L_{Ω_C} on $\mathbb{C}[G]$. In the basis G, it is an integer matrix, since for every $s \in G$,

$$L_{\Omega_C}(s) = \Omega_C s = \sum_{g \in C} gs. \tag{2.19}$$

Every V_i occurs in the regular representation by Eq. (2.5). On such a copy, L_{Ω_C} acts as the scalar $\omega_i(\Omega_C) = |C|\chi_i(c)/n_i$ by Eq. (2.12). This scalar is therefore an eigenvalue of the integer matrix L_{Ω_C}, and Proposition 2.1.10 proves the assertion. □

The division by n_i is justified by realizing the quotient as an eigenvalue, not by any general closure under division. This is the key point in the following theorem.

Theorem 2.1.2. *The degree n_i of every irreducible complex representation of a finite group G divides $|G|$.*

Proof. We use character orthogonality to express the rational number $|G|/n_i$ as a sum of products of algebraic integers. Proposition 2.1.9 will then finish the proof. Since $\langle \chi_i, \chi_i \rangle = 1$ and $\chi_i(g)^* = \chi_i(g^{-1})$, grouping the orthogonality sum by conjugacy classes gives

$$\frac{|G|}{n_i} = \frac{1}{n_i} \sum_{g \in G} \chi_i(g^{-1})\chi_i(g) = \sum_C \left(\frac{|C|\chi_i(c)}{n_i} \right) \chi_i(c^{-1}), \qquad c \in C. \tag{2.20}$$

The first factor in each summand is an algebraic integer by Proposition 2.1.11; the second is an algebraic integer by Exercise 2.1.4. Proposition 2.1.8 therefore makes the whole sum an algebraic integer. It is also the rational number $|G|/n_i$, so Proposition 2.1.9 implies $|G|/n_i \in \mathbb{Z}$. □

This is a useful consistency check when constructing a character table or identifying a symmetry multiplet: every candidate irreducible dimension must divide the group order, and the dimensions must also satisfy the sum-of-squares formula. The divisibility statement concerns irreducible degrees, not the dimension of an arbitrary reducible space or the total degeneracy of an accidentally coincident energy. For spinful or projective symmetry representations, one must first identify the appropriate ordinary representation group before applying this theorem.

Corollary 2.1.2. *If G is abelian, every irreducible complex representation of G has degree* 1.

Proof. Every conjugacy class is a singleton, so Theorem 1.3.5 gives $|G|$ irreducible representations. Their positive degrees satisfy $\sum_{i=1}^{|G|} n_i^2 = |G|$, so every n_i must equal 1. □

This corollary also follows directly from Schur's lemma and does not require the divisibility theorem. It is the finite-group reason that commuting translation symmetries can be labeled by scalar phases. Higher dimensional symmetry-enforced multiplets require additional structure, such as noncommuting point symmetries or antiunitary operations.

Example 2.1.2. *The group D_4 has order 8 and five conjugacy classes, so it has five irreducible degrees $n_1, \dots, n_5$ with $\sum_{i=1}^{5} n_i^2 = 8$. Since all degrees are positive, four must be 1 and one must be 2. These degrees divide 8, as required by Theorem 2.1.2, and agree with Table 1.1. Here the class count and sum-of-squares formula already determine the degrees; the divisibility theorem provides an additional check.*

Proposition 2.1.12. *Let N_{CD}^{E} be the class multiplication coefficients in Eq. (2.15). Every central character satisfies*

$$\omega_i(\Omega_C)\omega_i(\Omega_D) = \sum_E N_{CD}^{E}\omega_i(\Omega_E). \tag{2.21}$$

Equivalently, for $c \in C$, $d \in D$, and a representative $x_E \in E$,

$$\frac{|C|\chi_i(c)}{n_i}\frac{|D|\chi_i(d)}{n_i} = \sum_E N_{CD}^{E}\frac{|E|\chi_i(x_E)}{n_i}. \tag{2.22}$$

Proof. Apply the algebra homomorphism ω_i to Eq. (2.15); Eq. (2.21) follows. Substituting Eq. (2.12) gives Eq. (2.22). □

In the class-sum basis, multiplication by Ω_C is an integer matrix with entries obtained by counting products. In the idempotent basis, it is diagonal because $\Omega_C p_i = \omega_i(\Omega_C)p_i$. Thus the character table, with the factors $|C|/n_i$ in Eq. (2.12), connects a combinatorial basis to a basis of simultaneous eigenvectors. This gives useful constraints on proposed character tables and another way to construct operators that distinguish symmetry types.

Exercise 2.1.5. *For the five conjugacy classes $c_1, \dots, c_5$ of D_4 in Table 1.1, compute the products of the associated class sums. Begin with $\Omega_{c_3}^2 = 2\Omega_{c_1} + 2\Omega_{c_2}$ and verify the other products by counting pairs as in Exercise 2.1.3. Use the character table to check Eq. (2.21) for each irreducible representation. Explain why class sums can distinguish inequivalent irreducible types but cannot separate equivalent copies of the same type.*

2.2 Induced representations

In the preceding sections, we studied a representation after the group action had been given. We now ask how to construct a representation of G from one of its subgroups H. The construction is called induction. Its main idea is to place a copy of an H-representation over each left coset of H, and let G move these copies while keeping track of the action inside each copy. In solid state physics, this describes how a local orbital generates symmetry-related orbitals, or how a representation at one momentum generates states at all symmetry-related momenta. Throughout this section and the next one, the groups are finite and the representations are finite dimensional over $\mathbb{C}$.

2.2.1 Tensor product of modules

We begin with the tensor product, which allows us to state induction without choosing coset representatives. There are two different tensor products below: the tensor product over $\mathbb{C}$ combines vector spaces, whereas the tensor product over an algebra imposes additional relations between two module actions.

Definition 2.2.1. *Let V and W be vector spaces over $\mathbb{C}$. Their tensor product $V \otimes_{\mathbb{C}} W$ is the vector space generated by symbols $v \otimes w$, subject to bilinearity: $(v_1+v_2)\otimes w = v_1\otimes w + v_2 \otimes w$, $v\otimes(w_1+w_2) = v\otimes w_1 + v\otimes w_2$, and $(cv)\otimes w = v\otimes(cw) = c(v\otimes w)$ for $c \in \mathbb{C}$.*

If $\{v_i\}_{i=1}^{m}$ and $\{w_j\}_{j=1}^{n}$ are bases, then $\{v_i \otimes w_j\}_{i,j}$ is a basis of $V \otimes_{\mathbb{C}} W$. In particular,

$$\dim(V \otimes_{\mathbb{C}} W) = (\dim V)(\dim W). \tag{2.23}$$

Exercise 2.2.1. *Prove the basis assertion. Bilinearity proves spanning; to prove linear independence, apply the bilinear coordinate maps $(v, w) \mapsto v_i^*(v)w_j^*(w)$ to a proposed relation, where v_i^* and w_j^* are the dual bases.*

Definition 2.2.2. *Let V and W be G-modules. Their tensor product representation is $V \otimes_{\mathbb{C}} W$, with action*

$$g(v \otimes w) = (gv) \otimes (gw), \qquad g \in G. \tag{2.24}$$

We usually write $V \otimes W$ when the ground field is understood.

Proposition 2.2.1. *If V and W have characters χ_V and χ_W, then*

$$\chi_{V\otimes W}(g) = \chi_V(g)\chi_W(g). \tag{2.25}$$

Proof. Choose bases in which g has matrices A and B. On the basis $v_i \otimes w_j$, the diagonal coefficient is $A_{ii}B_{jj}$. Summing over i, j gives $\operatorname{tr}(A \otimes B) = (\operatorname{tr} A)(\operatorname{tr} B)$ and proves Eq. (2.25). □

Tensor products occur when a state has several kinds of degrees of freedom, such as orbital and spin degrees of freedom. They also describe operators and selection rules: an operator from V to W transforms in $W \otimes V^\vee$, where $V^\vee$ is defined below.

Definition 2.2.3. *The dual representation $V^\vee$ is the dual vector space, with action*

$$(g\alpha)(v) = \alpha(g^{-1}v), \qquad \alpha \in V^\vee, \quad v \in V. \tag{2.26}$$

Exercise 2.2.2. *Verify that Eq. (2.26) defines a representation and prove $\chi_{V^\vee}(g) = \chi_V(g^{-1}) = \chi_V(g)^*$. Show that $w \otimes \alpha \mapsto [v \mapsto \alpha(v)w]$ identifies $W \otimes V^\vee$ with $\operatorname{Hom}_{\mathbb{C}}(V, W)$, whose action is $F \mapsto \rho_W(g)F\rho_V(g)^{-1}$. Deduce that the invariant tensors correspond exactly to the intertwining operators.*

Definition 2.2.4. *Let A be an algebra over $\mathbb{C}$, let M be a right A-module, and let N be a left A-module. The tensor product $M \otimes_A N$ is the quotient of $M \otimes_{\mathbb{C}} N$ by the subspace generated by $(ma)\otimes n - m\otimes(an)$ for $m \in M$, $n \in N$, and $a \in A$. Thus it satisfies the balancing relation*

$$(ma) \otimes_A n = m \otimes_A (an). \tag{2.27}$$

The right action on M and the left action on N are essential when A is not commutative. In our application, $\mathbb{C}[G]$ is a right $\mathbb{C}[H]$-module by right multiplication, and an H-representation W is a left $\mathbb{C}[H]$-module. Left multiplication by G on $\mathbb{C}[G]$ commutes with its right $\mathbb{C}[H]$-action, so it descends to $\mathbb{C}[G] \otimes_{\mathbb{C}[H]} W$.

2.2.2 Induced representations

Definition 2.2.5. *Let $H \subset G$ and let W be an H-module. The induced representation of W from H to G is*

$$\operatorname{Ind}_H^G W = \mathbb{C}[G] \otimes_{\mathbb{C}[H]} W, \qquad g'(g \otimes w) = (g'g) \otimes w. \tag{2.28}$$

We omit the subscript on $\otimes$ inside this induced module. Its meaning is always fixed by Eq. (2.28); in particular, $gh \otimes w = g \otimes hw$ for $h \in H$.

Proposition 2.2.2. *Let $r_1, \dots, r_m$ represent the left cosets G/H, so $G = \bigsqcup_{i=1}^m r_i H$. Every element of $\operatorname{Ind}_H^G W$ can be written uniquely as $\sum_i r_i \otimes w_i$ with $w_i \in W$. Consequently,*

$$\dim \operatorname{Ind}_H^G W = [G : H] \dim W. \tag{2.29}$$

If $gr_i = r_j h$ with $h \in H$, the action is

$$g(r_i \otimes w) = r_j \otimes (hw). \tag{2.30}$$

Proof. The idea is to separate the motion between cosets from the action of H inside a coset. Every group element has a unique expression $r_i h$, and Eq. (2.27) gives $r_i h \otimes w = r_i \otimes hw$. This proves existence of the stated expression. To prove uniqueness, define a map to $W^{\oplus m}$ by sending $r_i h \otimes w$ to the vector with hw in its ith component and zero elsewhere, and extend linearly. The expressions $r_i hh' \otimes w$ and $r_i h \otimes h'w$ have the same image, so the map respects the balancing relation. Its inverse sends $(w_1, \dots, w_m)$ to $\sum_i r_i \otimes w_i$. This proves uniqueness and Eq. (2.29). Finally, $g(r_i \otimes w) = gr_i \otimes w = r_j h \otimes w = r_j \otimes hw$, proving Eq. (2.30). □

Proposition 2.2.2 turns the abstract definition into a practical construction of matrices: each column block has one nonzero block, given by the appropriate matrix of h. Changing the representatives changes the basis, not the representation. If H fixes a chosen atomic site, the cosets describe its symmetry orbit, and W describes the local orbitals at that site. This finite-group construction will also be used for momentum orbits.

Example 2.2.1. *For $H = G$, the map $g \otimes w \mapsto gw$ identifies $\operatorname{Ind}_G^G W$ with W. For $H = \{e\}$ and $W = \mathbb{C}$, Eq. (2.28) gives the canonical representation $\mathbb{C}[G]$. More generally, inducing the trivial representation of H gives the permutation representation on G/H: the basis vector $r_i \otimes 1$ corresponds to the coset $r_i H$.*

Proposition 2.2.3 (Function model)**.** *The representation $\operatorname{Ind}_H^G W$ is isomorphic to the space of functions $f : G \to W$ satisfying*

$$f(xh) = h^{-1}f(x), \qquad (gf)(x) = f(g^{-1}x), \qquad x, g \in G, \quad h \in H. \tag{2.31}$$

Proof. A function with the stated covariance is determined by its values at $r_1, \dots, r_m$. Associate $r_i \otimes w$ with the function $f_{i,w}$ supported on $r_i H$ and given by $f_{i,w}(r_i h) = h^{-1} w$. This is a vector space isomorphism by Proposition 2.2.2. If $g r_i = r_j h_0$, then $g f_{i,w}$ is supported on $r_j H$, and $(g f_{i,w})(r_j) = f_{i,w}(r_i h_0^{-1}) = h_0 w$. This is exactly Eq. (2.30), so the isomorphism intertwines the actions. □

Exercise 2.2.3. *Let $K \subset H \subset G$ and let U be a K-module. Prove transitivity of induction,*

$$\mathrm{Ind}_H^G(\mathrm{Ind}_K^H U) \cong \mathrm{Ind}_K^G U, \tag{2.32}$$

by constructing the map $g \otimes (h \otimes u) \mapsto gh \otimes u$ and its inverse. Check both balancing relations explicitly. Also show that replacing r_i by $r_i h_i$ in Proposition 2.2.2 amounts to replacing the vector w_i in that summand by $h_i^{-1} w_i$.

2.2.3 The character of an induced representation; the reciprocity formula

The coset description immediately suggests how to compute a character: a summand contributes to the trace only if the group element sends that coset to itself.

Theorem 2.2.1. *Let W be an H-module with character ψ. Then*

$$\chi_{\mathrm{Ind}_H^G W}(g) = \frac{1}{|H|} \sum_{\substack{x \in G \\ x^{-1} g x \in H}} \psi(x^{-1} g x). \tag{2.33}$$

Proof. Choose left coset representatives r_i and use the direct sum $\mathrm{Ind}_H^G W = \bigoplus_i (r_i \otimes W)$. By Eq. (2.30), g maps the ith summand to the summand over $g r_i H$. If $g r_i H \neq r_i H$, its matrix block is off the diagonal and contributes zero to the trace. If $g r_i H = r_i H$, then $r_i^{-1} g r_i \in H$, and its diagonal block is the action of $r_i^{-1} g r_i$ on W. Therefore

$$\chi_{\mathrm{Ind}_H^G W}(g) = \sum_{\substack{i \\ r_i^{-1} g r_i \in H}} \psi(r_i^{-1} g r_i). \tag{2.34}$$

For $x = r_i h$ in a contributing coset, $x^{-1} g x = h^{-1}(r_i^{-1} g r_i) h$. Since ψ is a class function on H, all $|H|$ such elements give the same value. Replacing the coset sum by the sum over $x \in G$ proves Eq. (2.33). □

This theorem often avoids constructing induced matrices. It also shows that the induced character vanishes on any conjugacy class of G that does not meet H. For the trivial representation of H, Eq. (2.34) counts fixed cosets, as expected for a permutation representation.

Definition 2.2.6. *For class functions on a finite group L, write*

$$\langle \alpha, \beta \rangle_L = \frac{1}{|L|} \sum_{l \in L} \alpha(l)^* \beta(l). \tag{2.35}$$

For a class function ψ on H, define $\mathrm{Ind}_H^G \psi$ by the right hand side of Eq. (2.33). For a G-module V, its restriction $\mathrm{Res}_H^G V$ is the same vector space with only the action of H retained, and its character is $\chi_V|_H$.

Theorem 2.2.2 (Frobenius reciprocity)**.** *Let ψ be a class function on H and let χ be a class function on G. Then*

$$\langle \mathrm{Ind}_H^G \psi, \chi \rangle_G = \langle \psi, \chi|_H \rangle_H. \tag{2.36}$$

In particular, if W and V are irreducible representations of H and G, respectively, the multiplicity of V in $\mathrm{Ind}_H^G W$ equals the multiplicity of W in $\mathrm{Res}_H^G V$.

Proof. We substitute Eq. (2.33) into the inner product, then use $h = x^{-1} g x$ to replace the constrained sum over (g, x) by an unconstrained sum over $(x, h) \in G \times H$. This gives

$$\langle \mathrm{Ind}_H^G \psi, \chi \rangle_G = \frac{1}{|G||H|} \sum_{x \in G} \sum_{h \in H} \psi(h)^* \chi(x h x^{-1}) = \frac{1}{|H|} \sum_{h \in H} \psi(h)^* \chi(h). \tag{2.37}$$

The last equality uses that χ is constant on conjugacy classes of G. The final expression is the right hand side of Eq. (2.36). For characters, orthogonality identifies these inner products with the stated multiplicities; the order of the two characters does not change a multiplicity, which is a nonnegative integer. □

The usefulness of Frobenius reciprocity is that restriction is usually easier to compute than induction. To find which full-group symmetry types can be built from a given local orbital, one may restrict each candidate symmetry type to the site symmetry group and count the required local type. The same relation later connects symmetry labels at momenta whose little groups are nested.

Proposition 2.2.4 (Frobenius reciprocity, module form)**.** *There is a natural vector space isomorphism*

$$\mathrm{Hom}_G(\mathrm{Ind}_H^G W, V) \cong \mathrm{Hom}_H(W, \mathrm{Res}_H^G V). \tag{2.38}$$

Proof. The idea is that a G-equivariant map from the induced representation is determined by its values on the copy $e \otimes W$. Given such a map F, define $\phi_F(w) = F(e \otimes w)$. For $h \in H$, the balancing relation gives $\phi_F(hw) = F(e \otimes hw) = F(h \otimes w) = h\phi_F(w)$, so ϕ_F is H-equivariant. Conversely, given an H-equivariant map $\phi : W \to V$, define

$$F_\phi(g \otimes w) = g\phi(w). \tag{2.39}$$

This respects the balancing relation because $gh\phi(w) = g\phi(hw)$, and it is G-equivariant because $F_\phi(g'g \otimes w) = g'F_\phi(g \otimes w)$. Finally, $\phi_{F_\phi}(w) = \phi(w)$, and $F_{\phi_F}(g \otimes w) = gF(e \otimes w) = F(g \otimes w)$. Thus the two constructions are inverse. □

Exercise 2.2.4. *Let 1_H be the trivial character of H and let V have character χ. Define $P_H = |H|^{-1} \sum_{h \in H} \rho_V(h)$. Show that P_H projects onto the subspace V^H of H-invariant vectors. Deduce*

$$\langle \mathrm{Ind}_H^G 1_H, \chi \rangle_G = \frac{1}{|H|} \sum_{h \in H} \chi(h) = \dim V^H. \tag{2.40}$$

If a localized orbital is unchanged by every site symmetry, explain how this formula determines its multiplicity in a candidate full-group representation.

2.2.4 Restriction to subgroups

Suppose we first induce from H to G, then restrict to another subgroup K. The copies of W over G/H split into K-orbits. Double cosets describe exactly these orbits, and the stabilizer of a coset determines the representation attached to it.

Definition 2.2.7. *For subgroups $H, K \subset G$, a double coset is $KxH = \{kxh : k \in K, h \in H\}$. The set of double cosets is denoted by $K\backslash G/H$. For $x \in G$, write ${}^xH = xHx^{-1}$ and define the conjugate representation ${}^xW$ on the vector space W by*

$$\rho_{{}^xW}(l) = \rho_W(x^{-1}lx), \qquad l \in xHx^{-1}. \tag{2.41}$$

Theorem 2.2.3 (Mackey restriction formula)**.** *Choose one representative x of each double coset in $K\backslash G/H$, and put $K_x = K \cap xHx^{-1}$. Then*

$$\mathrm{Res}_K^G \mathrm{Ind}_H^G W \cong \bigoplus_{x \in K\backslash G/H} \mathrm{Ind}_{K_x}^K \left(\mathrm{Res}_{K_x}^{xHx^{-1}} ({}^xW) \right). \tag{2.42}$$

Here and below, a sum over double cosets means a sum over a chosen set of representatives.

Proof. We first separate the induced module into the parts supported on different double cosets. We then identify each part with a representation induced from the stabilizer of xH. Since left multiplication by K and right multiplication by H preserve KxH, we have a decomposition into bimodules,

$$\mathbb{C}[G] = \bigoplus_{x \in K\backslash G/H} \mathbb{C}[KxH]. \tag{2.43}$$

Tensoring over $\mathbb{C}[H]$ with W therefore gives a direct sum of K-modules. For a fixed x, consider

$$\Phi_x : \mathbb{C}[K] \otimes_{\mathbb{C}[K_x]} {}^xW \longrightarrow \mathbb{C}[KxH] \otimes_{\mathbb{C}[H]} W, \qquad k \otimes w \longmapsto kx \otimes w, \tag{2.44}$$

where ${}^xW$ on the left is restricted to K_x. If $l \in K_x$, write $h = x^{-1}lx \in H$. The two images of the balancing relation agree: $klx \otimes w = kxh \otimes w = kx \otimes hw$, while l acts on ${}^xW$ as h by Eq. (2.41). Thus Φ_x is well-defined. It is K-equivariant, and it is surjective because every tensor $kxh \otimes w$ equals $kx \otimes hw$. To verify injectivity rather than merely assert it, choose representatives $k_1, \dots, k_t$ of K/K_x. The cosets $k_i xH$ are distinct: $k_i xH = k_j xH$ is equivalent to $k_j^{-1}k_i \in K_x$. They exhaust the left H-cosets in KxH. The source and target of Φ_x consequently both have dimension $[K : K_x] \dim W$, by the same coset-basis argument as Proposition 2.2.2. A surjective map between these finite dimensional spaces is an isomorphism. Combining these isomorphisms with Eq. (2.43) proves the formula. □

The intersection K_x is the subgroup of K that leaves the coset xH fixed. Thus Mackey's formula says that each K-orbit contributes its own induced representation. This is useful when a crystal symmetry is reduced: symmetry-related sites or momenta that previously belonged to one orbit may split into several orbits under the remaining subgroup.

Corollary 2.2.1 (Branching rule)**.** *Let V_i be an irreducible G-module with character χ_i, and let W_α be an irreducible H-module with character ψ_α. The multiplicity of W_α in $\mathrm{Res}^G_H V_i$ is*

$$b_{\alpha i} = \langle \psi_\alpha, \chi_i|_H \rangle_H = \langle \mathrm{Ind}^G_H \psi_\alpha, \chi_i \rangle_G. \tag{2.45}$$

It is also the multiplicity of V_i in $\mathrm{Ind}^G_H W_\alpha$.

Proof. The first equality is the character multiplicity formula, and the second is Frobenius reciprocity. □

Branching determines how a symmetry multiplet can split when only H remains a symmetry. It constrains the allowed splitting but does not determine the resulting energies. Along a symmetry line in momentum space, the same calculation gives compatibility relations between labels at an endpoint and labels on the line, provided the representations are compared within the appropriate common subgroup.

Exercise 2.2.5. *Let $G = D_4$ and $H = \langle s \rangle$ for a reflection s. Restrict the two-dimensional irreducible representation in Table 1.1 to $H \cong C_2$, and find the multiplicities of its even and odd representations. Explain why a perturbation preserving only this reflection can split the doublet. Use Eq.* (2.45) *to recover the multiplicity of the doublet in each representation induced from these one-dimensional H-representations.*

2.2.5 Mackey's irreducibility criterion

We now combine restriction and reciprocity to test irreducibility of an induced representation. The key observation is that its character norm is a sum of nonnegative overlap multiplicities, one for each double coset.

Theorem 2.2.4 (Mackey's irreducibility criterion)**.** *Let W be an irreducible H-module and put $H_x = H \cap xHx^{-1}$. Then $\mathrm{Ind}^G_H W$ is irreducible if and only if, for every $x \notin H$,*

$$\mathrm{Hom}_{H_x}\left(\mathrm{Res}^H_{H_x} W, \mathrm{Res}^{xHx^{-1}}_{H_x}({}^xW)\right) = 0. \tag{2.46}$$

Equivalently, the two restricted representations have no irreducible constituent in common. It is enough to check one representative of each double coset $HxH \neq H$.

Proof. Set $U = \mathrm{Ind}^G_H W$. By Theorem 1.3.3, U is irreducible exactly when $\langle \chi_U, \chi_U \rangle_G = 1$. First apply Frobenius reciprocity to write this norm as $\langle \chi_W, \chi_U|_H \rangle_H$. Next apply Theorem 2.2.3 with $K = H$. Finally apply Frobenius reciprocity to each resulting induced character. This gives

$$\langle \chi_U, \chi_U \rangle_G = \sum_{x \in H\backslash G/H} \langle \chi_W|_{H_x}, \chi_{{}^xW}|_{H_x} \rangle_{H_x}. \tag{2.47}$$

For the last use of reciprocity one may interchange the two characters, since their inner product is a real nonnegative integer. More explicitly, if two completely reducible modules have multiplicities a_j and b_j of the same irreducible types, their inner product and the dimension of their intertwiner space both equal $\sum_j a_j b_j$. Thus every summand in Eq. (2.47) is the dimension of the space in Eq. (2.46). The double coset H is represented by $x = e$, and contributes $\langle \chi_W, \chi_W \rangle_H = 1$. The norm equals one precisely when all other summands vanish. Choosing any particular $x \notin H$ as the representative of its double coset gives the formulation for every $x \notin H$. □

The criterion identifies the possible obstruction to irreducibility: the original representation and a copy moved by an element outside H may still overlap when compared on their common symmetry subgroup. In the little group construction, distinct translation characters will force these overlaps to vanish, making the criterion particularly simple to apply.

Corollary 2.2.2. *If H is normal in G and W is irreducible, then $\mathrm{Ind}^G_H W$ is irreducible if and only if ${}^xW \not\cong W$ for every $x \notin H$.*

Proof. Normality gives $H_x = H$. The conjugate ${}^xW$ remains irreducible, so Schur's lemma identifies Eq. (2.46) with non-isomorphism of W and ${}^xW$. □

Exercise 2.2.6. *Apply Mackey's criterion to $H = \{e\}$ and show that the canonical representation $\mathbb{C}[G]$ is irreducible exactly when $G = \{e\}$. More generally, show that $\mathrm{Ind}^G_H 1_H$ is reducible whenever H is a proper subgroup, by locating its invariant vector in the coset model.*

Exercise 2.2.7. *Suppose H is normal in G and W is irreducible. Show that ${}^x({}^yW) = {}^{xy}W$ at the level of the actions defined in Eq.* (2.41)*, and that ${}^hW \cong W$ for $h \in H$. Deduce that $G_W = \{g \in G : {}^gW \cong W\}$ is a subgroup containing H. This is the inertia group, which measures how far induction from H is from satisfying Corollary 2.2.2.*

2.3 Examples of induced representations

The abstract results of Section 2.2 become especially useful when G has a normal subgroup. We first show how an irreducible representation restricts to that subgroup and how to reconstruct it from one isotypic component. We then treat $G = A \rtimes H$ with A abelian. In this case the constituents of A are one-dimensional characters, and the construction becomes the finite-group version of the little group method used for crystal momentum.

2.3.1 Normal subgroups; applications to the degrees of the irreducible representations

Let N be normal in G. Eq. (2.41) now gives an N-representation ${}^gW$ for every $g \in G$. If ψ is the character of W, its conjugate character is

$$({}^g\psi)(n) = \psi(g^{-1}ng), \qquad n \in N. \tag{2.48}$$

Thus G acts on the isomorphism classes of irreducible N-representations. The inverse in Eq. (2.48) is consistent with the fact that applying g to a subspace of type W produces a subspace of type ${}^gW$.

Definition 2.3.1. *For an irreducible N-module W, its inertia group is*

$$G_W = \{g \in G : {}^gW \cong W\} = \{g \in G : {}^g\psi = \psi\}. \tag{2.49}$$

It contains N, by Exercise 2.2.7.

Proposition 2.3.1 (Restriction to a normal subgroup)**.** *Let V be an irreducible G-module, and let W be an irreducible constituent of $\mathrm{Res}^G_N V$. Choose representatives $g_1 = e, g_2, \dots, g_r$ of G/G_W. There is a positive integer e_W such that*

$$\mathrm{Res}^G_N V \cong \bigoplus_{i=1}^{r} ({}^{g_i}W)^{\oplus e_W}, \qquad \dim V = e_W [G : G_W] \dim W. \tag{2.50}$$

The conjugates appearing in this sum are pairwise non-isomorphic.

Proof. We use the canonical isotypic decomposition of V as an N-module. The proof has two steps: show that G permutes the isotypic components, and then use irreducibility to show that it permutes them transitively. For an irreducible N-module U, let $V[U]$ be the sum of all N-submodules of V isomorphic to U. If $Y \subset V$ is such a submodule, then

$$n(gy) = g(g^{-1}ng)y, \qquad n \in N, \quad y \in Y. \tag{2.51}$$

Normality ensures $g^{-1}ng \in N$, so gY has type ${}^gU$. Applying this argument to g^{-1} as well proves $gV[U] = V[{}^gU]$. The sum of all components in the orbit of W is therefore a nonzero G-invariant subspace. Irreducibility of V makes that sum all of V. The orbit is indexed by G/G_W, so there are exactly $r = [G : G_W]$ distinct types. The invertible map $v \mapsto g_i v$ identifies $V[W]$ with $V[{}^{g_i}W]$ as vector spaces. Every conjugate of W has dimension $\dim W$. Hence the number of copies in each component is the same integer $e_W = \dim V[W] / \dim W$. This proves Eq. (2.50). □

This is the basic restriction statement of Clifford theory. It says more than complete reducibility: all constituents lie in one orbit and have equal multiplicities. When N is a translation group, its characters label momenta. The orbit becomes a star of momenta, and e_W counts the internal states at each momentum in an irreducible full-group representation. Equal multiplicities do not imply $e_W = 1$.

Corollary 2.3.1. *With the notation above, if χ and ψ are the characters of V and W, respectively, then*

$$\chi(n) = e_W \sum_{i=1}^{r} \psi(g_i^{-1}ng_i), \qquad n \in N. \tag{2.52}$$

Proof. Take characters in Eq. (2.50). □

Proposition 2.3.2 (Reconstruction from an isotypic component)**.** *Under the assumptions of Proposition 2.3.1, put $I = G_W$ and $V_0 = V[W]$. Then V_0 is an irreducible I-module and*

$$V \cong \mathrm{Ind}^G_I V_0. \tag{2.53}$$

Proof. By Eq. (2.51), I preserves V_0. Suppose $0 \neq Y \subset V_0$ is I-invariant. The sum $\bigoplus_i g_i Y$ lies in the direct sum of the distinct N-isotypic components $g_i V_0$. It is G-invariant: if $gg_i = g_j h$ with $h \in I$, then $gg_i Y = g_j hY = g_j Y$. Since V is irreducible, this sum equals V. Intersecting it with V_0 gives $Y = V_0$, so V_0 is irreducible as an I-module. The map $g \otimes v \mapsto gv$ from $\mathrm{Ind}^G_I V_0$ to V is well-defined and G-equivariant. It maps the coset summands bijectively onto the direct summands $g_i V_0$, so it is an isomorphism. □

Reconstruction explains why the inertia group, rather than N alone, is the appropriate subgroup from which to induce. The remaining problem is to understand the irreducible representations of I whose restriction to N consists of copies of W. This problem is particularly simple when W is one-dimensional and extends to I, but the existence of such an extension needs a proof.

Proposition 2.3.3. *Let W be an irreducible N-module. Then $\operatorname{Ind}_N^G W$ is irreducible if and only if $G_W = N$. When this holds, its degree is $[G:N]\dim W$. For any irreducible G-module V, the multiplicity of V in $\operatorname{Ind}_N^G W$ equals the multiplicity of W in $\operatorname{Res}_N^G V$:*

$$\langle \chi_{\operatorname{Ind}_N^G W}, \chi_V \rangle_G = \langle \chi_W, \chi_V|_N \rangle_N. \tag{2.54}$$

Proof. The irreducibility statement is Corollary 2.2.2, and its dimension is Eq. (2.29). The equality of multiplicities is Frobenius reciprocity. In particular, when W occurs in V, both multiplicities equal the integer e_W in Eq. (2.50). □

We next prove a divisibility result that provides a useful check on possible irreducible degrees. The proof can be given with the ordinary representation theory already developed; no assumption about extending an invariant character is needed.

Lemma 2.3.1. *Let Z be a central subgroup of a finite group L. The degree d of an irreducible representation of L divides $[L:Z]$.*

Proof. By Schur's lemma, $z \in Z$ acts as a scalar $\lambda(z)$. For each positive integer m, let L^m act on $V^{\otimes m}$ by letting the jth group factor act on the jth tensor factor. Its character is $\chi(g_1)\cdots\chi(g_m)$, whose squared norm on L^m is $\langle \chi, \chi \rangle_L^m = 1$, so this representation is irreducible of degree d^m. The central subgroup

$$Z_m = \{(z_1, \ldots, z_m) \in Z^m : z_1 \cdots z_m = e\} \tag{2.55}$$

has order $|Z|^{m-1}$ and acts trivially, because its scalar action is $\lambda(z_1)\cdots\lambda(z_m) = 1$. Thus the representation descends to L^m/Z_m. Theorem 2.1.2 gives

$$d^m \;\Big|\; |L^m/Z_m| = |L|[L:Z]^{m-1}, \qquad m \geq 1. \tag{2.56}$$

Fix a prime p, and let a, b, and c be the exponents of p in d, $[L:Z]$, and $|L|$, respectively. Eq. (2.56) implies $ma \leq c + (m-1)b$ for every m. If $a > b$, this fails for sufficiently large m. Hence $a \leq b$ for every prime, proving $d \mid [L:Z]$. □

Theorem 2.3.2 (Itô's theorem)**.** *Let A be an abelian normal subgroup of G. Then the degree of every irreducible representation of G divides $[G:A]$.*

Proof. Let V be irreducible and choose a character λ of A occurring in V. Since A is abelian, all its irreducible representations are one-dimensional. Put $I = G_\lambda$ and let V_λ be the λ-isotypic component. Propositions 2.3.1 and 2.3.2 show that V_λ is irreducible under I and that $\dim V = [G:I]\dim V_\lambda$. To control $\dim V_\lambda$, we make the scalar action of A into a central subgroup of a quotient. Set $B = \ker\lambda \subset A$. Invariance of λ under I makes B normal in I, and B acts trivially on V_λ. Thus V_λ descends to an irreducible representation of I/B. Moreover, for $i \in I$ and $a \in A$, invariance gives $\lambda(iai^{-1}a^{-1}) = 1$. Hence $iai^{-1}a^{-1} \in B$, so A/B is central in I/B. Lemma 2.3.1 now gives $\dim V_\lambda \mid [I/B : A/B] = [I:A]$. Therefore

$$\dim V = [G:I]\dim V_\lambda \;\Big|\; [G:I][I:A] = [G:A]. \tag{2.57}$$

□

The theorem bounds irreducible degeneracies using an abelian normal subgroup, often a much smaller quotient than G itself. The proof also shows why an invariant character need not extend to the inertia group: A acts by scalars on V_λ, but those scalars need not be removable by a one-dimensional representation of I. The quotient I/A may then act projectively. For semidirect products with abelian normal factor, the extension used below can be constructed explicitly.

Exercise 2.3.1. *Assume that $\lambda : A \to \mathbb{C}^\times$ extends to a one-dimensional representation $\widetilde{\lambda}$ of $I = G_\lambda$. Show that $\widetilde{\lambda}^{-1} \otimes V_\lambda$ is an irreducible ordinary representation of I/A, and give a shorter proof of Itô's theorem in this case. To see why extension is a real assumption, take $G = Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ and $A = \{\pm 1\}$. Show that the character $\lambda(-1) = -1$ is G-invariant but has no one-dimensional extension to Q_8, since $[i,j] = -1$ and every one-dimensional representation kills commutators.*

Example 2.3.1. *Let $D_4 = \langle r, s : r^4 = e, s^2 = e, srs^{-1} = r^{-1}\rangle$, and let $A = \langle r\rangle \cong C_4$. Its characters are $\lambda_m(r) = \exp(2\pi im/4)$, with $m = 0, 1, 2, 3$. Conjugation by s exchanges λ_m and λ_{-m}, so the orbits are $\{\lambda_0\}$, $\{\lambda_2\}$, and $\{\lambda_1, \lambda_3\}$. The inertia group of λ_1 is A, so $\mathrm{Ind}_A^{D_4}\lambda_1$ is irreducible of degree two. Eq.* (2.33) *gives*

$$\chi(r^j) = e^{\pi ij/2} + e^{-\pi ij/2} = 2\cos(\pi j/2), \qquad \chi(sr^j) = 0. \tag{2.58}$$

This is the two-dimensional representation in Table 1.1.

Exercise 2.3.2. *Use the coset representatives e, s to construct the matrices of r and s in Example 2.3.1, and verify their defining relations. Show that each of λ_0 and λ_2 extends to two one-dimensional representations of D_4, distinguished by the sign assigned to s. Recover the four one-dimensional representations in Table 1.1.*

2.3.2 Semidirect products by an abelian group

Suppose $G = A \rtimes H$, where A is finite abelian. Thus A is normal, $A \cap H = \{e\}$, and every group element has a unique expression ah with $a \in A$, $h \in H$. Multiplication is

$$(a_1h_1)(a_2h_2) = a_1(h_1a_2h_1^{-1})h_1h_2. \tag{2.59}$$

Definition 2.3.2. *The dual group of A is $\widehat{A} = \mathrm{Hom}(A, \mathbb{C}^\times)$, whose elements are all the irreducible characters of A. The group H acts on it by*

$$(h\lambda)(a) = \lambda(h^{-1}ah). \tag{2.60}$$

The little group of λ in H is its stabilizer $H_\lambda = \{h \in H : h\lambda = \lambda\}$. Its inertia group in G is $G_\lambda = A \rtimes H_\lambda$.

The equality for G_λ follows because conjugation by an element of the abelian group A acts trivially on A. The little group therefore records precisely which elements of H preserve the chosen character.

Proposition 2.3.4. *The character λ extends to a one-dimensional representation of G_λ by*

$$\widetilde{\lambda}(ah) = \lambda(a), \qquad a \in A, \quad h \in H_\lambda. \tag{2.61}$$

Proof. The factorization ah is unique, so the formula is well defined. For $h_1 \in H_\lambda$, invariance gives $\lambda(h_1a_2h_1^{-1}) = \lambda(a_2)$ because $h_1^{-1} \in H_\lambda$. Using Eq. (2.59), we obtain

$$\widetilde{\lambda}((a_1h_1)(a_2h_2)) = \lambda(a_1)\lambda(a_2) = \widetilde{\lambda}(a_1h_1)\widetilde{\lambda}(a_2h_2). \tag{2.62}$$

Thus the extension is a homomorphism. □

This explicit extension is the step that uses the semidirect product assumption. An arbitrary group extension with abelian normal subgroup need not have this property. In particular, applying the construction to a nonsymmorphic space group requires keeping its additional translation phases; they lead to projective representations of the little co-group. Let σ be a representation of H_λ. We inflate it to G_λ by $\sigma(ah) = \sigma(h)$, so A acts trivially, and form $\widetilde{\lambda} \otimes \sigma$. Explicitly its action is $ah \mapsto \lambda(a)\sigma(h)$. It is irreducible exactly when σ is irreducible, because multiplication by a nonzero scalar does not change invariant subspaces.

Theorem 2.3.3 (Little group construction for $A \rtimes H$)**.** *Choose one representative λ from each H-orbit in $\widehat{A}$. For every irreducible representation σ of H_λ, set*

$$V_{\lambda,\sigma} = \mathrm{Ind}_{A\rtimes H_\lambda}^{A\rtimes H}(\widetilde{\lambda} \otimes \sigma). \tag{2.63}$$

These representations are irreducible, pairwise non-isomorphic, and exhaust all irreducible representations of G. Their dimensions are

$$\dim V_{\lambda,\sigma} = [H : H_\lambda] \dim \sigma. \tag{2.64}$$

If arbitrary orbit representatives are allowed, then $V_{\lambda',\sigma'} \cong V_{\lambda,\sigma}$ precisely when $\lambda' = h\lambda$ for some $h \in H$ and $\sigma' \cong {}^h\sigma$, where $H_{\lambda'} = hH_\lambda h^{-1}$ and ${}^h\sigma(l) = \sigma(h^{-1}lh)$.

Proof. The proof has three parts. Distinct A-characters make Mackey's overlap spaces vanish, proving irreducibility. Conversely, the weight spaces of any irreducible G-module identify its little group data. Finally, those same weight spaces distinguish the resulting representations. First, fix λ, σ, put $K = A \rtimes H_\lambda$, and let $W = \widetilde{\lambda} \otimes \sigma$. If $x \notin K$, then $x\lambda \neq \lambda$. The subgroup $K \cap xKx^{-1}$ contains A, because A is normal and contained in K. On W, an element $a \in A$ acts as the scalar $\lambda(a)$, whereas on ${}^xW$ it acts as $(x\lambda)(a)$. Any intertwiner F between the two restricted modules therefore satisfies $\lambda(a)F = (x\lambda)(a)F$ for every $a \in A$. Choose a for which the characters differ; then $F = 0$. Mackey's criterion proves that $V_{\lambda,\sigma}$ is irreducible. Notice that the scalar

character on each copy is λ, while the character of the whole restriction $W|_A$ is $(\dim\sigma)\lambda$. Second, let V be an irreducible G-module. Since A is abelian, it decomposes into weight spaces,

$$V = \bigoplus_{\mu\in\widehat{A}} V_\mu, \qquad V_\mu = \{v \in V : av = \mu(a)v \text{ for every } a \in A\}. \tag{2.65}$$

Eq. (2.51) gives $hV_\mu = V_{h\mu}$. Proposition 2.3.1 says that the nonzero weights form one H-orbit. Choose its fixed representative λ. By Proposition 2.3.2, V_λ is irreducible under $K = A \rtimes H_\lambda$, and $V \cong \mathrm{Ind}_K^G V_\lambda$. On V_λ, A acts as λ, and the restriction of the action to H_λ defines a representation σ. A subspace of V_λ is K-invariant if and only if it is H_λ-invariant, so σ is irreducible. The factorization ah gives $V_\lambda \cong \widetilde{\lambda}\otimes\sigma$, proving exhaustiveness. Third, a G-isomorphism preserves every A-weight space. Hence isomorphic induced representations have the same weight orbit. When the representative λ is fixed, the λ-weight space in Eq. (2.63) is exactly the summand over the coset K; its H_λ-representation is σ. An isomorphism therefore forces $\sigma \cong \sigma'$. If λ is replaced by $h\lambda$, the corresponding weight space is hV_λ, and the action of $l \in hH_\lambda h^{-1}$ is transported to $\sigma(h^{-1}lh)$ by Eq. (2.51). This gives the stated conjugacy condition. Conversely, conjugating the inducing data by h gives an isomorphic induced module: the map on tensors is $g\otimes w \mapsto gh\otimes w$, with the source induced from hKh^{-1}. Lastly, Eq. (2.29) and $[G : K] = [H : H_\lambda]$ prove Eq. (2.64). □

The theorem separates two distinct pieces of symmetry information. The orbit of λ tells us which momenta are related by the full group, and σ describes the internal symmetry at one chosen momentum. The factor $[H : H_\lambda]$ counts different momentum sectors; it should not be interpreted as a degeneracy within a single Bloch fiber. For a Hamiltonian commuting with the group, the irreducible little group dimension constrains the degeneracy at that momentum, while the full orbit relates energies at different momenta.

Corollary 2.3.2. *The restriction of $V_{\lambda,\sigma}$ to A and its character on A are*

$$\mathrm{Res}_A^G V_{\lambda,\sigma} \cong \bigoplus_{\mu\in H\lambda} \mu^{\oplus \dim\sigma}, \qquad \chi_{\lambda,\sigma}(a) = (\dim\sigma)\sum_{\mu\in H\lambda}\mu(a). \tag{2.66}$$

Proof. Choose coset representatives h_i of H/H_λ. On the summand $h_i \otimes W$, an element $a \in A$ acts by the scalar $\lambda(h_i^{-1}ah_i) = (h_i\lambda)(a)$. These are the distinct characters in the orbit, and every summand has dimension $\dim\sigma$. □

Example 2.3.2 (The dihedral group)**.** *Let $D_n = C_n \rtimes C_2$ have order $2n$, with $C_n = \langle r\rangle$ and $srs^{-1} = r^{-1}$. Its translation-like characters are $\lambda_m(r) = \exp(2\pi im/n)$ for $m = 0,\dots,n-1$, and $s\lambda_m = \lambda_{-m}$. Each two-element orbit $\{\lambda_m, \lambda_{-m}\}$ has trivial little group and gives one two-dimensional irreducible representation. The fixed character $m = 0$, and also $m = n/2$ when n is even, has little group C_2. Each fixed character gives two one-dimensional representations, according to the sign chosen for $\sigma(s)$. Thus there are two one-dimensional and $(n-1)/2$ two-dimensional irreducible representations for odd n, and four one-dimensional and $n/2-1$ two-dimensional irreducible representations for even n.*

Exercise 2.3.3. *Use $n = 4$ to recover all five representations in Table 1.1. For general n, verify that the sum of the squared degrees is $2n$. Explain why the two members of a generic orbit describe two momentum sectors rather than two independent little group labels at one momentum.*

Exercise 2.3.4. *Prove Itô's theorem directly for $G = A \rtimes H$ using Eq. (2.64) and the fact that $\dim\sigma$ divides $|H_\lambda|$. As a separate completeness check on the little group construction, show that*

$$\sum_{\lambda\in\widehat{A}/H}\ \sum_{\sigma\in\mathrm{Irr}(H_\lambda)} (\dim V_{\lambda,\sigma})^2 = |A||H|, \tag{2.67}$$

where $\widehat{A}/H$ denotes a set of orbit representatives. Use $\sum_\sigma(\dim\sigma)^2 = |H_\lambda|$ and the orbit-stabilizer formula.

Exercise 2.3.5. *Let $A = C_N$ be the translation group of a periodic chain with N sites and lattice spacing a, and let $H = C_2$ act by inversion. Use the translation convention $\lambda_k(t_{ja}) = e^{-ikja}$, with $k = 2\pi m/(Na)$ modulo $2\pi/a$. Find all inversion orbits and show that the little group is nontrivial only at $k = 0$ and, when N is even, $k = \pi/a$. Identify the little group labels at those points. Explain why inversion symmetry gives $E(k) = E(-k)$ while it alone does not force a twofold degeneracy at either inversion-invariant momentum.*

Chapter 3

Applications in solid state physics

In the preceding sections, we mainly studied finite groups and finite dimensional representations. In this section, we explain how these results enter the study of a crystal. The first step is to decompose the translation action, which introduces crystal momentum and the Brillouin zone. The second step is to study the symmetries that preserve a fixed momentum, where finite group representation theory becomes useful again. We then include antiunitary symmetries and use the resulting symmetry constraints in effective Hamiltonians. The standing assumptions of finite groups and finite dimensional spaces are relaxed whenever a translation group or a Hilbert space of an infinite crystal is considered. Unless spin is explicitly included, spatial symmetries below act by ordinary, single-valued unitary representations.

3.1 Translation group and the Brillouin zone

Let $a_1, \dots, a_d$ be linearly independent vectors in $\mathbb{R}^d$. The associated Bravais lattice and its translation group are

$$L = \mathbb{Z}a_1 \oplus \cdots \oplus \mathbb{Z}a_d, \qquad T = \{t_R : R \in L\}, \qquad t_R t_S = t_{R+S}. \tag{3.1}$$

Thus $T \cong L \cong \mathbb{Z}^d$ is abelian, but infinite. We already know that irreducible representations of a finite abelian group are one dimensional. For translations the analogous labels will be unitary characters, and the collection of these characters will be the Brillouin zone.

Definition 3.1.1. *The reciprocal lattice of L is*

$$L^* = \{K \in \mathbb{R}^d : e^{iK\cdot R} = 1 \text{ for all } R \in L\}. \tag{3.2}$$

The reciprocal basis $b_1, \dots, b_d$ is defined by $a_i \cdot b_j = 2\pi\delta_{ij}$.

Proposition 3.1.1. *The reciprocal lattice is $L^* = \mathbb{Z}b_1 \oplus \cdots \oplus \mathbb{Z}b_d$.*

Proof. Write $K = \sum_j \alpha_j b_j$ and $R = \sum_j n_j a_j$. Then

$$K \cdot R = 2\pi \sum_{j=1}^{d} \alpha_j n_j. \tag{3.3}$$

If all α_j are integers, Eq. (3.3) gives $K \in L^*$. Conversely, if $K \in L^*$, taking $R = a_j$ gives $e^{2\pi i\alpha_j} = 1$, so every α_j is an integer. □

The factor 2π is the usual physics convention: a plane wave $e^{iK\cdot x}$ is periodic under all lattice translations precisely when $K \in L^*$.

Theorem 3.1.1 (Characters of the translation lattice)**.** *Every unitary character of L has the form*

$$\chi_k(R) = e^{-ik\cdot R}, \qquad k \in \mathbb{R}^d, \tag{3.4}$$

and $\chi_k = \chi_{k'}$ if and only if $k - k' \in L^*$. Consequently, the character group is $\widehat{L} \cong \mathbb{R}^d/L^* \cong U(1)^d$.

Proof. The idea is that a homomorphism from the free abelian group L is completely determined by its values on the primitive translations. Let $\chi : L \to U(1)$ be a character and choose real numbers θ_j such that $\chi(a_j) = e^{-i\theta_j}$. For $R = \sum_j n_j a_j$, the homomorphism property gives

$$\chi(R) = \prod_{j=1}^{d} \chi(a_j)^{n_j} = e^{-i\sum_j n_j\theta_j}. \tag{3.5}$$

Set $k = \sum_j (\theta_j/2\pi) b_j$. Then $k \cdot a_j = \theta_j$, so Eq. (3.5) is precisely Eq. (3.4). Conversely, that formula defines a character for every k. Finally, $\chi_k = \chi_{k'}$ means that $e^{-i(k-k')\cdot R} = 1$ for every $R \in L$, which is equivalent to $k - k' \in L^*$ by Eq. (3.2). Pointwise multiplication of characters corresponds to addition of momenta modulo L^*, giving the asserted group isomorphism. The topology is also the usual torus topology, since a character is specified by its d values in $U(1)$. □

This is the lattice example of Pontryagin duality [16, 11]. The general theorem is not needed for the preceding proof. Its useful message here is that momentum is the variable dual to translation: two wavevectors represent the same crystal momentum exactly when translations cannot distinguish them.

Proposition 3.1.2. *Every finite dimensional irreducible unitary representation of L is one dimensional and is given by Eq.* (3.4).

Proof. Since L is abelian, each $\rho(R)$ commutes with the entire representation. The proof of Schur's lemma in Proposition 1.3.3 uses irreducibility and finite dimensionality, but does not require the group to be finite. Hence $\rho(R) = \chi(R)\,\mathrm{Id}_V$ for every R. Every line in V is then invariant, so irreducibility gives $\dim V = 1$. Unitarity implies $|\chi(R)| = 1$, and Theorem 3.1.1 applies. □

Without unitarity, finite dimensional irreducible representations are still one dimensional, but their generator eigenvalues can be arbitrary nonzero complex numbers. Unitarity is what restricts the physically relevant translation characters to the torus.

Definition 3.1.2. *The Brillouin zone, viewed intrinsically, is the torus $\widehat{L} = \mathbb{R}^d/L^*$. Its points are called crystal momenta or quasimomenta. A fundamental domain for L^* in $\mathbb{R}^d$ is a choice of wavevector representatives for this torus. The first Brillouin zone is conventionally the Wigner–Seitz cell*

$$B_1 = \{k \in \mathbb{R}^d : |k| \le |k - K| \text{ for every } K \in L^*\}. \tag{3.6}$$

The closed cell in Eq. (3.6) has repeated representatives on its boundary. These must be identified, or a boundary convention must be chosen, to obtain exactly one representative per momentum. In particular, opposite faces of a drawn Brillouin zone are not different physical momentum boundaries. Crystal momentum already exists for an infinite lattice; periodic boundary conditions merely replace this torus by a finite grid.

Exercise 3.1.1. *Show that a character χ_k descends to L/NL if and only if $Nk \in L^*$. Deduce that*

$$\widehat{L/NL} \cong \frac{(1/N)L^*}{L^*}, \qquad k = \sum_{j=1}^{d} \frac{m_j}{N} b_j, \qquad m_j \in \{0, 1, \dots, N-1\}. \tag{3.7}$$

Explain why there are N^d momenta and why these grids become dense in the Brillouin torus as $N \to \infty$.

We next explain how these characters decompose the Hilbert space of a crystal. For translations we use the convention $(t_R\psi)(x) = \psi(x - R)$; thus $e^{ik\cdot x}$ has eigenvalue $e^{-ik\cdot R}$. This fixes the sign in Eq. (3.4) and all subsequent Fourier formulas.

Definition 3.1.3. *Let W be the finite dimensional internal Hilbert space of one unit cell, containing its orbital, sublattice, and, when applicable, spin degrees of freedom. The tight-binding Hilbert space is*

$$\mathcal{H} = \ell^2(L) \otimes W = \left\{ \psi : L \to W : \sum_{R \in L} \|\psi(R)\|^2 < \infty \right\}, \qquad (t_S\psi)(R) = \psi(R - S). \tag{3.8}$$

Equivalently, $t_S(|R\rangle \otimes w) = |R + S\rangle \otimes w$.

Theorem 3.1.2 (Bloch–Fourier decomposition)**.** *Let $d\mu$ be the normalized Haar measure on $\widehat{L}$. The Fourier transform, initially defined for finitely supported functions by*

$$(\mathcal{F}\psi)(k) = \sum_{R \in L} e^{-ik\cdot R} \psi(R), \tag{3.9}$$

extends uniquely to a unitary map $\mathcal{F} : \mathcal{H} \to L^2(\widehat{L}, d\mu; W)$. It diagonalizes every translation:

$$(\mathcal{F} t_S \mathcal{F}^{-1} f)(k) = e^{-ik\cdot S} f(k). \tag{3.10}$$

Equivalently, the translation representation has the direct-integral decomposition

$$\mathcal{H} \cong \int_{\widehat{L}}^{\oplus} \mathcal{H}_k \, d\mu(k), \qquad \mathcal{H}_k \cong W. \tag{3.11}$$

Proof. We first prove that the transform preserves the inner product, then show that its range is the whole target space, and finally compute the translation action. Write $k = \sum_j (\theta_j/2\pi) b_j$ with $0 \le \theta_j < 2\pi$. In these coordinates $d\mu = \prod_j d\theta_j/(2\pi)$. Integrating each exponential separately gives the character orthogonality relation

$$\int_{\widehat{L}} e^{ik\cdot(R-R')}\, d\mu(k) = \delta_{R,R'}. \tag{3.12}$$

For finitely supported ψ, all sums are finite, so we may expand the norm and use Eq. (3.12) to obtain

$$\int_{\widehat{L}} \|(\mathcal{F}\psi)(k)\|^2\, d\mu(k) = \sum_{R,R'\in L} \int_{\widehat{L}} e^{ik\cdot(R-R')} \langle \psi(R), \psi(R')\rangle\, d\mu(k) = \sum_{R\in L} \|\psi(R)\|^2. \tag{3.13}$$

Here the inner product is conjugate linear in its first argument. Finitely supported functions are dense in $\mathcal{H}$, so this isometry extends uniquely to $\mathcal{H}$ and its range is closed. Its range contains all W-valued trigonometric polynomials. Such polynomials are dense in L^2 of the torus: scalar trigonometric polynomials are uniformly dense in continuous functions, and continuous functions are dense in L^2. These standard Fourier-analysis density facts show that the range is also dense, hence is the whole target. This last step is needed; an isometry alone need not be onto. For a finitely supported ψ, changing variables $R' = R - S$ gives

$$(\mathcal{F} t_S \psi)(k) = \sum_{R\in L} e^{-ik\cdot R} \psi(R-S) = e^{-ik\cdot S} \sum_{R'\in L} e^{-ik\cdot R'} \psi(R'). \tag{3.14}$$

Continuity extends this identity to all $\psi \in \mathcal{H}$, proving Eq. (3.10). The direct integral in Eq. (3.11) is another notation for the resulting space of square-integrable W-valued functions. □

The theorem is the Plancherel theorem for L, tensored with W [16, 11]. Unlike the finite-group decompositions in the preceding sections, Eq. (3.11) is not a direct sum of normalizable eigenspaces at individual momenta. A single point of the torus has measure zero. Formally one can write

$$|k, w\rangle = \sum_{R\in L} e^{ik\cdot R} |R\rangle \otimes w, \qquad t_S |k, w\rangle = e^{-ik\cdot S} |k, w\rangle, \tag{3.15}$$

but for $w \neq 0$ this vector is not square summable. It is a generalized Bloch state, while a physical normalizable state is a wave packet. For a finite periodic crystal, the direct integral becomes an ordinary finite direct sum and normalized fixed-momentum states do exist.

Exercise 3.1.2. *Verify the inverse formula $\psi(R) = \int_{\widehat{L}} e^{ik\cdot R} (\mathcal{F}\psi)(k)\, d\mu(k)$. Then show directly from Eq.* (3.8) *that no nonzero vector of the infinite-lattice Hilbert space is a simultaneous eigenvector of all translations: its norm in each cell would be independent of the cell.*

Theorem 3.1.3 (Translation-invariant Hamiltonians)**.** *Let H be a bounded operator on $\mathcal{H}$ commuting with every translation. Then there is an essentially bounded measurable family of operators $H(k)$ on W, unique up to sets of measure zero, such that*

$$(\mathcal{F} H \mathcal{F}^{-1} f)(k) = H(k) f(k). \tag{3.16}$$

If H is self-adjoint, $H(k)$ is Hermitian for almost every k. In particular, if

$$H(|R\rangle \otimes w) = \sum_{S\in L} |R+S\rangle \otimes A_S w, \qquad \sum_{S\in L} \|A_S\| < \infty, \tag{3.17}$$

then its Bloch Hamiltonian is the continuous matrix function

$$H(k) = \sum_{S\in L} e^{-ik\cdot S} A_S. \tag{3.18}$$

In this case self-adjointness is equivalent to $A_{-S} = A_S^\dagger$, and $H(k)$ is Hermitian for every k. Finite-range hopping satisfies the summability assumption automatically.

Proof. The general assertion says that an operator commuting with all scalar Fourier multipliers must itself be a matrix multiplier. We give the argument so that the distinction between bounded and absolutely summable hopping is explicit. Choose an orthonormal basis of W and consider a scalar matrix entry B of $\mathcal{F} H \mathcal{F}^{-1}$. Eq. (3.10) implies that B commutes with multiplication by every trigonometric polynomial. Put $h = B1$. For every such polynomial p, we have $Bp = ph$, and boundedness gives

$$\int_{\widehat{L}} |p(k)|^2 |h(k)|^2\, d\mu(k) \le \|B\|^2 \int_{\widehat{L}} |p(k)|^2\, d\mu(k). \tag{3.19}$$

Uniform density extends this inequality to continuous p. Approximating nonnegative continuous test functions by squares of continuous functions shows that $|h| \le \|B\|$ almost everywhere; otherwise the inequality fails on a set where $|h| > \|B\|$. Thus multiplication by h is bounded and agrees with B on the dense set of trigonometric polynomials, so the two operators are equal. Applying this argument to each of the finitely many matrix entries gives Eq. (3.16). The adjoint of a matrix multiplier is pointwise multiplication by the adjoint matrix, which proves the self-adjointness statement.

For the hopping model, Eq. (3.17) is equivalently $(H\psi)(R) = \sum_S A_S \psi(R-S)$. Fourier transformation and Eq. (3.10) give

$$(\mathcal{F}H\psi)(k) = \sum_{S\in L} A_S e^{-ik\cdot S}(\mathcal{F}\psi)(k) = H(k)(\mathcal{F}\psi)(k). \tag{3.20}$$

Absolute summability justifies the operator-norm convergence of this sum and makes Eq. (3.18) uniformly convergent, hence continuous. Comparing matrix elements between cells gives $H = H^\dagger$ if and only if $A_{-S} = A_S^\dagger$. Substituting this equality into Eq. (3.18) gives $H(k)^\dagger = H(k)$. □

The eigenvalues $E_1(k), \dots, E_{\dim W}(k)$ of $H(k)$ are the energy bands, counted with multiplicity. This is why an infinite translation-invariant tight-binding problem reduces to finite matrices parametrized by momentum. For an arbitrary bounded translation-invariant operator, the fiber matrix is only measurable; an absolutely convergent hopping series must not be assumed without a locality or summability hypothesis. In the physical finite-range case the fibers are continuous, so discussing a particular high-symmetry momentum is unambiguous. For a continuum Hamiltonian $H = -\Delta + V(x)$ with a real periodic potential satisfying suitable regularity assumptions, the analogous Bloch–Floquet decomposition has infinite dimensional fibers of functions on one unit cell. Generalized eigenfunctions have the form

$$\psi_{n,k}(x) = e^{ik\cdot x}u_{n,k}(x), \qquad u_{n,k}(x+R) = u_{n,k}(x), \qquad \psi_{n,k}(x+R) = e^{ik\cdot R}\psi_{n,k}(x). \tag{3.21}$$

One may use quasiperiodic boundary conditions for $\psi_{n,k}$ or conjugate by $e^{ik\cdot x}$ and work with periodic functions, in which case the fiber operator is $(-i\nabla + k)^2 + V$. The analytic questions about operator domains and completeness belong to Bloch–Floquet theory [10, 5, 13]; the translation character is the same as in the tight-binding case.

Example 3.1.1 (One-dimensional tight-binding chain)**.** *Let $L = a\mathbb{Z}$ and $W = \mathbb{C}$. Then $L^* = (2\pi/a)\mathbb{Z}$, so the Brillouin torus is a circle, represented for example by $-\pi/a < k \le \pi/a$. With real parameters ϵ, t, consider*

$$H = \epsilon \sum_{n\in\mathbb{Z}} |n\rangle\langle n| - t\sum_{n\in\mathbb{Z}} \big(|n+1\rangle\langle n| + |n\rangle\langle n+1|\big). \tag{3.22}$$

Here $|n\rangle$ denotes the orbital in cell na, so $A_0 = \epsilon$ and $A_{\pm a} = -t$. Eq. (3.18) immediately gives the single band

$$E(k) = H(k) = \epsilon - te^{-ika} - te^{ika} = \epsilon - 2t\cos(ka). \tag{3.23}$$

Its periodicity $E(k+2\pi/a) = E(k)$ expresses equality of translation characters, not an additional point symmetry.

Exercise 3.1.3. *For the chain in Example 3.1.1, impose periodic boundary conditions on $N \ge 3$ cells. Construct the normalized eigenstates at $k_m = 2\pi m/(Na)$, $m = 0, \dots, N-1$, and verify that their energies are given by Eq. (3.23). Explain the normalization factor and compare with the nonnormalizable states in Eq. (3.15).*

Translation symmetry has now supplied the momentum label. We next determine how the remaining crystal symmetries act on these labels and on the internal states at a fixed label.

3.2 Space group and irreducible Brillouin zone

Definition 3.2.1. *In Seitz notation an element of the Euclidean group is $g = \{p|\tau\}$, where $p \in O(d)$ and $\tau \in \mathbb{R}^d$, acting by $gx = px + \tau$. Composition and inversion are given by*

$$\{p|\tau\}\{q|\mu\} = \{pq|\tau + p\mu\}, \qquad \{p|\tau\}^{-1} = \{p^{-1}| - p^{-1}\tau\}. \tag{3.24}$$

In particular, $t_R = \{1|R\}$ is a pure translation.

Exercise 3.2.1. *Derive both formulas in Eq. (3.24) by applying the corresponding affine maps to a point x. Explain why the translation part of a product is generally not $\tau + \mu$.*

Definition 3.2.2. *A space group $\mathcal{G}$ is a discrete subgroup of the Euclidean group whose full subgroup of pure translations is a Bravais lattice:*

$$T = \mathcal{G} \cap \{\{1|R\} : R \in \mathbb{R}^d\} = \{t_R : R \in L\}. \tag{3.25}$$

The point homomorphism is $\pi : \mathcal{G} \to O(d)$, $\pi(\{p|\tau\}) = p$, and its image $P = \pi(\mathcal{G})$ is the point group of the crystal.

The word "full" matters: a convenient larger unit cell may have a translation sublattice smaller than the actual translation group. Here L always denotes all pure translations in $\mathcal{G}$. The point group of a crystal may likewise be smaller than the group of all orthogonal symmetries of its Bravais lattice, because the motif inside a unit cell can break lattice symmetries.

Proposition 3.2.1. *The translation subgroup is normal in $\mathcal{G}$, and*

$$\{p|\tau\}t_R\{p|\tau\}^{-1} = t_{pR}, \qquad pL = L \quad (p \in P). \tag{3.26}$$

Moreover, $\ker \pi = T$*, so $\mathcal{G}/T \cong P$.*

Proof. Eq. (3.24) gives $\{p|\tau\}\{1|R\}\{p^{-1}|-p^{-1}\tau\} = \{1|pR\}$. This is a pure translation in $\mathcal{G}$, so $pL \subset L$. Applying the same argument to p^{-1} gives equality. The assertion about the kernel follows directly from the definition of π, and the quotient assertion follows from the first isomorphism theorem. □

Theorem 3.2.1. *The point group P is finite.*

Proof. The idea is that an orthogonal map preserving a lattice has only finitely many choices for the images of a lattice basis. For each primitive vector a_j, Eq. (3.26) implies $pa_j \in L$, while orthogonality gives $|pa_j| = |a_j|$. A bounded set contains only finitely many points of L, so there are finitely many possibilities for each pa_j. A linear map is determined by the images of the d basis vectors; hence there are finitely many possibilities for p. □

Proposition 3.2.1 and Theorem 3.2.1 explain the basic structure of a space group:

$$1 \longrightarrow T \longrightarrow \mathcal{G} \xrightarrow{\pi} P \longrightarrow 1. \tag{3.27}$$

Its infinite normal subgroup is handled by Fourier analysis; its finite quotient is handled by representation theory. The broader crystallographic structural results are the Bieberbach theorems [2, 3], but the elementary argument above suffices for the facts needed here.

Exercise 3.2.2. *Let A_p be the matrix of $p \in P$ in the lattice basis and let $Q = (a_i \cdot a_j)$. Show that $A_p \in GL(d, \mathbb{Z})$ and $A_p^T Q A_p = Q$. Use positive definiteness of Q to give another proof of Theorem 3.2.1.*

Definition 3.2.3. *A space group is symmorphic if Eq.* (3.27) *splits: there exists a group homomorphism $s : P \to \mathcal{G}$ with $\pi \circ s = \mathrm{Id}_P$. Otherwise it is nonsymmorphic.*

Proposition 3.2.2. *For a symmorphic space group, every element can be written uniquely as $t_R s(p)$, and this gives an isomorphism $\mathcal{G} \cong L \rtimes P$ with multiplication*

$$(R, p)(R', p') = (R + pR', pp'). \tag{3.28}$$

Proof. If $\pi(g) = p$, then $gs(p)^{-1} \in \ker \pi = T$, so $g = t_R s(p)$ for some $R \in L$. Applying π proves uniqueness of p, and then of R. Finally, use Eq. (3.26) and the homomorphism property of s:

$$t_R s(p) t_{R'} s(p') = t_R\big(s(p) t_{R'} s(p)^{-1}\big) s(p) s(p') = t_{R+pR'} s(pp'). \tag{3.29}$$

This is Eq. (3.28). □

This proposition connects space groups directly to the semidirect products studied in the preceding section. For a nonsymmorphic group, representatives of point operations cannot all be chosen to multiply exactly as the point group. Screw rotations and glide reflections are familiar manifestations: their fractional translations cannot all be removed by changing the origin. The resulting translation errors will become phases in Bloch representations.

Exercise 3.2.3. *If a splitting s exists, show that the finite group $s(P)$ fixes a point of real space. A useful choice is the average of the orbit of any point. After moving the origin to this fixed point, show that $s(p) = \{p|0\}$. This proves the equivalence between the splitting definition and the usual description of a symmorphic group.*

Proposition 3.2.3. *The point group preserves L^* and therefore acts on the Brillouin torus by $[k] \mapsto [pk]$. The reciprocal-space action is characterized by*

$$(pk) \cdot R = k \cdot p^{-1}R. \tag{3.30}$$

Proof. Since p is orthogonal, Eq. (3.30) is the ordinary action on vectors. If $K \in L^*$ and $R \in L$, then $p^{-1}R \in L$, so $e^{i(pK)\cdot R} = e^{iK\cdot p^{-1}R} = 1$. Hence $pL^* \subset L^*$; applying the argument to p^{-1} gives equality. Thus replacing k by $k + K$ replaces pk by $pk + pK$, which is the same torus point. □

Proposition 3.2.4. *Let U_g implement $g = \{p|\tau\} \in \mathcal{G}$. A generalized Bloch state with character χ_k is mapped to one with character χ_{pk}. In fiber notation, U_g induces a unitary map $U_g(k) : \mathcal{H}_k \to \mathcal{H}_{pk}$.*

Proof. Use the conjugation identity to move a translation past U_g. For a state ψ_k with $t_R\psi_k = e^{-ik\cdot R}\psi_k$, we find

$$t_R U_g\psi_k = U_g t_{p^{-1}R}\psi_k = e^{-ik\cdot p^{-1}R}U_g\psi_k = e^{-i(pk)\cdot R}U_g\psi_k. \tag{3.31}$$

This is the claimed character. Equivalently, the same computation in the Fourier representation gives the stated map between fibers, understood almost everywhere for a merely measurable direct integral and pointwise in the usual continuous Bloch models. □

The point part p determines the new momentum, while the translation part can still affect the internal symmetry matrix. For a scalar continuum wavefunction, $(U_g\psi)(x) = \psi(p^{-1}(x - \tau))$, and substitution of Eq. (3.21) gives

$$(U_g\psi_k)(x) = e^{-i(pk)\cdot\tau}e^{i(pk)\cdot x}u_k\big(p^{-1}(x - \tau)\big). \tag{3.32}$$

The last factor remains L-periodic because $p^{-1}L = L$. Thus fractional translations are invisible to the momentum label but not to the Bloch symmetry action.

Definition 3.2.4. *The star, little co-group, and little group at k are respectively*

$$Pk = \{[pk] : p \in P\}, \qquad P_k = \{p \in P : pk - k \in L^*\}, \qquad \mathcal{G}_k = \pi^{-1}(P_k). \tag{3.33}$$

The star is an orbit of momenta; the little co-group is its stabilizer in P; the little group contains the full space-group operations preserving the fiber at k.

Proposition 3.2.5. *The little group has quotient $\mathcal{G}_k/T \cong P_k$, and*

$$1 \longrightarrow T \longrightarrow \mathcal{G}_k \longrightarrow P_k \longrightarrow 1, \qquad |Pk| = [P : P_k] = [\mathcal{G} : \mathcal{G}_k]. \tag{3.34}$$

Proof. The restriction of π to $\mathcal{G}_k = \pi^{-1}(P_k)$ is onto P_k and has kernel T. This proves the exact sequence. The remaining assertions are the orbit–stabilizer formula and the correspondence of cosets under π. □

A generic momentum has trivial little co-group. High-symmetry points, lines, and planes have larger stabilizers. The definition uses equality modulo L^*, which is essential at zone boundaries: an operation can preserve a crystal momentum even when it changes the chosen wavevector representative.

Definition 3.2.5. *A small representation at k is a finite dimensional representation $\sigma : \mathcal{G}_k \to GL(V)$ satisfying*

$$\sigma(t_R) = e^{-ik\cdot R}\,\mathrm{Id}_V \qquad (R \in L). \tag{3.35}$$

An irreducible small representation gives an irreducible symmetry label within a Bloch fiber. In the applications below these representations are unitary.

Theorem 3.2.2 (Little-group method)**.** *Let σ be an irreducible unitary small representation at k. Then $\mathrm{Ind}_{\mathcal{G}_k}^{\mathcal{G}}\sigma$ is irreducible and has dimension $|Pk|\dim\sigma$. Every finite dimensional irreducible unitary representation of $\mathcal{G}$ is obtained this way. Choosing one k from each star, the inequivalent irreducible small representations at these chosen momenta give inequivalent induced representations.*

Proof. The proof has the same structure as induction from an abelian normal subgroup: decompose into translation-character spaces, observe that the full group permutes them, and study the stabilizer of one space. Let V carry a finite dimensional irreducible unitary representation ρ of $\mathcal{G}$. The d commuting unitary primitive translations are simultaneously diagonalizable, so

$$V = \bigoplus_{k\in S} V_k, \qquad V_k = \{v \in V : \rho(t_R)v = e^{-ik\cdot R}v \text{ for all } R \in L\}, \tag{3.36}$$

where S is a finite set of distinct points of the Brillouin torus. Eq. (3.31) shows that $\rho(g)V_k = V_{pk}$. Thus the sum of all character spaces in a single star is $\mathcal{G}$-invariant. Irreducibility implies that S is one star.

Fix $k \in S$ and let $g_1, \dots, g_m$ represent $\mathcal{G}/\mathcal{G}_k$, with $g_1 = 1$. The restriction of ρ to $\mathcal{G}_k$ on V_k is a small representation σ. If $0 \neq W \subset V_k$ were a proper $\mathcal{G}_k$-invariant subspace, then $\bigoplus_i \rho(g_i)W$ would be a nonzero proper $\mathcal{G}$-invariant subspace: multiplication by any g permutes the cosets and acts on each copy through an element of $\mathcal{G}_k$. This contradicts irreducibility, so σ is irreducible. The map from the induced representation to V that sends the copy $g_i \otimes v$ to $\rho(g_i)v$ respects this same coset action. Its image is the direct sum of all the distinct spaces $V_{p_i k}$, hence it is an isomorphism.

Conversely, start from an irreducible small representation σ. The induced space is a direct sum of $m = |Pk|$ copies of its space, and translations act on these copies by the distinct characters in the star. Any invariant subspace of a finite dimensional unitary representation is invariant under the joint spectral projections of the translations, so it is the sum of its intersections with these character spaces. If it is nonzero, one intersection is nonzero; irreducibility of the corresponding conjugate small representation makes that intersection the whole character space. Transitivity on the star then gives every character space. Hence the induced representation is irreducible. Finally, an intertwiner preserves translation-character spaces, so different stars cannot be equivalent, and at the fixed representative k it restricts to an intertwiner of the small representations. This proves the classification and the dimension formula. □

This is the finite dimensional form of the little-group method associated with Wigner and Mackey [18, 15, 6]. It explains why the induction constructions of the preceding sections are useful in crystals. The factor $|Pk|$ counts different momenta in the star; it is not the number of degenerate states at one momentum. Degeneracy at a fixed k is controlled by the dimensions of its irreducible small representations. In the infinite crystal these finite dimensional representations organize generalized Bloch states; they are not finite dimensional subspaces of normalizable states at sharp momentum.

Definition 3.2.6. *An irreducible Brillouin zone is a choice of one representative from every orbit of P on $\mathbb{R}^d/L^*$. It may be pictured as a fundamental region in a chosen Brillouin cell, together with conventions for its boundary identifications.*

Proposition 3.2.6. *Suppose a translation-invariant Hamiltonian is invariant under the full space group. Then the fiber symmetry maps satisfy*

$$H(pk)U_g(k) = U_g(k)H(k), \qquad g = \{p|\tau\} \in \mathcal{G}. \tag{3.37}$$

Consequently, $H(pk)$ and $H(k)$ have the same spectrum, including multiplicities. For $g \in \mathcal{G}_k$, the operator $U_g(k)$ commutes with $H(k)$.

Proof. Restrict the identity $HU_g = U_g H$ to a Bloch fiber and use Proposition 3.2.4. This gives Eq. (3.37). Since $U_g(k)$ is unitary between the two fibers, the Hamiltonians are unitarily equivalent. The last statement follows when pk and k are the same torus point. □

This proposition has two different uses. First, it reduces the computation of band energies to an irreducible Brillouin zone: ordered eigenvalues obey $E_n(pk) = E_n(k)$. Second, at a fixed k each energy eigenspace carries a small representation, so an irreducible small representation of dimension d_α forces a multiplet of at least that dimension. With several equivalent copies, the Hamiltonian can mix the copies, just as in the block decomposition of a symmetric Hamiltonian studied earlier. For merely measurable fibers these statements hold almost everywhere; pointwise symmetry labels use a model with well-defined fiber matrices at the momentum in question. Boundary identifications need care, but being on the boundary of a chosen fundamental region does not itself imply a larger little co-group. Some boundary points are merely duplicated representatives. The stabilizer must always be computed from Eq. (3.33).

Example 3.2.1 (Square lattice)**.** *Let $L = a\mathbb{Z}e_x \oplus a\mathbb{Z}e_y$, so $L^* = (2\pi/a)\mathbb{Z}e_x \oplus (2\pi/a)\mathbb{Z}e_y$. Assume the crystal has the full square point group D_4, generated by $C_4(k_x, k_y) = (-k_y, k_x)$ and $m_x(k_x, k_y) = (k_x, -k_y)$. A first Brillouin cell is the closed square $|k_x|, |k_y| \leq \pi/a$ with opposite edges identified, and an irreducible region is $0 \leq k_y \leq k_x \leq \pi/a$. The usual special momenta are*

$$\Gamma = (0,0), \qquad X = (\pi/a, 0), \qquad M = (\pi/a, \pi/a), \qquad P_\Gamma = P_M = D_4, \quad P_X \cong D_2. \tag{3.38}$$

The star of X consists of $(\pi/a, 0)$ and $(0, \pi/a)$ modulo L^*. At M, changing a component from π/a to $-\pi/a$ changes the vector by a reciprocal vector, which explains why all of D_4 stabilizes it. A generic interior point of the triangle has trivial stabilizer; a generic point on one of its three edges has an order-two stabilizer.

If the space group is the symmorphic group $L \rtimes D_4$ and the states are single valued, small representations at Γ and M are labeled by ordinary irreducible representations of D_4, while those at X are labeled by irreducible representations of D_2. The two-dimensional irreducible representation of D_4 can enforce a doublet when it occurs in an energy eigenspace. The fact that the star of X has two points does not force a doublet at X: the spinless irreducible representations of D_2 are one dimensional. Little-group tables organize precisely these momentum-dependent labels [6, 1].

Exercise 3.2.4. *List all elements of D_4 that stabilize X in Eq. (3.38). Determine the stabilizers at a generic point of each of the segments ΓX, XM, and $M\Gamma$. For a symmorphic spinless model, explain which of these groups can force a multidimensional irreducible multiplet at that momentum.*

In a nonsymmorphic group, the little co-group alone is insufficient unless we also remember phases from the translation extension. We now derive these phases instead of treating them as extra rules.

Definition 3.2.7. *Choose a representative $s(p) = \{p|\tau_p\} \in \mathcal{G}_k$ for each $p \in P_k$, with $s(1) = 1$. The translation error of this choice is*

$$R(p,q) = \tau_p + p\tau_q - \tau_{pq} \in L, \qquad s(p)s(q) = t_{R(p,q)}s(pq). \tag{3.39}$$

For a small representation σ, put $D_k(p) = \sigma(s(p))$. Its multiplication rule is

$$D_k(p)D_k(q) = \omega_k(p,q)D_k(pq), \qquad \omega_k(p,q) = e^{-ik\cdot R(p,q)}. \tag{3.40}$$

A matrix assignment satisfying such a multiplication law is called a projective representation, and ω_k is its factor system.

To check that Eq. (3.39) is well defined, observe that $s(p)s(q)s(pq)^{-1}$ has trivial point part and belongs to $\mathcal{G}$, hence is a translation in L. The Seitz product gives its vector as displayed. Applying Eq. (3.35) then gives Eq. (3.40). Thus the small representation of the full little group is an ordinary representation; it is the reduced matrices indexed only by P_k that are projective.

Proposition 3.2.7. *The factor system satisfies the exact cocycle identity*

$$\omega_k(p,q)\omega_k(pq,r) = \omega_k(q,r)\omega_k(p,qr), \qquad p,q,r \in P_k. \tag{3.41}$$

If the representatives are replaced by $s'(p) = t_{B_p}s(p)$ with $B_p \in L$, then

$$\omega'_k(p,q) = \frac{\beta(p)\beta(q)}{\beta(pq)}\omega_k(p,q), \qquad \beta(p) = e^{-ik\cdot B_p}. \tag{3.42}$$

The two factor systems therefore describe equivalent projective representations, with $D'_k(p) = \beta(p)D_k(p)$.

Proof. First compare the two ways to multiply $s(p)s(q)s(r)$. Eqs. (3.39) and (3.26) give

$$R(p,q) + R(pq,r) = pR(q,r) + R(p,qr). \tag{3.43}$$

Because $p \in P_k$, we have $p^{-1}k - k \in L^*$, and hence $e^{-ik\cdot pR} = e^{-ik\cdot R}$ for every $R \in L$. Applying $e^{-ik\cdot(\,\cdot\,)}$ to Eq. (3.43) therefore gives Eq. (3.41). This is an exact equality for each fixed choice of representatives, not an equality only up to equivalence.

For the second assertion, substitute $s'(p) = t_{B_p}s(p)$ into its product. Moving t_{B_q} past $s(p)$ gives

$$R'(p,q) = B_p + pB_q + R(p,q) - B_{pq}. \tag{3.44}$$

Apply the same character, again using $p \in P_k$, and obtain Eq. (3.42). Finally, applying σ directly to $s'(p)$ gives $D'_k(p) = \beta(p)D_k(p)$. □

A change of phases changes the factor system but not its projective equivalence class. A symmorphic group admits a homomorphic section, for which $R(p,q) = 0$ and ordinary little-co-group representations suffice. At $\Gamma = 0$, Eq. (3.40) also has $\omega_0 = 1$ even for a nonsymmorphic group. At other momenta, especially on zone boundaries, a nontrivial projective class can force multidimensional representations. Nontrivial phases by themselves do not prove a degeneracy: one must examine the resulting representation algebra. Spinful double-valued spatial representations contribute an additional spin factor, or equivalently require a double group, beyond the translation factor derived here.

Example 3.2.2 (Glide reflection)**.** *Consider a rectangular lattice with primitive vectors ae_x, be_y and a glide $g = \{m_y|(a/2)e_x\}$, where $m_y(x,y) = (x,-y)$. The Seitz rule gives $g^2 = t_{ae_x}$. The glide preserves a Bloch fiber only when $k_y = 0$ or π/b modulo $2\pi/b$. On these glide-invariant lines, a single-valued small representation satisfies*

$$U_g(k)^2 = e^{-ik_x a}I, \qquad \lambda_\pm(k_x) = \pm e^{-ik_x a/2}. \tag{3.45}$$

At $k_x = \pi/a$ its possible eigenvalues are $\pm i$. Each possibility can occur in a one-dimensional representation, so the relation $U_g^2 = -I$ alone does not force a doublet. The two continuously chosen eigenvalue branches exchange when k_x increases by $2\pi/a$, which becomes relevant when following bands around the Brillouin torus. Away from the glide-invariant lines, U_g maps two different fibers and is not an operator that can be diagonalized within one fiber.

Example 3.2.3 (Screw rotation)**.** *Let $g = \{C_n|c/n\}$, where c is a lattice vector along the rotation axis and $C_n c = c$. Then $g^n = t_c$. At a momentum satisfying $C_n k = k$ modulo L^*, its possible single-valued screw eigenvalues are*

$$U_g(k)^n = e^{-ik\cdot c} I, \qquad \lambda_j(k) = \exp\left(-\frac{ik\cdot c}{n} + \frac{2\pi i j}{n}\right), \qquad j = 0, \ldots, n-1. \tag{3.46}$$

At $k \cdot c = \pi$ modulo 2π, these are the nth roots of -1. As for the glide, this changes the allowed symmetry eigenvalues, while degeneracy requires further information about the other symmetries and their relations.

Exercise 3.2.5. *Let two unitary little-group matrices satisfy $AB = -BA$ and commute with $H(k)$. If v is a simultaneous eigenvector of $H(k)$ and A, show that Bv has the same energy but the opposite eigenvalue of A, and hence is linearly independent of v. Explain why this algebra forces a doublet, whereas $A^2 = -I$ alone does not. Such anticommutation can arise when two space-group products differ by a lattice translation with Bloch phase -1.*

Exercise 3.2.6. *For a normalized section $s(1) = 1$, check that $\omega_k(1, p) = \omega_k(p, 1) = 1$. Starting from a projective representation with the factor system in Eq.* (3.40)*, show that*

$$\sigma(t_R s(p)) = e^{-ik\cdot R} D_k(p) \tag{3.47}$$

defines an ordinary small representation of $\mathcal{G}_k$. This verifies that no representation information is lost when translations are replaced by their phases.

Finally, induction also relates a localized orbital to its Bloch symmetry labels. This is a different starting point from induction at a fixed momentum: here we begin with a stabilizer in real space.

Definition 3.2.8. *For a site $q \in \mathbb{R}^d$, its site stabilizer and its translation-orbit stabilizer are*

$$\mathcal{G}_q = \{g \in \mathcal{G} : gq = q\}, \qquad \mathcal{G}_{[q]} = \{g \in \mathcal{G} : gq \in q + L\}. \tag{3.48}$$

Proposition 3.2.8. *The site stabilizer $\mathcal{G}_q$ is finite, $\mathcal{G}_{[q]} = T\mathcal{G}_q$, and $\mathcal{G}_{[q]}/T \cong \mathcal{G}_q$.*

Proof. The restriction of π to $\mathcal{G}_q$ is injective: a translation fixing q must be the identity. Thus $\mathcal{G}_q$ embeds in the finite group P. If $gq = q + R$, then $t_{-R}g$ fixes q, so $g \in T\mathcal{G}_q$; the reverse inclusion is immediate. Finally, $T \cap \mathcal{G}_q = \{1\}$, so the quotient is isomorphic to $\mathcal{G}_q$. □

Thus the exact site stabilizer is already finite and has no nontrivial lattice translations to quotient out. It fixes one physical site. The larger group $\mathcal{G}_{[q]}$ preserves all its lattice translates and becomes finite only after quotienting by T.

Definition 3.2.9. *Let σ be a finite dimensional unitary representation of $\mathcal{G}_q$ on V. The induced representation $\mathrm{Ind}_{\mathcal{G}_q}^{\mathcal{G}} \sigma$, with the square-summable Hilbert-space completion appropriate to the infinite orbit, is called a band representation.*

The space V describes the orbital degrees of freedom at q, such as an s orbital or a set of p orbitals. Induction places copies at every symmetry-related site and specifies how the full space group permutes and transforms them. Although the induced Hilbert space is infinite dimensional, the number of sites in this orbit per primitive cell is finite:

$$m = [\mathcal{G} : \mathcal{G}_{[q]}] = [P : \pi(\mathcal{G}_q)], \qquad \dim \mathcal{H}_k = m \dim V. \tag{3.49}$$

The Bloch transform therefore produces finite dimensional fibers. At each momentum their little-group representations decompose into irreducible small representations, supplying possible symmetry labels for bands built from these orbitals. Symmetry alone does not specify the hopping amplitudes or the band energies. Spinful orbitals require the corresponding double site and space groups when spatial rotations act double valued.

Exercise 3.2.7. *In a one-dimensional symmorphic chain with translations t_a and inversion $I : x \mapsto -x$, place an orbital at the inversion center $q = 0$ with parity $\eta = \pm 1$. Show that its induced Bloch state transforms from k to $-k$, and determine its inversion eigenvalues at $k = 0$ and $k = \pi/a$. Repeat for an orbital centered at $q = a/2$, whose site inversion is $t_a I$, using a Bloch basis with cell phases e^{ikna}. Explain how the site position changes the zone-boundary inversion eigenvalue although the local parity η is unchanged.*

3.3 Magnetic groups and corepresentations

Ordinary space groups describe spatial symmetries. A magnetic structure may also be invariant under a spatial operation followed by time reversal, even when neither operation is a symmetry separately. The new mathematical feature is that quantum time reversal is antiunitary: it conjugates scalar coefficients. We first introduce the corresponding group structure, then explain how antiunitarity changes representation theory, and finally apply it to Bloch states and degeneracies [17, 7, 9, 8].

Definition 3.3.1. *Let $1'$ be a formal operation satisfying $(1')^2 = 1$ and commuting with every spatial operation. A magnetic space group is a subgroup $M \subset E(d) \times \{1, 1'\}$ whose spatial projection is a space group. A magnetic point group is defined similarly using a finite subgroup of $O(d)$. We write an element as $\{p|\tau\}$ or $\{p|\tau\}'$, and define $\eta(m) = 1$ for an unprimed element and $\eta(m) = -1$ for a primed element. The unitary subgroup is $M_0 = \ker \eta$.*

The prime records time reversal and does not change the affine multiplication of spatial operations. If M contains a primed element a, then M_0 is normal of index two and

$$M = M_0 \sqcup aM_0, \qquad a^2 \in M_0. \tag{3.50}$$

The classical relation $(1')^2 = 1$ must be distinguished from the square of its quantum representative. For a spin-$\frac{1}{2}$ particle, the latter is $-I$.

Definition 3.3.2. *Magnetic space groups are divided into four types. Type I is an ordinary space group G with no primed operations. Type II is the grey group $G \times \{1, 1'\}$. For either of the remaining types, let $H \subset G$ be a subgroup of index two and set*

$$M = H \sqcup 1'(G \setminus H). \tag{3.51}$$

It is type III when H and G have the same translation subgroup, and type IV when the translation subgroup of H has index two in that of G. In type IV, the antiunitary coset contains a translation followed by time reversal. Magnetic point groups have types I, II, and III; type IV requires translations.

Type IV is useful for describing antiferromagnets: a translation can exchange oppositely oriented moments, while time reversal restores them. In three dimensions a magnetic moment is an axial vector. Under a spatial point operation p it transforms as $m \mapsto \det(p)pm$, and time reversal adds a minus sign.

Example 3.3.1. *Suppose a square geometry has point group C_{4v} and carries a ferromagnetic moment along $\hat{z}$. Rotations around $\hat{z}$ preserve the moment. A vertical mirror reverses the axial vector $\hat{z}$, but the mirror followed by time reversal preserves it. Assuming no additional order breaks these operations, the magnetic point group is $C_4 \sqcup 1'\sigma_v C_4$.*

Definition 3.3.3. *A map $A : V \to V$ is antilinear if $A(cv + w) = c^* A(v) + A(w)$. It is antiunitary if it is bijective and*

$$\langle Av, Aw \rangle = \langle w, v \rangle. \tag{3.52}$$

Here the inner product is linear in its second argument, as in Definition 1.1.6.

Exercise 3.3.1. *Show that the product of two antiunitary maps is unitary and that the product of a unitary map and an antiunitary map is antiunitary. In an orthonormal basis, let K denote complex conjugation. Show that every antiunitary map has the form $A = UK$ with U unitary, and that $A^2 = UU^*$, where U^* denotes entrywise complex conjugation, not the adjoint.*

Definition 3.3.4. *A strict unitary corepresentation of M assigns a unitary operator $D(m)$ to each $m \in M_0$ and an antiunitary operator to each $m \notin M_0$, with $D(m)D(n) = D(mn)$. It is irreducible if no nonzero proper complex subspace is invariant under every $D(m)$. A projective corepresentation instead satisfies*

$$D(m)D(n) = \omega(m, n)D(mn), \qquad \omega(m, n) \in U(1). \tag{3.53}$$

Associativity imposes a modified condition on ω. Define $z^{\eta(m)} = z$ when m is unitary and $z^{\eta(m)} = z^*$ when m is antiunitary, for $z \in U(1)$. Then

$$\omega(m, n)\omega(mn, r) = \omega(n, r)^{\eta(m)}\omega(m, nr). \tag{3.54}$$

The conjugation on the right is essential: $D(m)$ must pass through the scalar $\omega(n, r)$. Spinful electrons can be described by projective corepresentations of classical magnetic groups, or by strict corepresentations of suitable double magnetic groups. In the latter description, the central element representing a 2π rotation acts as $-I$.

Exercise 3.3.2. *Derive Eq.* (3.54) *by multiplying three symmetry operators in the two possible orders. If* $D'(m) = b(m)D(m)$ *with* $b(m) \in U(1)$*, show that the new factor system is* $\omega'(m,n) = b(m)b(n)^{\eta(m)}b(mn)^{-1}\omega(m,n)$*. Explain why rephasing an antiunitary operator does not change its square.*

Proposition 3.3.1. *Let* S_y *be the spin angular momentum operator with its usual physical units. In the standard* S_z *basis, time reversal may be represented as*

$$\Theta_s = e^{-i\pi S_y/\hbar}K, \qquad \Theta_s^2 = (-1)^{2s}I. \tag{3.55}$$

Proof. In this basis $S_y^* = -S_y$, so $e^{-i\pi S_y/\hbar}$ is real. Therefore $\Theta_s^2 = e^{-i2\pi S_y/\hbar}$. The eigenvalues of $S_y/\hbar$ are $-s, -s+1, \ldots, s$, and every corresponding exponential is $(-1)^{2s}$. □

For a spinless state, $\Theta = K$ and $\Theta^2 = I$. For one spin-$\frac{1}{2}$ doublet, an equivalent phase convention is $\Theta = i\sigma_y K$, with $\Theta^2 = -I$. The phase choice does not affect the square. We now explain how this distinction fits into the general classification.

Theorem 3.3.1 (Wigner corepresentation classification)**.** *Let* $M = M_0 \sqcup aM_0$ *be finite, and work with strict corepresentations. Let* ρ *be an irreducible unitary representation of* M_0 *on* V*, of dimension* d*, and set* $\rho^a(h) = \rho(a^{-1}ha)^*$*. There are three possibilities.*
1. If $\rho^a \not\simeq \rho$*, one obtains an irreducible corepresentation of dimension* $2d$*, whose restriction to* M_0 *is* $\rho \oplus \rho^a$*; this is case (c).*
2. If $\rho^a \simeq \rho$*, choose an antiunitary* $B : V \to V$ *satisfying* $B\rho(h)B^{-1} = \rho(aha^{-1})$*. There is a sign* $\varepsilon = \pm 1$ *such that*

$$B^2 = \varepsilon\rho(a^2). \tag{3.56}$$

For $\varepsilon = +1$*,* ρ *extends to an irreducible corepresentation of dimension* d*; this is case (a). For* $\varepsilon = -1$*, the irreducible corepresentation has dimension* $2d$ *and restricts to* $\rho \oplus \rho$*; this is case (b). Every irreducible corepresentation is obtained in this way, with the pair* ρ, ρ^a *counted only once in case (c).*

Proof. The idea is to restrict to M_0 and follow what the antiunitary element does to the irreducible summands. It either exchanges two different types or preserves one type. In the second situation, Schur's lemma reduces the obstruction to a single sign.

If W carries ρ, then $D(a)W$ carries ρ^a: in the basis $D(a)v_i$, the matrix of h is $\rho(a^{-1}ha)^*$. Applying this twice returns to the original type because $a^2 \in M_0$. Thus an irreducible corepresentation has either one isotypic component or two exchanged components. In the latter case, $W \oplus D(a)W$ is already invariant under M, so irreducibility forces it to be the whole space. This gives case (c). Explicitly, on $V \oplus \overline{V}$ one can take

$$D(h) = \begin{pmatrix} \rho(h) & 0 \\ 0 & \rho^a(h) \end{pmatrix}, \qquad D(a)\begin{pmatrix} v \\ w \end{pmatrix} = \begin{pmatrix} \rho(a^2)\overline{w} \\ \overline{v} \end{pmatrix}. \tag{3.57}$$

These operators satisfy the conjugation relation and $D(a)^2 = D(a^2)$. A subspace invariant under M_0 is a sum of some of the two inequivalent irreducible summands, and $D(a)$ exchanges them. Hence the corepresentation is irreducible.

Suppose now that $\rho^a \simeq \rho$. An intertwiner gives an invertible antilinear map implementing $h \mapsto aha^{-1}$; its polar decomposition, or Schur's lemma applied to its adjoint times itself, allows it to be normalized to an antiunitary B. Both B^2 and $\rho(a^2)$ implement conjugation by a^2, so Schur's lemma gives $B^2 = c\rho(a^2)$ with $|c| = 1$. Moreover B commutes with $\rho(a^2)$. Comparing BB^2 with B^2B therefore gives $c^* = c$, hence $c = \varepsilon = \pm 1$. Any other normalized choice differs by a phase, so the sign is independent of that choice. If $\varepsilon = +1$, take $D(a) = B$ on V. If $\varepsilon = -1$, no extension on a single copy can satisfy $D(a)^2 = \rho(a^2)$; on two copies set

$$D(h) = \begin{pmatrix} \rho(h) & 0 \\ 0 & \rho(h) \end{pmatrix}, \qquad D(a)(v,w) = (Bw, -Bv). \tag{3.58}$$

Its square is $\mathrm{diag}(\rho(a^2), \rho(a^2))$, as required.

For completeness, the multiplicity space explains why no further dimensions are needed. On an isotypic space $V \otimes \mathbb{C}^r$, write the antiunitary action as $B \otimes J$, where J is antiunitary on the multiplicity space. The square condition is $J^2 = \varepsilon I$. If $J^2 = I$, the real fixed vectors of J span the complex space, so there is an invariant complex line. If $J^2 = -I$, each nonzero vector u and Ju span an invariant complex plane and are linearly independent, since $Ju = cu$ would imply $-u = |c|^2 u$. Orthogonal complements of these invariant spaces are invariant because J is antiunitary. Consequently an irreducible multiplicity space has dimension one in case (a) and two in case (b). In case (b) the plane has no invariant line, which also proves the irreducibility of Eq. (3.58). □

The sign in Eq. (3.56) compares B^2 with $\rho(a^2)$; it is not a claim that the physical operator always squares to $+I$ or $-I$. This distinction matters for magnetic translations and nonsymmorphic symmetries. A useful computational form of the classification is Herring's criterion:

$$\frac{1}{|M_0|}\sum_{b\in aM_0}\chi_\rho(b^2)=\begin{cases}1, & \text{case (a)},\\ -1, & \text{case (b)},\\ 0, & \text{case (c)}.\end{cases} \tag{3.59}$$

We quote this character identity without proof [17, 7, 8]. It applies directly to strict representations of the finite group under consideration. For a projective description one must include its factor system, or first pass to a suitable extension; substituting projective characters into Eq. (3.59) without this modification is not valid.

Exercise 3.3.3. *For a grey group with central $a = 1'$ and $a^2 = e$, apply Eq.* (3.59) *to a one-dimensional real character and to a nonreal character of C_3. Identify cases (a) and (c). Verify the conjugation and square relations in Eqs.* (3.57) *and* (3.58).

We next return to a magnetic space group. Its unitary translation subgroup is $T_0 = \{t_R : R \in L\} \subset M_0$, where L is the magnetic lattice. In an antiferromagnet this can be a sublattice of the nonmagnetic crystal's translation lattice, giving a larger real-space unit cell and a smaller Brillouin zone.

Proposition 3.3.2. *Let $\psi \in \mathcal{H}_k$ denote a Bloch fiber state, with translation action $t_R\psi = e^{-ik\cdot R}\psi$. If m has point part p_m and unitary sign $\eta(m)$, then*

$$D(m) : \mathcal{H}_k \longrightarrow \mathcal{H}_{\eta(m)p_m k}. \tag{3.60}$$

Proof. Use spatial conjugation first, and only then move the Bloch scalar through $D(m)$. Since $m^{-1}t_R m = t_{p_m^{-1}R}$,

$$t_R D(m)\psi = D(m)t_{p_m^{-1}R}\psi = D(m)\big(e^{-ik\cdot p_m^{-1}R}\psi\big) = e^{-i\eta(m)(p_m k)\cdot R}D(m)\psi. \tag{3.61}$$

The last equality uses antilinearity when $\eta(m) = -1$, and orthogonality of p_m. □

Thus ordinary time reversal sends k to $-k$, while a primed spatial operation sends k to $-p_m k$. This determines whether an antiunitary symmetry relates different momenta or acts within a single Bloch fiber.

Definition 3.3.5. *The magnetic little group and magnetic little co-group are*

$$M_k = \{m \in M : \eta(m)p_m k = k \bmod L^*\}, \qquad \overline{M}_k = M_k/T_0. \tag{3.62}$$

A small corepresentation is a corepresentation of M_k in which $D_k(t_R) = e^{-ik\cdot R}I$. The elements of $\overline{M}_k$ are the pairs $(\eta(m), p_m)$ arising from $m \in M_k$, not arbitrary orthogonal operations satisfying the momentum condition.

Proposition 3.3.3. *The sequence $1 \to T_0 \to M_k \to \overline{M}_k \to 1$ is exact. Choose representatives $s(\alpha) = \{p_\alpha|\tau_\alpha\}^{\epsilon_\alpha}$ of the finite quotient and write $s(\alpha)s(\beta) = t_{R(\alpha,\beta)}s(\alpha\beta)$. The operators on a Bloch fiber obey*

$$D_k(s(\alpha))D_k(s(\beta)) = \omega_k(\alpha,\beta)D_k(s(\alpha\beta)), \qquad \omega_k(\alpha,\beta) = \omega_{\text{spin}}(\alpha,\beta)e^{-ik\cdot R(\alpha,\beta)}. \tag{3.63}$$

Here $\omega_{\text{spin}} = 1$ in the strict spinless setting, and the full factor system satisfies Eq. (3.54).

Proof. The kernel of $m \mapsto (\eta(m), p_m)$ consists exactly of the unitary translations, and the image defines $\overline{M}_k$. The product of two representatives differs from the representative of their product by a lattice translation. Acting with that translation on $\mathcal{H}_k$ gives the phase in Eq. (3.63); a projective spin action contributes the additional factor. Associativity gives the twisted cocycle identity as before. □

This reduces the symmetry labels at a fixed momentum to a finite problem. A continuum Bloch fiber need not be finite dimensional, but an isolated finite-dimensional eigenspace, or a finite collection of bands, carries these labels. Both spin and fractional translations can affect the squares of antiunitary operators.

Theorem 3.3.2 (Antiunitary degeneracy and Kramers' theorem)**.** *Let A be an antiunitary symmetry preserving a Hilbert space on which a self-adjoint Hamiltonian H acts, and suppose $AH = HA$. If $A^2 = \lambda I$ with $|\lambda| = 1$, then necessarily $\lambda = \pm 1$. If $\lambda = -1$, every energy eigenstate ψ has an orthogonal partner $A\psi$ at the same energy, and every finite-dimensional energy eigenspace has even dimension.*

Proof. There are two separate points: commutation preserves energy, while the negative square makes the partner independent. First, comparing AA^2 with A^2A gives $\lambda^* A = \lambda A$, so λ is real and equals ± 1. If $H\psi = E\psi$, then $H(A\psi) = A(E\psi) = EA\psi$ because E is real. By Eq. (3.52),

$$\langle A\psi, A^2\psi\rangle = \langle A\psi, \psi\rangle, \qquad \langle A\psi, A^2\psi\rangle = -\langle A\psi, \psi\rangle. \tag{3.64}$$

Hence $\langle A\psi, \psi\rangle = 0$. The plane spanned by $\psi, A\psi$ is invariant under A. Its orthogonal complement within the eigenspace is also invariant by antiunitarity, so repeating the argument decomposes a finite-dimensional eigenspace into orthogonal pairs. □

A scalar antiunitary square on an invariant fiber cannot be an arbitrary phase. When a space-group relation gives $A^2 = t_R$, the phase $e^{-ik\cdot R}$ must reduce to $+1$ or -1 at momenta where A preserves that fiber. If A^2 is not scalar, its action must be analyzed on the relevant states instead. Also, $A^2 = I$ does not exclude degeneracy; it only means that this negative-square argument does not force it. Case (c) of Theorem 3.3.1 can still pair distinct unitary representation types.

Definition 3.3.6. *A time-reversal-invariant momentum, or TRIM, satisfies $k = -k \bmod L^*$. If $b_1, \dots, b_d$ is a reciprocal basis, the distinct TRIMs are*

$$k_\nu = \frac{1}{2}\sum_{i=1}^{d} \nu_i b_i \bmod L^*, \qquad \nu_i \in \{0, 1\}. \tag{3.65}$$

Example 3.3.2. *In a basis with spinless time reversal K, the condition is $H(k)^* = H(-k)$. At a TRIM one can choose a basis of the fiber in which $H(k)$ is real symmetric; its eigenvalues need not coincide. For spinful time reversal $\Theta = i\sigma_y K$, the condition is $(i\sigma_y)H(k)^*(-i\sigma_y) = H(-k)$, with identity factors on other degrees of freedom. Theorem 3.3.2 forces pairs at the TRIMs. Away from a TRIM, time reversal relates the spectra at k and $-k$ but does not by itself force degeneracy within either fiber. In one dimension with $L = a\mathbb{Z}$, the TRIMs are 0 and π/a modulo $2\pi/a$.*

Example 3.3.3. *Let P be inversion with $P^2 = I$, and suppose P commutes with spinful time reversal. Then $P\Theta$ fixes every momentum and $(P\Theta)^2 = -I$. Thus a system preserving both symmetries has double degeneracy at every k. The combined symmetry alone is sufficient if these algebraic properties hold. In the spinless case $(P\Theta)^2 = I$, so this argument does not force a pair; a real basis can be chosen locally, but a globally smooth real basis need not exist.*

Example 3.3.4 (Antiferromagnetic translation)**.** *Consider an alternating magnetic order with neighboring sites separated by a. The unitary translation is t_{2a} and the combined antiunitary symmetry is $A = \Theta t_a$, with Θ commuting with translations. The magnetic reciprocal lattice is $(\pi/a)\mathbb{Z}$, and A fixes only $k = 0$ and $k = \pi/(2a)$ modulo that lattice. If $\Theta^2 = sI$, where $s = \pm 1$, then*

$$D_k(A)^2 = se^{-2iak} I. \tag{3.66}$$

For $s = +1$, the square is negative at the magnetic zone boundary and positive at the center. For $s = -1$, it is negative at the center and positive at the boundary. The degeneracy forced by A alone therefore occurs at different momenta in the two cases. The $s = +1$ case is an algebraic model for appropriate integer-spin or spinless degrees of freedom in a time-reversal-breaking background; it should not be interpreted as an ordinary real scalar potential acquiring an alternating spin moment.

Example 3.3.5. *Suppose a glide or screw followed by time reversal gives an antiunitary symmetry A with $A^2 = t_R$ in the spinless description. At a momentum fixed by its action $k \mapsto -p_A k$, a phase $e^{-ik\cdot R} = -1$ enforces a pair by Theorem 3.3.2. For spinful particles, compute the full square including the spin rotation and time reversal; their signs can multiply or cancel. The classical spatial square alone is insufficient.*

Exercise 3.3.4. *Prove Eq.* (3.65) *and count the TRIMs in d dimensions. For a rectangular two-dimensional lattice, let $A = \Theta M_x$, where M_x sends (k_x, k_y) to $(-k_x, k_y)$. Find the lines fixed by A, and explain why locating these lines does not determine degeneracy until A^2 is known.*

Exercise 3.3.5. *Verify Eq.* (3.66) *at both invariant momenta. Explain why its generic complex value at another momentum is consistent with Theorem 3.3.2: at that momentum A maps between different fibers. For an antiunitary operator that does preserve a finite-dimensional space, use $A = UK$ and determinants to give a second proof that $A^2 = -I$ requires even dimension.*

Exercise 3.3.6. *Assume a finite-dimensional energy eigenspace is irreducible under a magnetic corepresentation. Show directly, by considering spectral subspaces, that any Hermitian operator commuting with all unitary and antiunitary symmetries is scalar on it. Explain why this conclusion does not require the complex-linear form of Schur's lemma to apply to arbitrary operators commuting with antiunitary symmetries.*

3.4 The $\vec{k} \cdot \vec{p}$ expansion

We now combine character theory with the symmetry of Bloch states. Choose a momentum k_0 and write $k = k_0 + q$. A local effective Hamiltonian describes a chosen finite-dimensional space of bands near k_0 as a power series in q. Group theory determines its allowed terms; a microscopic calculation or experiment determines their coefficients. We first explain the microscopic origin of the expansion, then turn it into a problem of finding invariant tensors.

Definition 3.4.1. *Let E be a finite-dimensional subspace of states at k_0 invariant under the magnetic little group. Assume the retained bands are separated from omitted bands near k_0, so that a smooth local effective description is possible. A $\vec{k}\cdot\vec{p}$ Hamiltonian is a Hermitian matrix-valued expansion*

$$H_{\text{eff}}(q)=\sum_{n\geq 0}H^{(n)}(q),\qquad H^{(n)}(tq)=t^nH^{(n)}(q),\qquad H^{(n)}(q)^\dagger=H^{(n)}(q) \tag{3.67}$$

for real q,t. A truncated expansion describes the Hamiltonian to its stated order; convergence of an infinite series is not assumed unless analyticity is established.

Examples of E include a nondegenerate band, an orbital doublet, and a Kramers pair. The choice must include every symmetry partner and every nearby band whose energy separation is too small to justify eliminating it. We write $P_{k_0}=\mathcal{G}_{k_0}/T$ for the finite unitary little co-group. Its matrices $D(g)$ on E can be projective, with phases arising from spin and fractional translations. The action on the small momentum is $q\mapsto p_gq$; for an antiunitary element a it is $q\mapsto -p_aq$.

Proposition 3.4.1. *Suppose a local choice of basis represents symmetry by constant matrices $D(g)$, or by sewing matrices whose scalar phases cancel in conjugation. Then the symmetry conditions are*

$$D(g)H_{\text{eff}}(q)D(g)^{-1}=H_{\text{eff}}(p_gq) \tag{3.68}$$

for unitary elements, and

$$U_aH_{\text{eff}}(q)^*U_a^{-1}=H_{\text{eff}}(-p_aq),\qquad D(a)=U_aK, \tag{3.69}$$

for antiunitary elements fixing k_0 modulo L^*.

Proof. A symmetry intertwines the Hamiltonian before and after its action on momentum. For a unitary symmetry, a state v at q is sent to $D(g)v$ at p_gq, so $H_{\text{eff}}(p_gq)D(g)v=D(g)H_{\text{eff}}(q)v$. This gives Eq. (3.68). For an antiunitary symmetry, replace $D(g)v$ by $U_a\overline{v}$ and p_gq by $-p_aq$, using Proposition 3.3.2. Moving complex conjugation through the Hamiltonian gives Eq. (3.69). □

These equations are the working rules of the method. Their basis assumption must be checked when constructing a model from Bloch eigenvectors. In a general local frame the sewing matrix $B_g(q)$ depends on q, and the exact unitary condition is

$$B_g(q)H_{\text{eff}}(q)B_g(q)^{-1}=H_{\text{eff}}(p_gq). \tag{3.70}$$

An overall scalar phase of $B_g(q)$ cancels, including many fractional-translation phases. A non-scalar dependence does not cancel: one must choose a compatible symmetry-adapted frame or expand $B_g(q)$ consistently along with $H_{\text{eff}}(q)$. Merely replacing an arbitrary sewing matrix by its value at $q=0$ is not an exact argument at all orders. All constant-matrix constructions below assume Eqs. (3.68) and (3.69) hold in the chosen effective basis.

Proposition 3.4.2. *For the spinless continuum Hamiltonian $H=p^2/(2m)+V(x)$, with $p=-i\hbar\nabla$ and $V(x+R)=V(x)$, the cell-periodic Bloch Hamiltonian is $H(k)=e^{-ik\cdot x}He^{ik\cdot x}$. It satisfies*

$$H(k)=\frac{(p+\hbar k)^2}{2m}+V(x),\qquad H(k_0+q)=H(k_0)+\frac{\hbar}{m}q\cdot(p+\hbar k_0)+\frac{\hbar^2|q|^2}{2m}I. \tag{3.71}$$

Proof. Apply the product rule to $p(e^{ik\cdot x}u)$ to obtain $e^{-ik\cdot x}pe^{ik\cdot x}=p+\hbar k$. The first identity follows, and expanding the square at k_0+q gives the second because q is a constant vector. □

At $k_0=0$ the linear term is $(\hbar/m)q\cdot p$, which explains the name. For orthonormal cell-periodic states $u_1,\dots,u_N$ spanning E, its matrix elements are

$$\big(H^{(1)}_{\text{eff}}(q)\big)_{mn}=\frac{\hbar}{m}\sum_i q_i\langle u_m,(p_i+\hbar k_{0,i})u_n\rangle. \tag{3.72}$$

Remote bands affect higher orders. To see where their contribution enters, let $\mathcal{P}$ project onto E and let $\mathcal{Q}=I-\mathcal{P}$. If $z-H_{\mathcal{Q}\mathcal{Q}}(q)$ is invertible, eliminating the $\mathcal{Q}$ component of an eigenvector gives the exact energy-dependent equation

$$H_{\text{eff}}(z,q)=H_{\mathcal{P}\mathcal{P}}(q)+H_{\mathcal{P}\mathcal{Q}}(q)\big(z-H_{\mathcal{Q}\mathcal{Q}}(q)\big)^{-1}H_{\mathcal{Q}\mathcal{P}}(q). \tag{3.73}$$

This is a Schur complement, not yet an energy-independent second-order Hamiltonian. Expanding it about the active energies, with a nonzero gap to remote states, yields perturbative corrections. This distinction makes clear which statement is exact and where an approximation enters [14, 12, 4, 19].

Exercise 3.4.1. *Derive Eq.* (3.73) *from the two block equations for an eigenvector. If* $H(0)|_E = E_0 I$ *and* $V^{(1)}(q)$ *is the linear perturbation, show that the remote-band contribution at second order is* $\mathcal{P}V^{(1)}\mathcal{Q}(E_0 - H_{\mathcal{Q}\mathcal{Q}}(0))^{-1}\mathcal{Q}V^{(1)}\mathcal{P}$. *Include the direct quadratic term from Eq.* (3.71) *and identify the assumption that fails if a remote level approaches* E_0.

We now formulate the symmetry problem using representations. Let Q be the complexification of the real vector space of momentum displacements. Homogeneous polynomials of degree n form $\mathrm{Sym}^n(Q^\vee)$, with action $(gf)(q) = f(p_g^{-1}q)$. Matrices on E form $\mathrm{End}(E) \cong E \otimes E^\vee$, with action $M \mapsto D(g)MD(g)^{-1}$. Thus matrix-valued polynomials carry the action

$$(gF)(q) = D(g)F(p_g^{-1}q)D(g)^{-1}. \tag{3.74}$$

Notice the inverse in the action on functions. It is the same pullback convention used in Example 1.2.9.

Theorem 3.4.1. *The space of complex degree-n polynomials satisfying Eq.* (3.68) *is* $[\mathrm{Sym}^n(Q^\vee) \otimes \mathrm{End}(E)]^{P_{k_0}}$. *Its dimension is*

$$N_n = \frac{1}{|P_{k_0}|} \sum_{g \in P_{k_0}} \chi_{\mathrm{Sym}^n(Q^\vee)}(g)|\chi_E(g)|^2. \tag{3.75}$$

The real dimension of its Hermitian subspace is also N_n. *The formula remains valid when* D *is a unitary projective representation with a common scalar factor system on* E.

Proof. The proof has two steps: translate the covariance equation into invariance, then count the invariant subspace by its character. By Eq. (3.74), $gF = F$ is equivalent to $D(g)F(q)D(g)^{-1} = F(p_g q)$. Therefore the required subspace is precisely the invariant tensor space. The averaging operator $|P_{k_0}|^{-1}\sum_g g$ is its projection, so its trace gives the dimension. Characters multiply under tensor products, and the conjugation action has character $\chi_E(g)\chi_E(g^{-1}) = |\chi_E(g)|^2$ for an ordinary unitary representation. More generally, conjugation by a unitary matrix has trace $\mathrm{tr}\, D(g)\, \mathrm{tr}\, D(g)^{-1} = |\,\mathrm{tr}\, D(g)|^2$. Scalar projective factors cancel in conjugation, so it is an ordinary representation of P_{k_0} even when D itself is projective. This proves Eq. (3.75).

Finally, taking the adjoint at real q preserves the invariant space. Any invariant polynomial is uniquely $F = A + iB$, with $A = (F + F^\dagger)/2$ and $B = (F - F^\dagger)/(2i)$ Hermitian invariant polynomials. Thus the complex space is the complexification of its Hermitian real subspace, which proves the dimension statement. □

The theorem is the character-theoretic answer to how many independent real coefficients a Hermitian local model can contain before antiunitary constraints are imposed. It does not assert that the coefficients are nonzero. Symmetry gives permission, not a microscopic value. Also, pairing multidimensional representations requires an invariant contraction of their components; the product of an arbitrary polynomial and an arbitrary matrix is not generally invariant.

Proposition 3.4.3. *For any matrix-valued homogeneous polynomial* F, *define*

$$(\Pi F)(q) = \frac{1}{|P_{k_0}|} \sum_{g \in P_{k_0}} D(g)F(p_g^{-1}q)D(g)^{-1}. \tag{3.76}$$

Then $\Pi^2 = \Pi$, *and its image is the space counted in Eq.* (3.75). *If* F *is Hermitian for real* q, *so is* ΠF.

Proof. The important point is that Eq. (3.74) is a group action. Hence $h\Pi F = |P_{k_0}|^{-1}\sum_g (hg)F = \Pi F$, by reindexing the sum. Conversely, averaging an invariant polynomial leaves it unchanged, proving both the image statement and $\Pi^2 = \Pi$. Unitary conjugation and a real average preserve Hermiticity. □

This is the same averaging trick used to prove complete reducibility. It now constructs Hamiltonian terms: apply Π to a basis of monomials times Hermitian matrices, discard zero results, and retain a linearly independent set. If antiunitary symmetries are present, impose Eq. (3.69) on that set as real linear conditions. One representative of the antiunitary coset, together with all unitary constraints, suffices.

Exercise 3.4.2. *Let* p_g *have eigenvalues* $\lambda_1, \ldots, \lambda_d$ *on* Q. *Show that* $\sum_{n\geq 0} \chi_{\mathrm{Sym}^n(Q^\vee)}(g)t^n = \prod_i (1 - t\lambda_i^{-1})^{-1}$. *Deduce* $\chi_{\mathrm{Sym}^2(Q^\vee)}(g) = [\chi_{Q^\vee}(g)^2 + \chi_{Q^\vee}(g^2)]/2$, *and use it to count quadratic terms without constructing every polynomial.*

At $q = 0$, Eq. (3.68) reduces to commutation with the little-group representation. If E is irreducible, Schur's lemma gives $H^{(0)} = E_0 I$. More generally, if $E = \bigoplus_\alpha (V_\alpha \otimes \mathbb{C}^{m_\alpha})$, then

$$H^{(0)} = \bigoplus_\alpha (I_{V_\alpha} \otimes A_\alpha), \qquad A_\alpha^\dagger = A_\alpha. \tag{3.77}$$

Equivalent copies can mix through A_α; symmetry does not generally assign an independent scalar to each arbitrarily chosen irreducible copy. After diagonalizing A_α, each of its eigenvalues occurs at least $\dim V_\alpha$ times. Antiunitary symmetry may impose further relations or doubling, as in Theorem 3.3.1.

Proposition 3.4.4 (Cyclic selection rule)**.** *Suppose a rotation C_n acts on two basis states with eigenvalues $e^{2\pi i j_\alpha/n}$ and $e^{2\pi i j_\beta/n}$. Set $q_\pm = q_x \pm i q_y$ and $M_{\alpha\beta} = |\alpha\rangle\langle\beta|$. A monomial $q_+^r q_-^s M_{\alpha\beta}$ satisfies the unitary rotation constraint if and only if*

$$r - s \equiv j_\alpha - j_\beta \pmod{n}. \tag{3.78}$$

For a spinful representation the j's may be half-integral, but their difference in the same spin sector is integral.

Proof. Conjugation sends $M_{\alpha\beta}$ to $e^{2\pi i(j_\alpha - j_\beta)/n} M_{\alpha\beta}$, while evaluation at rotated momentum gives $(pq)_+^r (pq)_-^s = e^{2\pi i(r-s)/n} q_+^r q_-^s$. Eq. (3.68) equates these phases, giving Eq. (3.78). □

This is conservation of angular momentum modulo the order of the discrete rotation. There is no contradiction with the dual action on polynomial functions: the preceding proof uses evaluation at pq, whereas Eq. (3.74) uses the pullback at $p^{-1}q$. A physical off-diagonal coupling must also include its Hermitian conjugate.

Example 3.4.1. *Consider a spinless doublet at Γ in two dimensions, transforming as the vector representation E of D_4, with basis (p_x, p_y). Use the character table already obtained in Table 1.1, and name its rows $A_1 = \chi_1$, $A_2 = \chi_2$, $B_1 = \chi_3$, $B_2 = \chi_4$, and $E = \chi_5$. The momentum displacement also transforms as E. Take generators*

$$D(C_4) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad D(m_x) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{3.79}$$

where m_x is reflection across the x axis. Character multiplication gives $\mathrm{End}(E) \cong A_1 \oplus A_2 \oplus B_1 \oplus B_2$, and conjugation by these generators identifies a Hermitian basis:

$$I \sim A_1, \qquad \sigma_y \sim A_2, \qquad \sigma_z \sim B_1, \qquad \sigma_x \sim B_2. \tag{3.80}$$

There is no E in this matrix space, so Eq. (3.75) gives no linear term. The quadratic polynomials form $\mathrm{Sym}^2(E) = A_1 \oplus B_1 \oplus B_2$, with bases $q_x^2 + q_y^2$, $q_x^2 - q_y^2$, and $2q_xq_y$. Consequently the most general model through quadratic order is

$$H_{\mathrm{eff}}(q) = E_0 I + a(q_x^2 + q_y^2) I + b(q_x^2 - q_y^2)\sigma_z + 2c q_x q_y \sigma_x + O(|q|^4), \tag{3.81}$$

with real coefficients. All odd orders vanish because C_2 sends q to $-q$ and acts as $-I$ on E. The eigenvalues of the displayed quadratic part are

$$E_\pm(q) = E_0 + a|q|^2 \pm \sqrt{b^2(q_x^2 - q_y^2)^2 + 4c^2 q_x^2 q_y^2}. \tag{3.82}$$

Thus the doublet is degenerate at Γ, and its generic leading splitting is quadratic. For nonzero b, c, the quadratic model has an isolated touching there; if a coefficient vanishes accidentally, further degeneracies or higher-order splitting must be examined. At fourth order, $q_xq_y(q_x^2 - q_y^2)$ transforms as A_2, so its product with σ_y is allowed by D_4. If spinless time reversal K is also imposed, this term is forbidden: its polynomial is even under $q \mapsto -q$, while $\sigma_y^* = -\sigma_y$.

Exercise 3.4.3. *Use Table 1.1 to verify the decompositions in Example 3.4.1 and obtain $N_0 = 1$, $N_1 = 0$, and $N_2 = 3$. Verify Eq. (3.80) using both generators. Apply Eq. (3.76) to $q_x^2\sigma_z$ and find the resulting quadratic invariant.*

Example 3.4.2. *For a single Kramers pair near a TRIM, write $H(q) = \epsilon(q)I + \sum_i d_i(q)\sigma_i$ with real functions. Since $\Theta = i\sigma_y K$ sends σ_i to $-\sigma_i$, Eq. (3.69) gives*

$$\epsilon(q) = \epsilon(-q), \qquad d_i(q) = -d_i(-q). \tag{3.83}$$

Thus time reversal allows linear spin splitting away from the TRIM. If inversion is also present and acts on this pair as $\pm I$, it gives $H(q) = H(-q)$, forcing every d_i to vanish. This is the local form of the $P\Theta$ degeneracy discussed above. The statement assumes an isolated active pair; a model with additional orbitals can have nontrivial inversion matrices and must be treated in that larger space.

Example 3.4.3 (Rashba coupling)**.** *For a two-dimensional spinful system with point group C_{4v} and no inversion, consider*

$$H_R(q) = \alpha(q_y\sigma_x - q_x\sigma_y), \qquad \alpha \in \mathbb{R}. \tag{3.84}$$

Under a proper rotation about z, both in-plane momentum and spin rotate, so this combination is unchanged. Under the vertical mirror $(q_x, q_y) \mapsto (q_x, -q_y)$, axial spin transforms as $(\sigma_x, \sigma_y) \mapsto (-\sigma_x, \sigma_y)$, which verifies the same covariance. Time reversal reverses both momentum and spin and also permits the term. Inversion reverses momentum but preserves axial spin, so it forbids Eq. (3.84). Rotations alone would also allow other couplings, such as $q_x\sigma_x + q_y\sigma_y$; the mirror is needed to exclude that one.

Example 3.4.4. *Suppose a two-band model has mirror $D(M) = \sigma_z$ and momentum action $(q_x, q_y) \mapsto (q_x, -q_y)$. Conjugation preserves I, σ_z and reverses σ_x, σ_y. Their coefficient functions must therefore be even and odd in q_y, respectively. Through first order the model is*

$$H(q) = (\epsilon_0 + v_0 q_x)I + (m + v_z q_x)\sigma_z + v_x q_y \sigma_x + v_y q_y \sigma_y + O(|q|^2). \tag{3.85}$$

All coefficients are real. If spinless time reversal K acts within this model, it additionally sets $v_0 = v_z = v_x = 0$, while m and v_y remain allowed. In particular, neither mirror nor this time reversal forces $m = 0$, so a crossing at the expansion point would require a further symmetry or parameter tuning.

Example 3.4.5. *Let $D(C_3) = \mathrm{diag}(\omega, 1)$ with $\omega = e^{2\pi i/3}$. Eq.* (3.78) *allows the Hermitian linear coupling*

$$H_{\mathrm{off}}(q) = \begin{pmatrix} 0 & v q_+ \\ v^* q_- & 0 \end{pmatrix}, \qquad v \in \mathbb{C}. \tag{3.86}$$

There is no symmetry requirement that the two diagonal energies agree at $q = 0$, since the states carry different one-dimensional irreducible representations. Also, at a TRIM this two-state space is not closed under spinless time reversal: the state with eigenvalue ω needs a partner with eigenvalue ω^*. One must add that state before imposing time reversal within a single local model.

Proposition 3.4.5. *At any momentum of a local model fixed by an antiunitary symmetry whose square on that fiber is $-I$, every energy level has even multiplicity. In particular, this holds along a line or plane fixed pointwise by such an operation, provided its square is $-I$ at each point of that line or plane.*

Proof. At each fixed momentum Eq. (3.69), or its sewing-matrix version, says that the antiunitary operator commutes with that fiber's Hamiltonian. Apply Theorem 3.3.2 on that fiber. □

Preserving a line as a set is insufficient: the operation must fix the momentum under discussion. Likewise, ordinary time reversal can be imposed within a single expansion only at a TRIM. Elsewhere it relates the models near k_0 and $-k_0$. For a nonsymmorphic antiunitary operation, recompute the square on each fiber; its value at k_0 alone does not establish degeneracy throughout a neighborhood. To interpret a two-band model, write $H(q) = h_0(q)I + \vec{h}(q) \cdot \vec{\sigma}$. The energies are $h_0(q) \pm |\vec{h}(q)|$, so a crossing requires all three components of $\vec{h}$ to vanish. In three dimensions, if a crossing exists at $q = 0$ and $h_i(q) = \sum_j v_{ij} q_j + O(|q|^2)$ with $\det v \neq 0$, it is an isolated linear Weyl point. In two dimensions two independent linear components give a cone, but a permitted constant third component is a mass that gaps it. The D_4 example instead forbids all linear splitting. The identity term shifts both bands and can tilt their dispersion, but does not determine whether their energies coincide.

Exercise 3.4.4. *For two C_4 eigenstates, find the smallest possible degree $r + s$ in Eq.* (3.78) *when $j_\alpha - j_\beta = 0, 1, 2, 3 \bmod 4$. Which mismatch permits both q_+^2 and q_-^2? Explain why an allowed coupling does not itself imply that the uncoupled bands meet.*

Exercise 3.4.5. *Verify Eq.* (3.83) *directly with the Pauli matrices. In the D_4 example, prove that $q_x q_y (q_x^2 - q_y^2)$ transforms as A_2, and show that D_4 together with K actually forbids every σ_y term, not just the fourth-order one. Hint: first use C_2 to fix momentum parity.*

Exercise 3.4.6. *For $H = v_x q_x \sigma_x + v_y q_y \sigma_y + m\sigma_z$, impose the mirror and spinless time reversal of Eq.* (3.85). *Which parameters survive? Does the resulting model necessarily describe an isolated Dirac point? Explain how checking the rank of the remaining linear terms prevents a misleading conclusion.*

In practice we choose the active band space and its little-group action, list monomials to the desired order, and combine them with a Hermitian matrix basis. Eq. (3.75) counts the allowed unitary terms, Eq. (3.76) constructs them, and Eq. (3.69) imposes magnetic symmetry. This connects the abstract decomposition of representations to a concrete local Hamiltonian while keeping the symmetry constraints separate from its material-dependent coefficients.

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